Differentiating AI Revisions from Human-Generated and AI-Generated Texts in Academic Writing

Unterscheidung von KI-Revisionen von menschlich und von KI-generierten Texten im akademischen Schreiben

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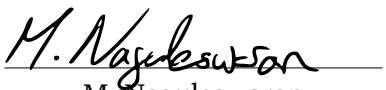
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Zusammenfassung

Künstliche Intelligenz (KI) wird zu einem großen Bestandteil des akademischen und professionellen Schreibens, und ihre Rolle als Sprachassistenz hat den Prozess und die Bearbeitung der Forschung verändert. Während KI-Tools große Vorteile bieten, indem sie Klarheit, Stil und Zugänglichkeit verbessern, können sie die traditionellen Grenzen der Autorenschaft verwischen und Bedenken hinsichtlich der akademischen Integrität und Originalität aufwerfen. Bestehende Forschungs- und Erkennungsmethoden konzentrieren sich in erster Linie auf die Unterscheidung zwischen vollständig von Menschen erstellten Texten (HGT) und vollständig von KI generierten Texten (AGT) und übersehen dabei die nuancierte Kategorie der von KI überarbeiteten Texte (ART). Die Unterscheidung von ART ist notwendig, um die Integrität der akademischen Authentizität aufrechtzuerhalten, eine faire Bewertung zu gewährleisten und das Vertrauen in akademische Texte zu erhalten. Um dies zu erreichen, wird in dieser Thesis *SciHRA-Detect* (Scientific Human, Revised, and Artifical Text Detection) vorgestellt, ein paralleler Datensatz, der Original-Forschungszusammenfassungen (HGT), KI-überarbeite Versionen (ART) und vollständig KI-generierte Version(AGT) derselben Arbeiten abgleicht. Anschließend wurden umfassende linguistische und stilistische Analysen durchgeführt und moderne Klassifizierungsmodelle auf ihre Fähigkeit hin untersucht, zwischen HGT, AGT und ART zu unterscheiden. Die Ergebnisse zeigen, dass Klassifikatoren, die nur auf Encoder-Modellen basieren, eine Genauigkeit von etwa 80% bei der Klassifizierung auf Satzebene erreichen, während fein abgestimmte Large Language Models (LLM) eine Verbesserung von fast 7% bieten, was den Wert einer erweiterten Modellanpassung für diese Aufgabe demonstriert. Diese Studie schließt mit der Diskussion von Einschränkungen und Vorschlägen für zukünftige Forschungen, wobei die Bedeutung einer robusten hybriden Texterkennung für die Gewährleistung von Fairness und Authentizität in akademischen Texten betont wird.

Abstract

As Artificial Intelligence (AI) becomes an integral part of academic and professional writing, its role as a language assistant has transformed the process and editing of research. While AI tools offer great benefits by increasing clarity, style and accessibility, they can blur the traditional authorship boundaries and raise concerns about academic integrity and originality. Existing research and detection tools primarily focus on separating fully Human-Generated Text (HGT) from fully AI-Generated Text (AGT), overlooking the nuanced category of AI-Revised Text (ART). Distinguishing ART is necessary to uphold the integrity of academic authenticity, ensure fair evaluation, and maintain trust in academic writing. To address this, the thesis introduces *SciHRA-Detect* (Scientific Human, Revised, and Artifical Text Detection), a parallel dataset aligning original research abstracts (HGT), AI-refined versions (ART), and fully AI-generated (AGT) versions of the same papers. Next, comprehensive linguistic and stylistic analyses were conducted, and state-of-the-art classification models were evaluated for their ability to distinguish among HGT, AGT, and ART. The results show that classifiers based on encoder-only models achieve around 80% accuracy on sentence-level classification, while fine-tuned Large Language Models (LLM) provide an improvement close to 7%, demonstrating the value of advanced model adaptation for this task. This study concludes by discussing limitations and proposing directions for future research, emphasizing the importance of robust hybrid text detection to ensure fairness and authenticity in academic writing.

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1. Introduction

1.1. Motivation

With the wide availability of Large Language Models (LLM) such as ChatGPT (OpenAI, 2025), many people are using such tools to refine their texts to avoid grammatical errors and improve style and clarity (Hwang et al., 2023). From students preparing academic papers to professionals writing documents, AI-driven tools are increasingly being used to refine language, clarify meaning, improve readability, and aid idea generation. This reflects the growing symbiosis between human intellect and AI in the domain of written communication.

The use of LLMs to refine language without altering the original meaning of human-authored text is generally accepted under prevailing standards of ethics. However, the worldwide introduction of content solely produced by AI has raised legitimate concerns regarding authenticity, academic integrity, and potential propagation of misinformation (Yan et al., 2024). Thus, the increasing reliance on AI in scholarly and professional writing contexts demands careful consideration and nuanced analysis (Miao et al., 2023).

Recent Natural Language Processing (NLP) tool research and development primarily focuses on binary classification with the aim of separating only fully AI-generated text (AGT) from purely human-generated text (HGT) (Pudasaini et al., 2025). This approach fails to account for mixed or collaboratively generated content – human text revised or polished by AI (ART) – which reduces its effectiveness in real-world contexts where AI is extensively utilized for language enhancement. Misclassification of ART as AGT (false positives) can lead to unfair judgments, such as desk rejections in scientific publishing, thereby posing challenges to the adoption of advanced NLP tools and impeding further technical development in the field (Pudasaini et al., 2025; Miao et al., 2023). There is a research gap in analyzing and detecting AI-revised text, differentiating it from purely human-generated and fully AI-generated text.

Addressing this gap is essential because the correct identification of ART is a unique and challenging task that has not been widely investigated in previous literature, where the greatest concern has usually been the identification of pure HGT and AGT (Zhang et al., 2024). This oversight creates a significant loophole in current detection methods and highlights the need for innovative approaches and targeted research to facilitate deeper understanding of mixed-authorship text in the age of AI (Zhang et al., 2024; Abdali et al., 2024).

1.2. Research Goal

In this work, we will present a comprehensive literature review of existing research on AI text detection. Then, we will address the research gap by creating a parallel paragraph/sentence-level dataset of HGT, AGT, and ART. This novel dataset is constructed as the foundation for using original research abstracts as HGT, generating GPT-4o (Hurst et al., 2024) refined abstracts (ART) from the HGT, and GPT-4o entirely AI-generated abstracts

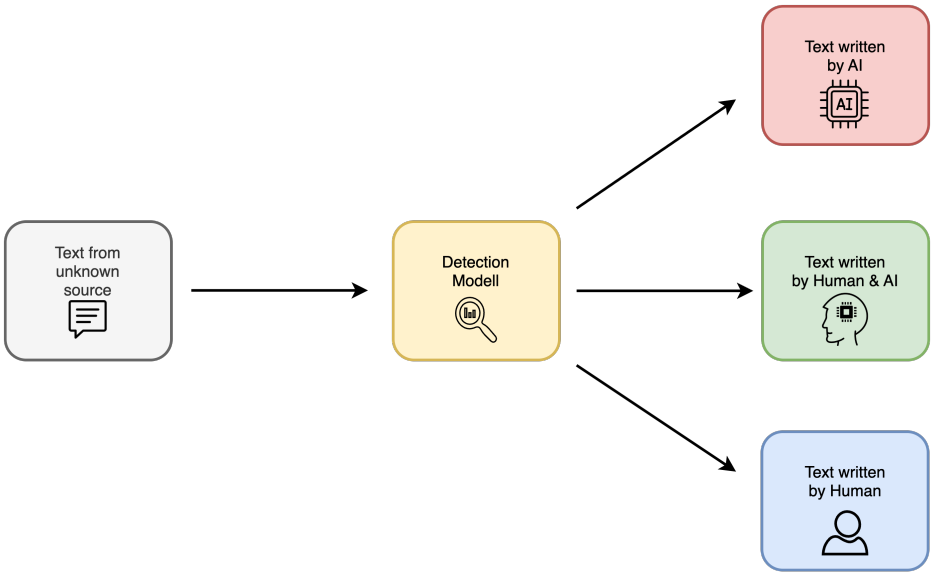


Figure 1.1.: Schematic overview of the text detection framework, where a detection model classifies text from an unknown source as written by Human (HGT), AI (AGT), or by Human & AI (ART)

(AGT) from the main text of the corresponding research papers. This approach ensures topic alignment across the samples, enabling a comparative analysis of the style, structure, and linguistic features of HGT, AGT, and ART (Yadagiri et al., 2024). In addition, this research investigates and compares a range of state-of-the-art methodologies for three-level classification of HGT, AGT, and ART at the paragraph/sentence level (Ruan et al., 2024; Bahad et al., 2024) using our new dataset. This includes fine-tuning LLMs for optimizing their performance in distinguishing between these text categories. Figure 1.1 illustrates the overall structure for AI text detection, showing the flow of text from an unknown source through a detection model to the determination of the text as either AI-written, human-written, or a hybrid of both.

Finally, this thesis aims to provide answers to the following essential research questions:

- How does ART differ from HGT and AGT?
- Can NLP models reliably detect ART?
- Is it feasible or useful to distinguish between HGT, AGT, and ART?

1.3. Structure of the Thesis

This thesis is organized into six chapters. Chapter 2, ”Related Work”, reviews existing AI text detection methods and datasets, highlighting limitations in analyzing hybrid texts. Chapter 3, ”Creation of SciHRA-Detect”, describes the creation of a parallel dataset of HGT, ART, and AGT, including data collection, cleaning, and prompt engineering. Chapter 4, ”Data Analysis”, presents a linguistic analysis of the dataset, examining lexical patterns, frequency distributions, and part-of-speech tag distributions among the three text types. Chapter 5, ”Experiment”, outlines the experimental setup and evaluation of the Baseline and fine-tuned Large Language Model classifiers for the three-level classification task. Chapter 6, ”Conclusion”, summarizes the key findings and addresses the research questions, while Chapter 7 discusses the ”Limitations and Future Work”.

2. Related Work

In this chapter, we provide an in-depth overview of the related work within the field of hybrid text analysis, including existing studies, methodologies, and datasets. We review recent research in AI-generated text detection, highlighting its limitations when handling hybrid text cases. Further, we review available datasets utilized in hybrid text detection and analyze to what degree they could be applied to real-life applications. Finally, we point out the necessity of a more structured parallel dataset and effective classification models to distinguish between HGT, AGT, and ART.

2.1. AI Text Detection Methods

Method	Key Approach	Limitations
DetectGPT (2023), DNA-GPT (2023)	Perturbation-based AI text detection	Surface-level analysis, misses hybrid integration
DIPPER (2023)	Paraphrasing to test model robustness	Vulnerable to minor text rewrites
Manual Fusion (2023; 2024)	Detects transition points between human and AI text	Oversimplifies hybrid text dynamics

Table 2.1.: Summary of key AI text detection approaches and their limitations on hybrid content.

DetectGPT, introduced by Mitchell et al. (2023), creates small, controlled perturbations with the T5 model (Raffel et al., 2020) to analyze the behavior of detection systems. It assumes that minor perturbations in machine-generated text would typically produce smaller base model log probabilities in comparison to the original sample. While such an approach is an improvement on the traditional log-probability-based methods, it relies on threshold-based logit regression in order to limit its robustness/endurance. Similarly, DNA-GPT (Yang et al., 2023) explores perturbation-based methods but on non-English texts and model sourcing. Both approaches, however, primarily treat mostly surface level alterations and not the interaction of hybrid texts.

DIPPER (Krishna et al., 2023) employs paraphrasing attacks to stress-test AI text detectors, such as DetectGPT and GPTZero. By generating semantically consistent but lexically diverse paraphrases, DIPPER significantly reduces detection performance (e.g., DetectGPT's accuracy decreases from 70.3% to 4.6% at a 1% false positive rate). This work points out the vulnerability of current detection models to paraphrasing but does not account for scenarios where human editing and AI inputs are blended together.

There have been studies that have examined manually fusing HGT and AGT to identify changeover points between the two (Dugan et al., 2023; Zeng et al., 2024). Such techniques attempt to pinpoint where human

authorship ends and AI takes over. But they tend to oversimplify the seamless and highly integrated nature of hybrid texts, giving little attentions to their stylistic and structural complexity.

These approaches are summarized in Table 2.1, which outlines the main techniques and their limitation.

2.2. Existing Datasets for Hybrid Text Detection

The availability of hybrid text dataset remains limited, with only a few notable contributions. Beemo (Artemova et al., 2024) is one of them, a benchmark that contains 19.600 texts generated by instruction-finetuned LLM and subsequently edited by professional annotators or other LLMs. This dataset demonstrates that expert-based edits can effectively bypass existing AI-generated text detection systems, emphasizing the difficulty of detection of humanized AI content with existing approaches.

The MixSet (Zhang et al., 2024) is the first dataset designed for three-class classification distinguishing between AGT, HGT, and hybrid texts (ART). Although MixSet highlights the challenges of accurately detecting hybrid texts, its non-parallel design, where HGT and AGT samples originate from separate datasets, limits its utility for comparative analysis. Consequently, most studies using MixSet treat hybrid texts as AGT in binary classification tasks, not taking full advantage of its three class structure.

Another contribution is LLM-DetectAIve (Abassy et al., 2024), whose binary classification approach is complemented by fine-grained multi-class classification of non-academic hybrid texts. Although this dataset provides another perspective on the task, no empirical comparative research has yet been conducted, leaving its practical impact on hybrid text detection largely unexplored.

A comparison of these datasets and their characteristics is listed in Table 2.2, highlighting key distinctions such as dataset size, availability of parallel text pairs, and whether the content originates from academic domains.

Dataset	**Size**	**Parallel**	**Academic Domain**
Beemo (2024)	19.600	No	No
MixSet (2024)	3.600	No	No
LLM-DetectAIve (2024)	Several hundred thousand	No	No
SciHRA-Detect (ours)	9.000 (abstracts) / 60,300 (sentences)	Yes	Yes

Table 2.2.: Comparison of datasets for hybrid text detection. Our dataset, SciHRA-Detect, is the only one designed for parallel analysis in the academic domain.

These limitations underscore the requirement for a well-structured parallel dataset to facilitate the study of hybrid text characteristics. A reasonably well-maintained dataset would enable researchers to explore differences between AGT, HGT, and hybrid texts in terms of linguistic features, structural differences, and stylistic variations. Further research is also needed to test whether hybrid text detection models could be trained using real-world applicable data, particularly in academic and professional settings. Such research would provide valuable insights into the real-world implications of hybrid text classification and its potential role in content verification and authorship attribution. Our proposed dataset, *SciHRA-Detect*, directly addresss these gaps by providing aligned abstract-level and sentence-level instances, built specifically from academic publications to include realistic use cases.

3. Creation of *SciHRA-Detect*

To successfully analyze ART in scientific writing, we developed a well-structured dataset *SciHRA-Detect* (Scientific Human, Revised, and Artifical Text Detection). Since our focus is academic writing, we specifically used research paper abstracts as our primary text structure. This chapter details the data preprocessing pipeline for creating the different text types: Human-Generated Text (HGT), AI-Revised Text (ART), and AI-Generated Text (AGT).

3.1. Data Collection

Data Sources

We compiled our dataset from five existing collections of research papers: NLPeer (Dycke et al., 2022), ArXiv[1], PubMed[2], and JSTOR[3]. To ensure the absence of AI influence in the original writing, we carefully selected papers published before 2020, predating the release of ChatGPT (Naveed et al., 2023). Table 3.1 presents an overview of our dataset composition.

Source	Number of Papers
NLPeer	141
ArXiv	1659
PubMed	300
JSTOR	900
Total	**3,000**

Table 3.1.: Composition of our dataset by source.

Domain Distribution

To ensure diversity, we balanced the dataset to 10 categories with 300 papers per category. Figure 3.1 illustrates the distribution of these categories.

[1] https://arxiv.org/
[2] https://pubmed.ncbi.nlm.nih.gov/
[3] https://constellate.org/

To achieve a balanced representation across domains, we combined computer science papers from NLPeer (141 papers) with computational language papers from ArXiv to reach the 300-paper threshold for the CS category. JSTOR contributed papers in Area Studies, Social Sciences, and Business & Economics. PubMed covered the Biological Sciences & Medicine domain (PMC). ArXiv provided papers for the remaining natural sciences categories, including Nonlinear Sciences (nlin), Condensed Matter (cond-mat), Mathematics (math), Statistics Applications (stat.AP), and Astrophysics (astro-ph).

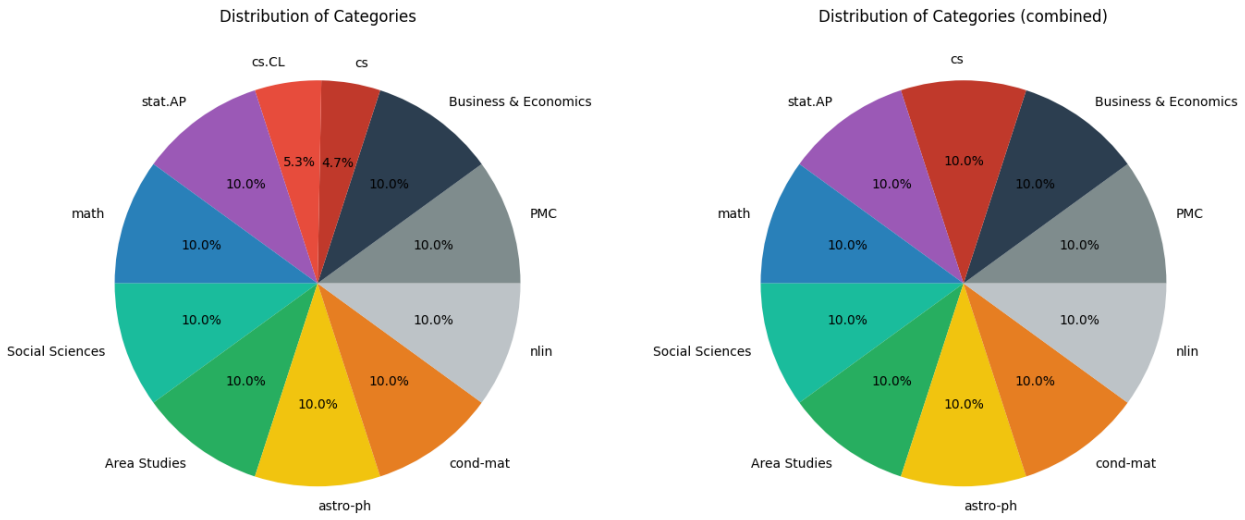


Figure 3.1.: Category distribution of the dataset before(left) and after (right) balancing, showing the equal representation of ten scientific domains used for analysis.

3.2. Dataset Cleaning

Cleaning was crucial to ensure text quality, balance, and usability for further processing. The abstract refers to the summary of the paper, while the full text includes the entire content of the paper, excluding the abstract part. The following steps were applied:

Length Filtering

We applied length filtering to remove papers with extremely short or long abstracts or full texts. such that the dataset possessed consistent quality and removed inconsistencies that could hinder further analysis.

- **Abstract/Fulltext Length**: Papers with extremely short or long words counts in both abstract and full text were removed using a 95% standard deviation threshold (see Figure 3.2 for an example).
- **Full Text Token Limit**: Full Text exceeding 20,000 tokens were discarded to maintain consistency across categories and prevent API limitations for AGT and ART generation.
- **Abstract Token Limit**: Removed abstracts with more than 455 tokens (350 words) to balance the dataset, particularly addressing longer abstracts from sources like JSTOR.

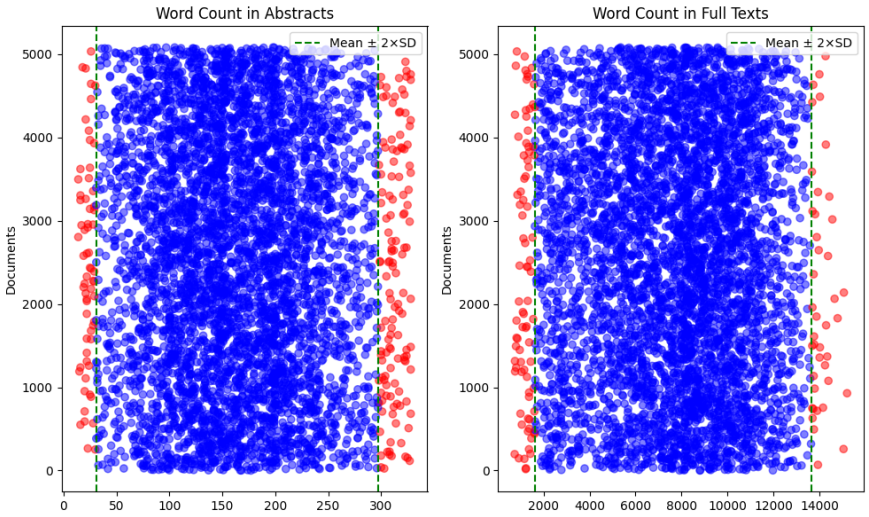


Figure 3.2.: Example of length filtering on the JSTOR dataset: Scatter plots of word counts in abstracts (left) and full texts (right), with outliers beyond 95% standard deviation threshold (red) identified and removed to ensure text quality and consistency.

Overlap Removal

We used the **Rouge-L similarity** (Lin, 2004) to filter papers with significant overlap between the abstract and the first paragraph of the full text, adjusting thresholds manually (see Table 3.2). This ensured that they didn't contain similar phrases, preventing biased AGT generation, as the AI should create the abstract independently.

Dataset	Rouge-L Threshold
ArXiv	70%
PubMed	60%
JSTOR	75%
NLPeer	40%

Table 3.2.: This table lists the manually selected Rouge-L similarity thresholds used to filter out papers with high textual overlap between abstract and the beginning of the full text for each dataset source.

Removing Noisy Text

We removed papers with lots of LaTeX commands, math placeholers or multilingual abstracts to improve overall data quality and reduce irrelevant or confusing content.

- Papers with excessive LaTeX commands (e.g., "`\usepackage`") were removed.
- Papers containing excessive placeholder math expressions (e.g., "`@math0`") exceeding 1000 instances were filtered out) to improve data quality.
- Multilingual papers (containing multiple language versions of abstracts) were discarded to avoid confusion in analysis.

3.3. Prompt Design for Dataset Creation

To generate ART and AGT, we used ChatGPT-4o (Hurst et al., 2024) API[1] with carefully designed prompts tailored for each text type.

Both prompts emphasized improving clarity, coherence, and alignment with academic standards while maintaining the original structure and meaning. The key difference between ART and AGT prompts lies in their roles: ART focuses on revision while AGT involves abstract generation from scratch based on full-text content. Additional guidelines for writing good abstracts were provided within the AGT prompt (The Writing Center, University of Wisconsin–Madison, 2025). These guidelines were to ensure the language model produces abstracts that closely adhere to academic convention and are in the style of HGT and ART.

AI-Revised Text (ART) Prompt

The input for ART was the original abstract. The AI acted as an **academic writing assistant** refining human-written abstracts:

AI-Revised Text (ART) Prompt

> You are an AI academic writing assistant specializing in revising academic abstracts. Your task is to revise and polish the provided abstract (100–250 words) to improve its clarity, coherence, and alignment with the highest academic standards for the domain ...

(See Appendix A for full prompt details.)

AI-Generated Text (AGT) Prompt

For AGT, the input was the full text without its abstract. The AI assumes the **role of the paper's author** and generates a concise abstract:

AI-Generated Text (AGT) Prompt

> You are an AI acting as the author of a research paper. Your task is to generate a concise, polished abstract (100–250 words) based on the provided full text ...
>
> 1. The context or background of the study, explaining its relevance or importance.
> 2. The research problem, objective, or central question addressed by the study.
> 3. The methods or approach used to conduct the research.
> 4. The key findings, results, or arguments derived from the study.
> 5. The implications, significance, or contributions of the research to the field.

(See Appendix A for full prompt details.)

[1] https://platform.openai.com/docs/models/gpt-4o

What is a Good Abstract?

A well-written abstract is a concise yet comprehensive summary of a research paper that enables readers to catch its meaning and relevance (Andrade, 2011). Depending on established guidelines and the specific requirements of academic writing, a good abstract typically contains some key elements, each fulfilling a distinct function. Such as stating the motivation/background, describing the objectivs and methods of the paper, presenting key results, and highlighting the implications/conclusions (Andrade, 2011; Dernoncourt and Lee, 2017). Since the LLM was generating an abstact from the body text and not from an existing abstract for the AGT, it needs some guidance to be made a summary text. We incorporated these guidelines into the AGT prompt so that the generated abstract would be more organized and similar in style to the HGT and ART versions. This consistent structure also enabled us to create a foundation for conducting more fine-grained sentence-level analysis in the later stages of our study.

3.4. Sentence-Level Dataset

After creating the Baseline model for the classification section, it was realized that the abstract derived from the three different types text is very simple. The model is overly accurate (see Table 5.2). To complicate this study, Sentence-Level Classification was implemented, for which there was a requirement to modify the dataset. We pulled out every sentence of every text type and labeled it according to its text type. This resulted in a dataset containing only sentences. However, this created a class imbalance, as ART had longer abstracts. Table 3.3 shows the count distribution of the newly formatted dataset across the different text types.

As with the abstract dataset, we split the data into training, testing and validation sets based on the article ID, such that text types from the same paper are in the same set. In this way, we ensure that, during the experiment, the models learn to distinguish not just based on content, but also on stylistic and linguistic features.

Text Type	**Count**
Human Generated Text (HGT)	18.314
AI Revised Text (ART)	16.939
AI Generated Text (AGT)	25.047
Total	**60.300**

Table 3.3.: Number of sentences for each text type in the dataset after splitting abstracts into individual sentences.

With the new approach and dataset, the accuracy of the baseline models has decreased.

4. Data Analysis

In this chapter, we analyze the final version of our dataset *SciHRA-Detect* to uncover linguistic, stylistic, and structural differences between every text type. We begin at abstract-level, followed by fine-grained sentence-level analysis using rhetorical sentence labels to investigate structural tendencies and transitions. Finally, we perform a domain-specific comparison analysis to identify potential unique patterns based on authorship type.

4.1. Lexical Analysis

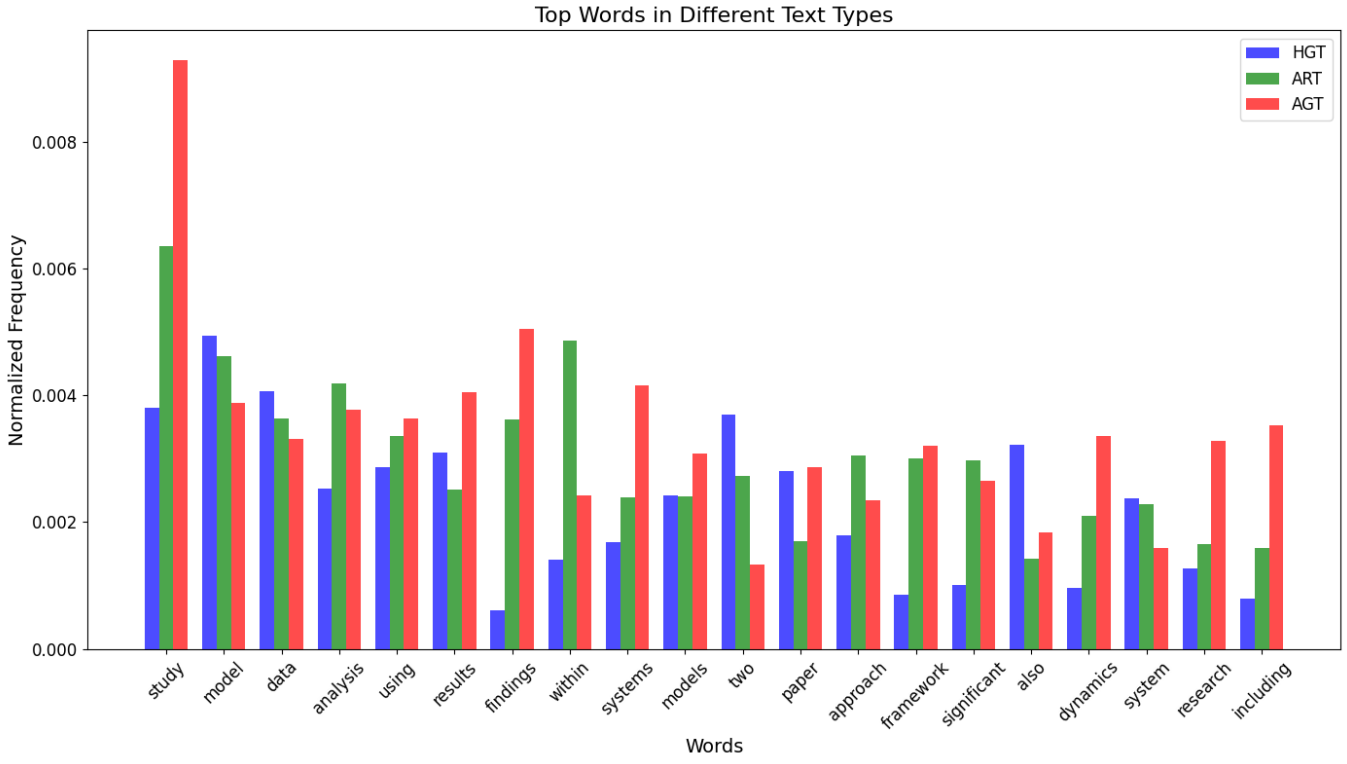


Figure 4.1.: Normalized frequency of the top common words in different text types.

We examined the ten most frequent words across different text types (see Figure 4.1). Our analysis revealed that the frequency of these words in HGT and ART tends to be quite similar. However, notable exceptions were observed, where substantial differences emerged, particularly for the terms ”study”, ”findings”, and ”within”, which appeared more frequently in ART and AGT. These lexical patterns can be categorized within scientific

discourse. This suggests that AI enhances HGT by incorporating a greater frequency of scientific vocabulary (Wordvice, 2023) while reducing the occurrence of non-scientific terms, because the length between HGT and ART is close. Examining the top ten words further indicated a tendency towards more scientific language in AI, a trend even more dominant in AGT. Specifically, the words ”study”, ”systems”, ”research”, and ”including” occurred significantly more frequently in AGT compared to the other text types.

Interestingly, HGT and ART both have relatively similar distributions for the most common words, with ART demonstrating a slight shift toward more formal scientific terminology compared to purely HGT. This pattern indicates that ART prefers to preserve much of the original human lexical distribution while selectively enhancing scientific terms.

The bigram frequency distribution (see Figure 4.2) provides yet stronger evidence for stylistic differences. Scientific process verbs paired with research nouns (e.g., ”study examines”) appear with substantially higher frequency in AGT (Wordvice, 2023). Specifically, ”study investigates” shows a normalized frequency approximately 20 times higher in AGT than in HGT, with ART falling between these extremes.

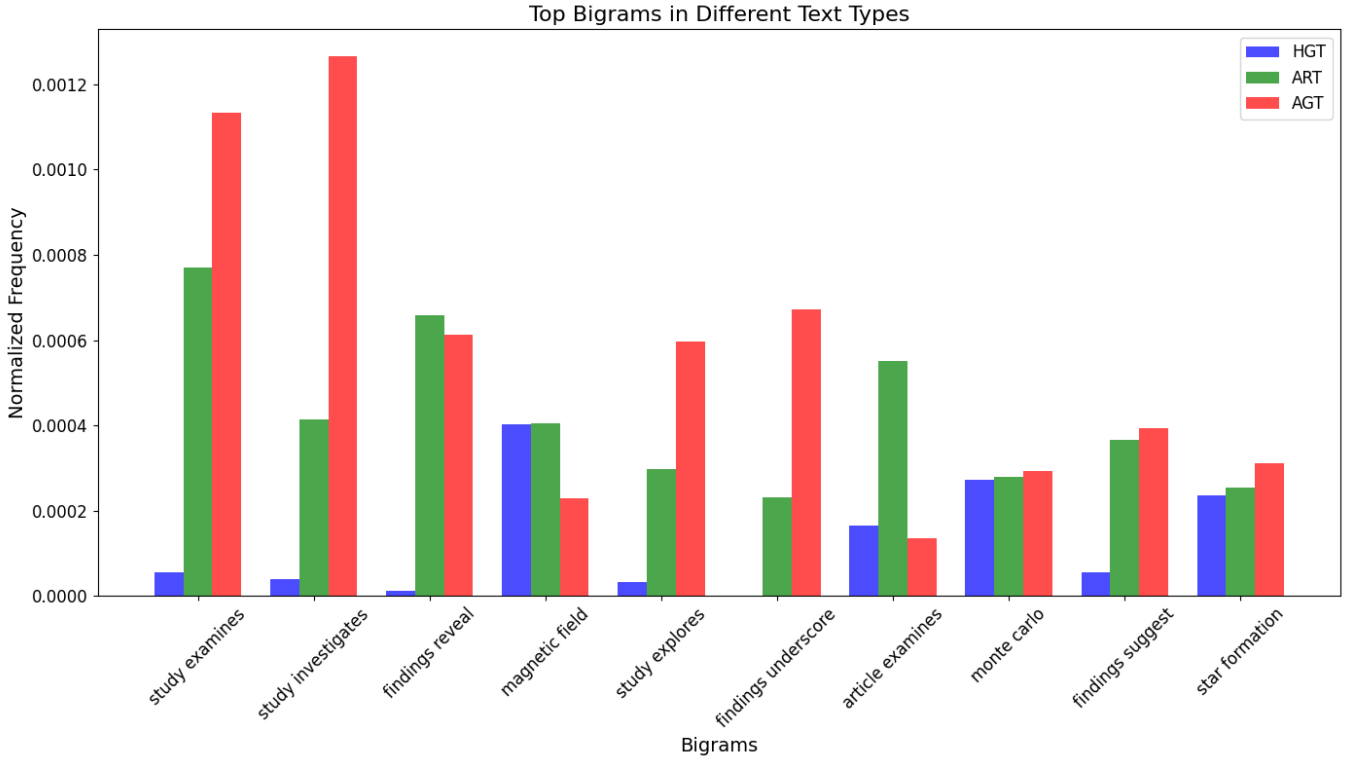


Figure 4.2.: Normalized frequency of the top bigrams in different text types.

These bigrams can be categorized into two functional groups (Wordvice, 2023):

- **Investigative expressions:** ”study examines”, ”study investigates”, and ”study explores” occur substantially more frequently in AGT than in HGT, with ART showing intermediate values.
- **Inferential expressions:** Phrases like ”findings reveal”, ”findings underscore”, and ”findings suggest” appear with greater frequency in AGT, demonstrating AI’s tendency toward explicit result discovery.

Technical term bigrams like ”magnetic field” and ”monte carlo” show more consistent usage across all text types, suggesting domain-specific terminology remains stable regardless of authorship. However, both AI texts demonstrate a clear preference for formal scientific expression patterns that explicitly frame research activities and findings.

We also analyzed the top trigrams across the dataset, which revealed a strong pronounced presence of domain-specific technical terms such as “natural language processing” and “monte carlo simulations.” Notably, both AGT and ART exhibited a significantly higher frequency of these specialized terms compared to HGT. Moreover, a comprehensive analysis of the most frequent unigrams within each domain was conducted. The detailed findings of these results are documented in Appendix B.

4.2. Frequency Analysis

4.2.1. Sentence / Text Structure Analysis

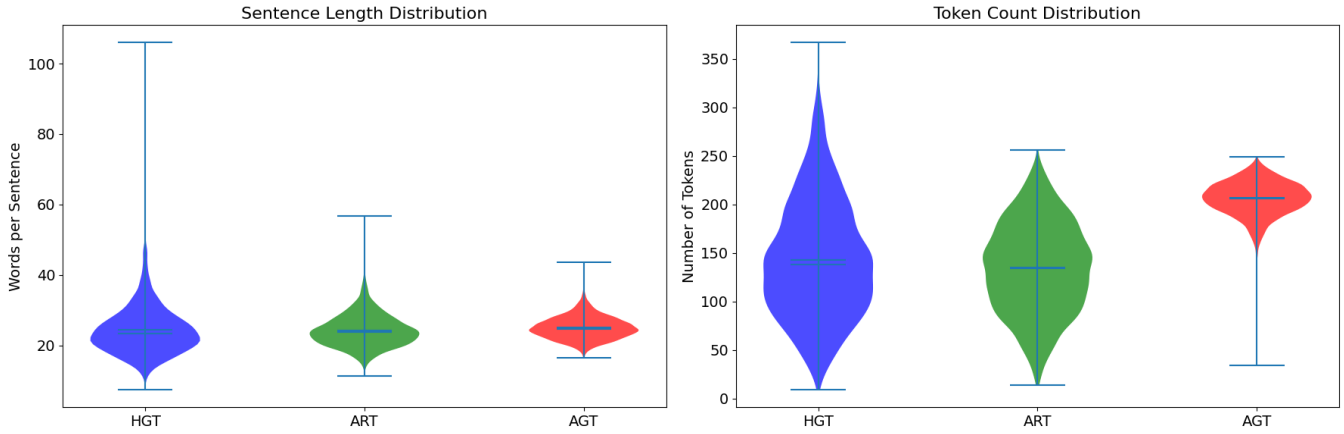


Figure 4.3.: This figure shows the distribution of sentence lengths (left) and token count of the entire abstract (right) for each text type.

The left plot in Figure 4.3 displays the distribution of sentence lengths for each text type, measured in words. In this plot, each data point represents the length of a single sentence, revealing Sentence length analysis reveals fundamental differences in syntactic construction across the text types. HGT has the widest variance in sentence length, with some sentences exceeding 100 words, yet most of them have 20-25 words per sentence. This widespread reflects the natural variability in human writing, with authors varying between concise statements and more complex constructions. Additionally, ART and AGT presented narrower distributions with smaller variances and fewer outliers (around 23–25 words per sentence), indicating more consistent sentence lengths. This constrained variance suggests that AI systems implement more consistent sentence structure with fewer very short and very long sentences. AI-revised text is in between, showing that the revision process partially normalizes human sentence length variation while retaining some of the original structural characteristics.

The right plot in Figure 4.3 shows the distribution of token counts for the entire abstracts and provides additional details on the overall text length pattern. Each data point corresponds to the total number of tokens present in a single abstract, where a token is defined as a word or punctuation mark according to standard

natural language processing tokenization. AGT has significantly less variance compared to the other types of texts, despite the prompt specifying the same word count for both the ART and AGT (Prompts are in Appendix A). HGT and ART are more similar in token count distribution. However, HGT exhibited more outliers, whereas ART had high alignment with HGT, with a tendency towards shorter texts. ART was closer to HGT in terms of token count but had shorter outputs. AGTs display a narrow concentration around 200–210 tokens with minimal variance. This indicates AGT tends toward a more even density of information representation, while human writers exercise greater freedom in text length.

4.2.2. Part-of-Speech Tag Distribution

Next, we estimated the Part-of-Speech (POS) tag distribution (Petrov et al., 2011). All the text forms show nouns (NN) as the most frequent POS category, although with some variations in frequency. HGT demonstrates the highest normalized frequency of nouns (approximately 0.34), followed by ART (0.30) and AGT (0.29). This suggests that human writers may rely more on nominal constructions when presenting scientific concepts. The distribution of adjectives (JJ) remains relatively consistent across all text types (approximately 0.22-0.23). However, plural nouns (NNS) show a clear progression, with AGT exhibiting the highest frequency (0.20), followed by ART (0.17), and HGT showing the lowest (0.15). This pattern reflects that more plural forms are used in AGT. The remaining top POS tags have roughly the same normalized value across all text types. We can safely conclude that the POS tag does show no substantial differences between the text types, except perhaps in NN or NNS. (See Appendix B for the Visual)

4.3. Stylistic & Linguistic Analysis

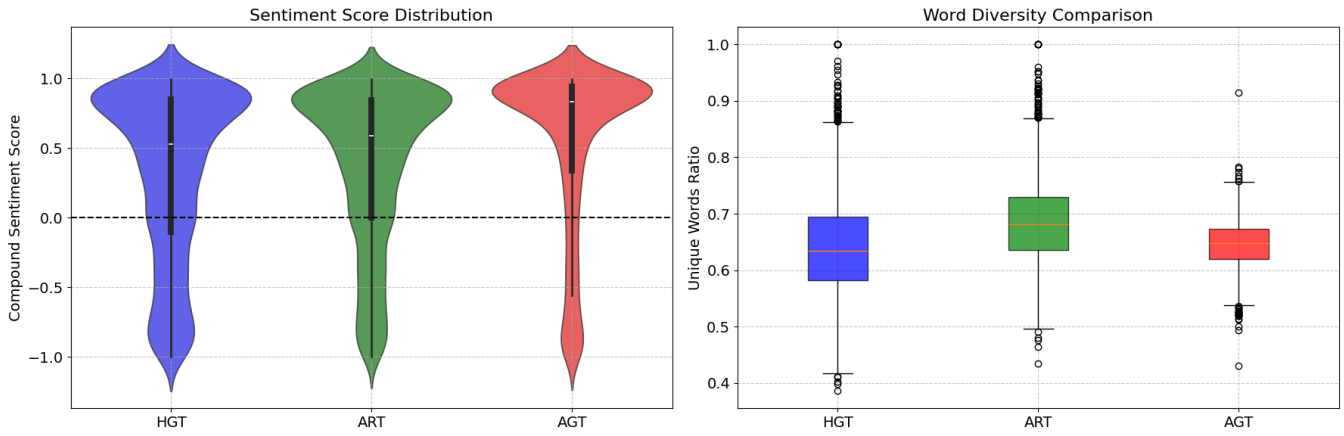


Figure 4.4.: This figure shows the sentiment score distributions (left) and word diversity ratios (right) across different text types.

4.3.1. Sentiment Analysis

Figure 4.4 (left plot) demonstrates the different sentiment profiles (Gujjar and Kumar, 2021; Pang et al., 2002) of HGT, AGT, and ART. The compound sentiment score for every abstract is graphed along the vertical axis.

The scores reflect the overall emotional tone of the texts, with positive value indicating positive sentiment, negative values indicating negative sentiment, and zero being neutral. The width of each violin represents the density of abstracts at different sentiment levels. All three types show generally positive scores, yet the AGT has a higher positive sentiment distribution with a more concentrated density of scores in the positive range (0.5 to 1.0) and fewer negative sentiment instances. The positive tendency in AGT suggests that AGT is less emotionally nuanced, leaning towards a more positive sentiment, possibly due to the fact that AGT was trained to be friendly and inoffensive (Krugmann and Hartmann, 2024). That is also why both AI texts tend to be more likely to possess a higher median sentiment score (AGT ca. 0.8, ART ca. 0.65, and HGT ca. 0.5). Human-generated text shows a more balanced distribution of sentiment scores with a greater dispersion in the negative range. ART shows a sentiment profile that is very similar to HGT. The outliers of ART and HGT are very similar, with the exception that there are slightly more positive-sounding abstracts in ART, which could indicate a limitation in the way current AI systems express critical or negative perspectives that are often essential in academic discourse.

4.3.2. Word Diversity Analysis

The boxplot shown on the right side of Figure 4.4 compared the word diversity across the different text type. In this analysis, word diversity is the ratio of unique words to the number of words in each abstract. The box shows the interquartile range, the line shows the median, and the outliers dots represents the overall spread.

Word diversity analysis indicates that ART has the highest median proportion of unique words (0.68), followed by AGT (0.65), whereas HGT possesses the lowest (0.63). However, AGT has the narrowest diversity range (approximately 0.55 to 0.75), reflecting a more uniform lexical structure. In contrast, HGT demonstrates the widest variability (between approximately 0.42 and 0.88), which may be representative of the natural word and style variation of human authors.

While both AI texts (ART and AGT) possess slightly greater mean scores in word variety, both have reduced variability compared to texts produced by humans. This reduced variability may be attributed to the lexical consistency of language models, which are motivated to prefer grammatical correctness and coherence over creative variation. This is consistent with results from Muñoz-Ortiz et al. (2024).

It's important to mention that ART combines the strengths of each approach, using the lexical richness of AGT without compromising the greater variability closer to HGT. This finding lends support to the hypothesis that AI revision has the potential to enhance human-written text through increasing lexical precision without entirely giving up stylistic variation.

4.3.3. Reabability Analysis

Readability metrics (Flesch, 1948) reveal substantial differences in how accessible each text type is to readers, with implications for academic communication.

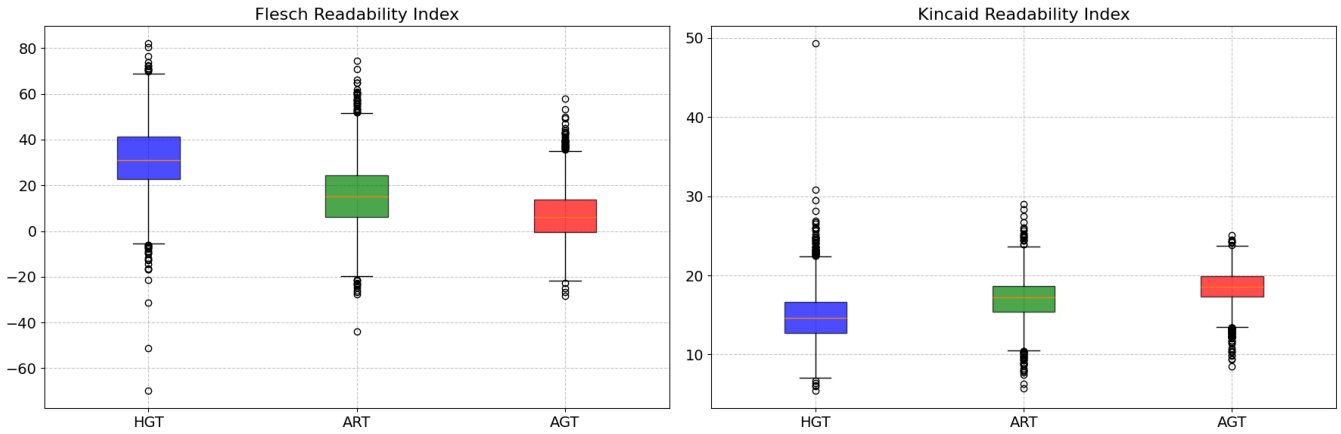


Figure 4.5.: This figure plots each text type's Flesch and Kincaid readability indices.

Flesch Readability Index

The Flesch Readability Index or Flesch Reading Ease Score (Flesch, 1948) is a measure of how easy a text is to read. In other words, it measures the accessibility. It is based on average sentence length and average number of syllables per word. Scores typically range from 0 to 100. Negative readability scores result when sentences are unusually long (i.e., high words-per-sentence) and words have unusually high complexity (i.e., high syllables-per-word).

The comparison in Figure 4.5 indicates that HGT have the highest median score (30), with ART (15), and AGT showing the lowest median score (5). Since higher Flesch scores represent easier readability, this suggests that human authors generally write more reader-friendly abstracts than their AI counterparts.

HGT have the highest (80) and the lowest (-70) outliers, yet the lower-end of the HGT variance tends to cluster nearer zero than the other text types. AGT and ART have lower-end variances that are further into the negative range (around -20), considerably lower than HGT. The negative scores reflect the diversity of human writing approaches, while AI systems may produce more consistently formal academic text.

Kincaid Readability Index

The Kincaid Readability Index or Flesch-Kincaid Grade Level (Flesch, 1948), which estimates the years of U.S. education needed to understand a text, shows an opposite trend. It is also calculated from sentence length and word complexity, but is scaled to correspond to educational grade levels. Higher Kincaid scores indicate more complex text requiring more education to comprehend. The score 8 can be equally set to the Harry Potter books, 15 is for an academic paper. AGT has the highest median score (19), followed by ART (17), with HGT registering the lowest score (15).

The consistently higher Kincaid scores for both AI texts suggest that they prefer more complex sentence structures, potentially making the text less accessible to broader audiences (see right plot in Figure 4.5). This aligns with research showing that AI essays employ a broader vocabulary and engage with complex language constructions more frequently than their human counterparts (Muñoz-Ortiz et al., 2024).

Readability Trade-offs

Seeing both indices, we can see that AI enhances the complexity and readability of the text. Humans tend to write in a style that is more accessible to a wider audience, to gain more readers, and make it easier to understand. The readability analysis also reveals an interesting trade-off in AI-generated content: while typically scoring higher on formal complexity metrics (Kincaid), it scores lower on accessibility metrics (Flesch). What this suggests is that AI-generated academic abstracts might be less accessible to broader audiences, but they maintain the formal academic tone appropriate to scholarly publications.

4.3.4. Common Replacement from HGT to ART

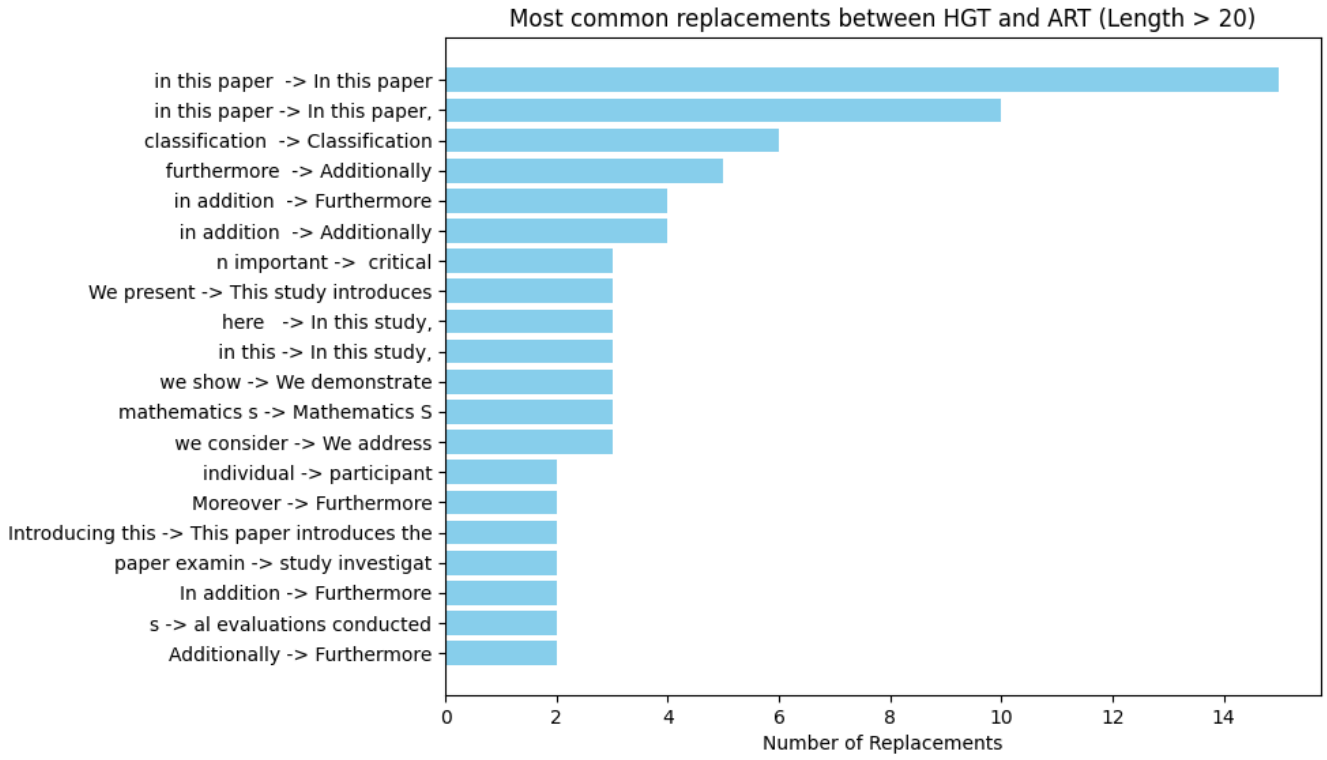


Figure 4.6.: Most common phrase replacements longer than 20 characters between HGT and ART, highlighting substantive linguistic and stylistic changes.

This section presents a detailed analysis of the transformations applied during the revision process from HGT to ART. By analyzing the purpose and nature of textual modifications, this section uncovers the strategies the AI applied in improving clarity, coherence, and academic rigor. The revisions have consistent improvements in sentence structure and academic tone.

One of the trends observed when manually comparing HGT and ART of the same paper is the reorganization of sentences during revision to improve logical flow and eliminate small redundancy. The AI achieved this by systematically refining less scholarly phrasing by merging or splitting sentences to better represent conventional norms in academic writing. For example, imprecise constructions like ”This study aims to examine” were

revised to more grammatically correct alternatives like ”This study explores”. These corrections not only refine correct syntactic errors but also enhance coherence and flow.

To further illustrate the character of such revisions, Figure 4.6 shows a bar chart of the most frequent observed replacements between text chunks larger than 20 characters. This threshold was selected to exclude substantive revisions from minor changes such as capitalization or spacing. Transitional phrases such as ”furthermore” and ”in addition” were replaced which provided more to a smoother and professional tone. Different verbs were used, and introductory phrases such as ”in this paper” and ”in this” were rephrased to meet conventional academic wording.

4.3.5. Addtional Metric Analysis

To assess more stylistic and linguistic differences among HGT, ART, and AGT, we conducted an extensive stylometric analysis. This section provides a comprehensive evaluation of key linguistic metrics, focusing on their role in distinguishing the three text types and drawing interpretive insights into each text type’s behavior. All metric values and results discussed in this section are sourced from Table 4.1.

Metrics	**HGT**	**ART**	**AGT**
Passive Voice Ratio	0.34	0.28	0.15
Ratio Was/Were	0.61	0.43	0.45
Subject Clauses	1.40	0.99	1.28
Hapax Legomena	67.91	74.94	109.96
MATTR	0.78	0.82	0.85
Dugast’s Uber Index	55.22	66.06	67.88
Yule’s K	0.0152	0.0135	0.0127
Shannon Entropy	6.05	6.15	6.65
Burstiness	0.36	0.28	0.26
Coherence Score	0.3941	0.4566	0.4532

Table 4.1.: This table presents the average values of selected stylometric and linguistic metrics comparing the three text types.

Voice and Verb Usage

One of the most distinctive stylistic features is the use of passive voice (Ahmad, 2012). In passive voice construction, the subject of the sentence receives the action rather than performing it (e.g., ”The experiment was conducted by the team” instead of the active ”The team conducted the experiment”). It is determined by identifying syntactic constructions known to be passive (e.g., ”nsubjpass”) using dependency parsing. An interesting decrease in passive constructions is observed from HGT (0.34) to ART (0.28), and further to AGT (0.15). This suggests a clear tendency of AI models, both in revising and generating text, to favor more direct expression. The ”was/were” ratio (Dublin City Univercity, 2020) indicates how often the auxiliary verbs “was” and “were” appear in a text relative to the total number of verbs. These verbs commonly occur in

passive constructions and act as a surface-level indicator for passive structures or past-tense formulations. The decreasing ratio from HGT (0.61) to ART (0.43) and AGT (0.45) aligns with the previous finding, that AI content leans towards more active phrasing.

Clause Structure

Subordinate clauses, also known as dependent clauses, are a group of words that add information to the main clause of a sentence. It cannot be an independent sentence on its own. It can be identified by seeing at the dependency parsing with the label called ”ccomp”. Subordinate clauses often indicate hierarchical sentence structuring and argumentational complexity. HGT (1.40) and AGT (1.28) show higher mean use of subordinate clauses per sentence than ART (0.99), suggesting that HGT and AGT retain more complex sentence structuring. In contrast, the reduced use of subordination in ART is very surprising.

Lexical Richness and Diversity

Lexical richness provides substantial insight into the stylistic variance between the three sources. Hapax legomena (Popescu and Altmann, 2008) counts the number of words that appear only once in a text. A higher count indicates greater lexical variety, as more unique words are used. The metric rises from HGT (67.91) to ART (74.94) and most significantly AGT (109.96), reflecting that AGT introduces more lexical variation. To assess the depth of vocabulary usage beyond simple frequency counts, we incorporate several established metrics:

- MATTR (Moving-Average Type Token Ratio) measures lexical diversity by calculating the average ratio of unique words to total words in a fixed sliding window (size=50) across the text (Covington and McFall, 2010). AGT scores the highest (0.85), which indicates consistent lexical variety throughout the text.
- Dugast’s Uber Index (Tweedie and Baayen, 1998) applies logarithmic scaling to relate vocabulary size to text length, with higher values indicating a denser and richer vocabulary. AGT (67.88) outperforming again, suggesting a richer, denser vocabulary.

We also use Yule’s K (Tanaka-Ishii and Aihara, 2015), a measure of vocabulary repetition, where lower values indicate higher diversity and less repetition. AGT shows the lowest K (0.0127), supporting the finding of greater lexical richness.

These measurements all confirm the pattern of AGTs having the most varied and non-repetitive vocabulary. ART consistently falls between HGT and AGT, highlighting its role as a more polished but hybrid form of the original human-authored content. However, it is important to note that increased diversity does not automatically lead to improved coherence or clarity (McNamara et al., 2010).

Entropy, Burstiness, and Structural Variability

Shannon entropy (Shannon, 1951) measures the unpredictability of word choice based on word frequency distributions. It is calculated by applying a standard entropy formula over the probabilities of individual word occurrences. The formula is:

$$H = -\sum p(x_i) * \log_2(p(x_i))$$

where $p(x_i)$ is the probability of each word in the text. A larger value indicates a more unpredictable, information-rich text. Once again, AGT scores highest (6.65), followed by ART (6.15) and HGT (6.05), possibly reflecting the widespread lexical choices made by generative models.

Burstiness, on the other hand, measures sentence length variance, reflecting the stylistic rhythm of the text. It is measured as the standard deviation of sentence length normalized by the average sentence length. HGT (0.36) show the greatest burstiness, highlighting natural fluctuations in structure and pacing. AGT (0.26) and ART (0.28) texts show more uniform sentence structures, a trademark of machine-generated rhythm.

Coherence and Flow

Finally, the most insightful differentiator is the coherence score, which provides a semantic view of textual flow. It's calculated by taking the average cosine similarity between adjacent sentence embeddings using a pre-trained language model. The pre-trained language model is all-MiniLM-L6-v2[1], which is a popular sentence-transformer perfectly suited for this task. AGT (0.4532) and ART (0.4566) outperform HGT (0.3941) in this metric, suggesting smoother transitions and stronger surface-level cohesion in AI-generated or revised texts. However, these higher scores can be deceptive. While the transitions are smooth, they may lack the argumentative nuance characteristic of human writing. HGT's slightly lower coherence score suggests a more organic, and sometimes less predictable, flow of ideas.

Conclusion

Every measure introduced makes a unique contribution to stylistic preference across each text type. Voice and clause structures reveal syntactic preferences, with AI preferring assertive and simplified constructions. Lexical richness and entropy determine AGT's diverse vocabulary, while burstiness and coherence determine rhythm and logical flow differences.

[1] https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2

4.4. Sentence-Level Analysis

To conduct a more nuanced and comparative analysis of scientific abstracts, we labeled each sentence using a fine-tuned SciBERT model (Beltagy et al., 2019). SciBERT, an encoder-only transformer-based language model pre-trained on a large corpus of scientific papers, is well-suited for tasks in this domain. The model was fine-tuned on domain-specific datasets, which were created from PubMed22k (Dernoncourt and Lee, 2017) and CS Abstracts (Gonçalves et al., 2020), and merged to create a comprehensive training corpus. The specific fine-tuning details and training parameters can be found in the Appendix E Table E.1. Each sentence was categorized into one of five functional labels: **BACKGROUND**, **OBJECTIVE**, **METHODS**, **RESULTS**, and **CONCLUSIONS**. This label schema follows standardized abstract writing guidelines (The Writing Center, University of Wisconsin–Madison, 2025) and enables fine-grained structural and functional comparison across different types of texts.

Label	Count
Background	30389
Objective	19334
Methods	80430
Results	78315
Conclusions	36649
Total	**245117**

Table 4.2.: This table shows the number of sentences assigned to each sentence label in the dataset used for the sentence classifier.

Given that the distribution Table 4.2 of these sentence labels is unbalanced in the training data, due to naturally occurring imbalances in abstract structures, we applied class weighting to decrease potential biases in the classification model. For future research, more balanced training datasets could potentially enhance classification performance and stability.

4.4.1. Label Frequency Distribution

The Figure 4.7 highlights a key observation: HGT and ART show remarkably similar distributions among all five sentence labels. This means that the AI revisions primarily affect lexical or syntactic features, while preserving the semantic and structural integrity of the original HGT. In contrast, AGT reveals distinct differences. Most obviously, AGT exhibits a significantly higher frequency of CONCLUSIONS-labeled sentences, indicating a strong focus on summarization and end statements. Additionally, METHODS are underrepresented in AGT compared to HGT and ART, suggesting that generative models might deprioritize procedural content. AGT also seems to include longer OBJECTIVE sections, possibly reflecting an overcompensation in framing or intent-setting. The BACKGROUND category is also underrepresented in AGT compared to the other text types. These results suggest that AGT has a tendency to prioritize conclusions rather than background/method information, reflecting its focus on goal-oriented content.

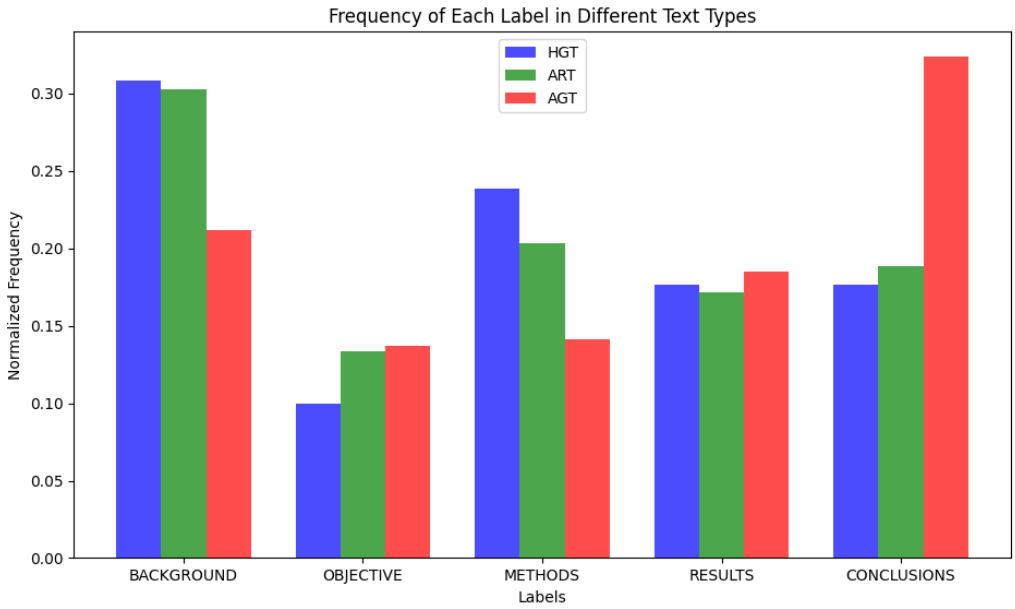


Figure 4.7.: This figure shows the distribution of sentence labels across different text types.

4.4.2. Top Label Transitions

To explore the structural flow of abstracts, we analyzed the top 10 most frequent label-to-label transitions for each text type (see Figure 4.8). In this context, a transition refers to the sequential movement from one sentence label to another within the abstract. Once again, HGT and ART remain closely aligned in transition patterns. A particularly striking difference is the appearance of the transition ”BACKGROUND → CONCLUSIONS” in HGT only. This unusual jump, bypassing METHODS and RESULTS, did not appear in the ART or AGT top 10 transitions, likely because these systems follow more rigid and conventional structuring. This human-authored transition may reflect flexibility or context-dependent summarization found in authentic scholarly writing. In AGT, the most dominant transition is ”CONCLUSIONS → CONCLUSIONS”, confirming the earlier observation that AGT abstracts emphasize summarization and conclusion. Other prominent transitions in AGT include ”RESULTS → CONCLUSIONS” and ”METHODS → RESULTS”, suggesting a streamlined narrative that prioritizes outcomes over methodological detail. These tendencies point to an output style in AGT that favors informativeness and clarity at the expense of comprehensive procedural reporting.

We analyzed label-3 transitions to capture deeper sequential structures (see Figure 4.9). AGT again shows a preference for CONCLUSIONS-dominant chains such as ”CONCLUSIONS → CONCLUSIONS → CONCLUSIONS”, which are rare in human writing. HGT, on the other hand, exhibits transitions like ”BACKGROUND → BACKGROUND → BACKGROUND”, reflecting more thorough context setting. ART maintains the same with HGT, with no noticeable differences between each other.

These patterns underscore how AGT often lacks the structural diversity and rhetorical complexity compared to human writing. ART closely mirrors HGT, reinforcing the argument that revision systems will preserve structure while modifying surface-level text features. Additional analysis of sentence label transitions across domains highlights the flow and structural patterns of abstracts. The observed transition patterns align with earlier results. Detailed transition analyses are included in Appendix B.

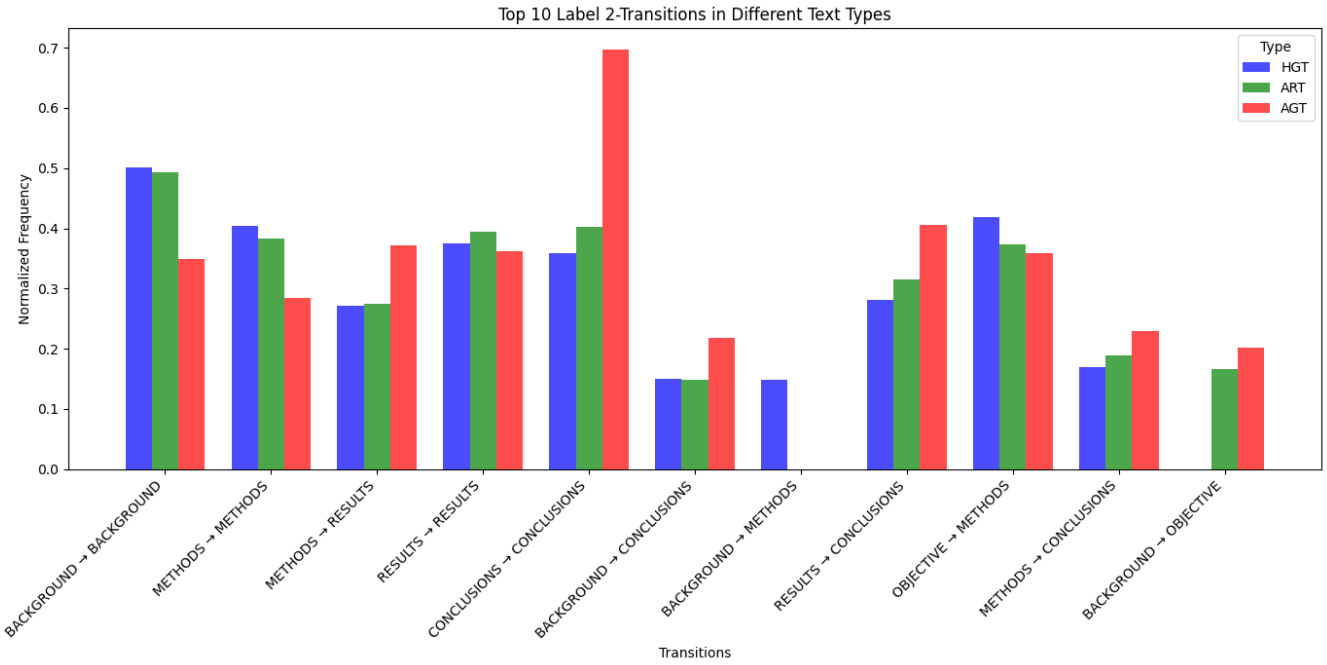


Figure 4.8.: This figure illustrates normalized the top 10 sentence labels of 2-transitions in different text types.

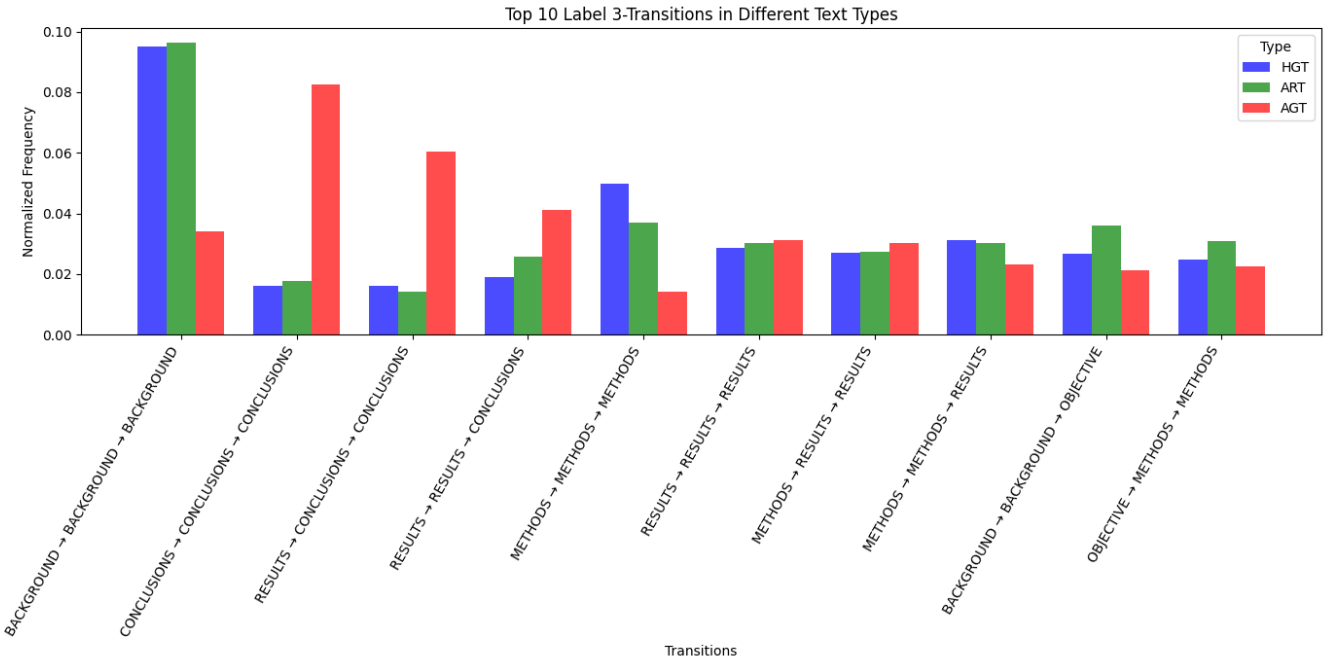


Figure 4.9.: This figure illustrates normalized the top 10 sentence labels of 3-transitions in different text types.

4.4.3. Divergence and Entropy-Based Metrics

Label	HGT vs. ART	HGT vs. AGT	ART vs. AGT
Background	0.83	1.37	0.85
Objective	1.05	1.53	0.99
Methods	0.90	1.62	1.17
Results	1.06	1.53	0.99
Conclusions	1.22	1.53	0.79

Table 4.3.: This table shows the KL divergence values for word distributions between pairs of text types across different sentence labels, presenting the differences in their probability distributions.

The KL divergence (Kullback and Leibler, 1951), also known as the Kullback-Leibler divergence or relative entropy, is a measure in information theory that quantifies the difference between two probability distributions. It measures how much information is lost when approximating one distribution with another. In the context of text analysis, it is typically used to compare word distributions between different types of texts to illustrate how stylistically or lexically distinct they are.

The KL divergence from P to Q $\Rightarrow$ KL(P || Q) is usually defined as follows:

$$KL(P||Q) = \sum P(x_i) * log(\frac{P(x_i)}{Q(x_i)})$$

where $P(x_i)$ is the probability of event x_i in distribution P and $Q(x_i)$ is the probability of event x_i in distribution Q. A high KL divergence indicates that the two distributions diverge significantly, while a low KL divergence suggests that they are more similar. As can be seen from Table 4.3, HGT and AGT show the greatest differences, indicating that they have different word distributions.

ART, on the other hand, is exactly in the middle, which means that it is the same as HGT and AGT. One difference is that ART and HGT have the biggest difference in Conclusions sentences, but it has the closest distribution with AGT. This suggests that AI has a strong influence on the structure of conclusion sentences, possibly indicating a characteristic language pattern unique to conclusion sentences.

In addition, we also estimated the KL divergence on the abstract level using a bag-of-words representation. Results, visualized in Figure 4.10, capture similar tendencies. HGT and AGT abstracts are the most divergent (e.g., KL(HGT‖AGT) = 1.28), while ART abstracts are nearest to HGT abstracts (e.g., KL(ART‖HGT) = 0.49). This corroborates our earlier findings, which show that ART preserves lexical characteristics of HGT to a significant degree, while AGT diverges both lexically and stylistically.

Pointwise Mutual Information (Fano and Hawkins, 1961) measures how much more likely it is for two events (Text Type and Sentence Label) to co-occur compared to pure chance. It's the association between Text Type and Sentence Label (see Table 4.4).

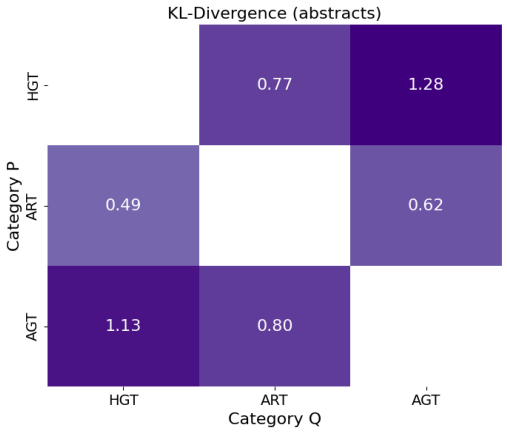


Figure 4.10.: KL-Divergence heatmap for abstracts.

Text Type	Background	Objective	Methods	Results	Conclusions
Human-Generated-Text	0.21	**-0.33**	**0.34**	-0.02	-0.45
AI-Revised-Text	0.18	0.10	**0.11**	-0.06	-0.35
AI-Generated-Text	**-0.33**	0.14	-0.41	0.05	**0.42**

Table 4.4.: This table reports the pointwise mutual information (PMI) values for the association between each text type and sentence label, quantifying how much more or less frequently each label occurs in a given text type compared to pure chance.

- When PMI > 0, the Sentence Label occurs more frequently in a given text type than by chance.
- When PMI < 0, the Sentence Label occurs less frequently in a given text type than by chance.

Humans prefer to write Methods and avoid Objectives and Conclusions. However, ART tends to modify Objective sentences, weakening the association for Methods compared to HGT. AGT uses more Conclusions and Objectives sentences, but way less sentences in Background and Methods. Given the near-zero values in Results sentences, this category is excluded from this analysis.

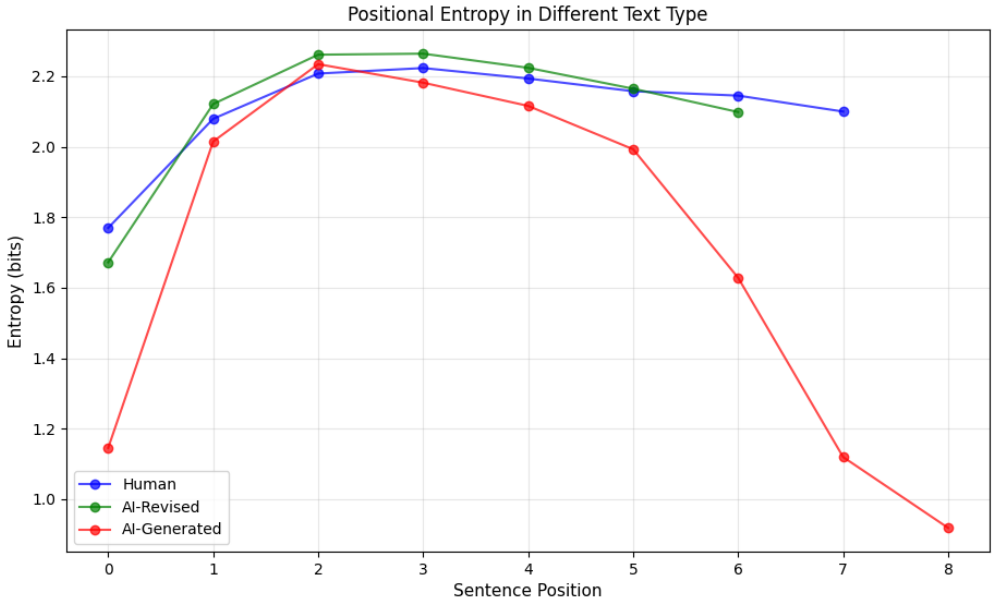


Figure 4.11.: Positional entropy of sentence labels across sentence positions in different text types. Higher entropy $\Rightarrow$ greater diversity of sentence labels, Lower entropy $\Rightarrow$ more rigid or templated structure.

Figure 4.11 shows positional entropy of sentence labels by abstract type. Positional entropy measures the diversity or unpredictability of sentence labels at specific positions within a sequence, such as the sentences of an abstract. For each sentence position, entropy is calculated based on the probability distribution of different labels appearing in that position across all texts of a given type. A curious pattern can be observed when looking at the entropy at positional 0. HGT possesses a moderate entropy value of 1.770, showing some

similarity in the first sentence label (e.g., starting with Background sentence) but without rigid adherence to a template. ART contains a slightly lower entropy of 1.672 in position 0. At drastic contrast, AGT contains a significantly lower entropy of 1.146 at position 0, reflecting somewhat higher tendency towards template conformity (e.g., beginning more often with an Objective sentence).

Positions 1-7 have significant variation in all abstract types, with entropy values mostly over 2.0. This implies that beyond the first sentence, there is no dominant structure. But with AGT, there is a significant drop in entropy levels in positions 6 to 8 (1.628 to 0.919). This decrease is a sign of a higher degree of structuring on the last sentences, maybe due to a template requiring the Conclusions sentences to dominate in these last positions. Therefore, the AGTs lack more variety at the end of the abstracts.

The mean agreement statistic in Table 4.5 provides further insight into the similarity of sentence label structures in the different text types. Mean agreement measures the average percentage of matching sentence labels at corresponding positions between pairs of text types. A higher mean agreement indicates greater structural similarity in sentence labels are sequenced, while a lower value reflects more divergent or varied structures.

HGT & ART	HGT & AGT	ART & AGT	ALL
63.76%	33.89%	36.23%	25.44%

Table 4.5.: Mean Agreement percentages between pairs of text types, indicating the similarity of sentence label sequences across abstracts.

The most highly agreeing is human and AI-revised writing with an average of 63.76%, which is in line with their same entropy trends in the Figure 4.11. The high agreement suggests that ART has the tendency to preserve the rhetorical structure of HGTs and only makes moderate revisions. In contrast, the mean agreement of AGT and HGT is much lower at 33.89%. AGT and ART also exhibit low agreement (36.23%), further suggesting the different structural strategies in AI generation compared to human writing or revision. The total consistency for all three types of texts is merely 25.44%, indicating the inherent disparities in sentence label distribution. Such low consistency for all types corroborates our earlier finding that AGT follows a more template-based structure, particularly at the beginning and end of abstracts, while HGT and ART are more variable in their abstract structures.

These agreement measures complement positional entropy visualization and provide quantitative verification that AI generation produces characteristically distinct sentence label patterns from human writing or human-AI collaborative revision.

4.4.4. Comparative Metric Analysis on each Sentence-Label

For this analysis, we identified the sentences with the same rhetorical function label within the abstract and compared them among the three different text types. We used some of the previously established metrics in Section 4.3.5, which focuses on readability, syntactic complexity, and lexical diversity.

It is important to note that, within abstracts, sentences bearing the same rhetorical function label are not necessarily contiguous. Evaluating the coherence between them is less meaningful. Instead, the focus is on isolated sentence characteristics within their rhetorical category.

Our primary interest lies in detecting more subtle shifts introduced during the AI revision process (ART), especially in comparison to the original human-authored sentences (HGT). To that end, Table 4.6 presents the most salient metric differences between HGT and ART for each label category. Since we found from the mean agreement in Table 4.5 that AGT is structured very differently compared to HGT and ART, our

focus lies primarily on HGT and ART. Nevertheless, we have included the values for AGT in the Table 4.6 for completeness.

Substantial differences emerge in readability across text types. For instance, in the Objective category, the Flesch-Kincaid Grade Level increases from 7.924 (HGT) to 10.579 (ART), and up to 14.966 for AGT, indicating a consistent trend toward more complex sentence construction. Similarly, in the Conclusion, ART shows a grade level increase of over 2 points compared to HGT, suggesting that AI revision tends to increase syntactic and lexical complexity, though to a lesser extent than full AI generation.

Conversely, Flesch Reading Ease scores drop markedly in both ART and AGT. In the Background section, the score decreases from 22.084 (HGT) to 9.828 (ART), reinforcing the pattern of increasing textual density and reduced readability.

The Unique Word Ratio increases from 0.441 (HGT) to 0.532 (ART) for Objective sentences. The same trend is observable with Hapax Legomena counts for Methods sentences and MATTR (Moving-Average Type-Token Ratio) for Objective sentences. These results indicate that ARTs introduce more diverse vocabulary than HGTs, but only within these specific sentence labels.

The Dugast's Uber Index, which captures lexical sophistication by accounting for vocabulary rarity and length, shows notable increases. For example, in the Background label, Dugast scores rise from 55.199 (HGT) to 67.832 (ART), and further to 91.686 (AGT). This pattern is consistent across multiple labels, suggesting that both ART and AGT tend to favor more lexically complex expressions.

In summary, ART is particularly extreme in terms of increased complexity and lexical variation for some sentence labels. These results are essential for understanding the stylistic transformations induced by AI revisions, which may have implications for readability, tone, and perceived authorship. For more specific analysis, we added the full Table for each sentence label in the Appendix C. All of these analysis results are consistent with the observation in Section 4.3.5.

Metric	**HGT**	**ART**	**AGT**	**Label**
Flesch Kincaid	7.924	10.579	14.966	Objective
Flesch Kincaid	8.954	11.176	18.344	Conclusion
Flesch Reading Ease	22.084	9.828	4.500	Background
Flesch Reading Ease	22.926	10.293	4.622	Methods
Unique Word Ratio	0.441	0.532	0.685	Objective
Hapax Legomena	21.092	19.679	23.543	Methods
Hapax Legomena	12.753	16.102	25.269	Objective
MATTR	0.099	0.151	0.425	Objective
Dugast's Uber Index	55.199	67.832	91.686	Background
Dugast's Uber Index	44.790	63.677	93.722	Objective
Dugast's Uber Index	49.423	68.091	97.016	Conclusion

Table 4.6.: This table presents the best metric values for each sentence label across the different text types.

4.4.5. Comparative Analysis across Domain-Specific Abstracts

This section compares the frequency distributions of sentence labels by domain, quantifies their domain-specificity using Kullback-Leibler (KL) divergence, and analyzes their associations with Pointwise Mutual Information (PMI) to reveal how sentence labels vary across domains and authorship.

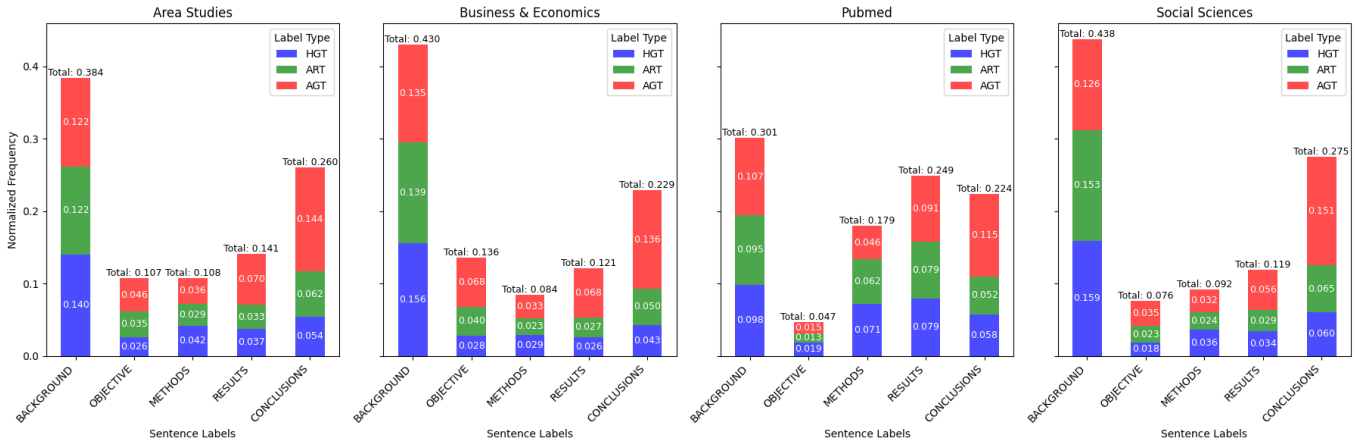
Frequency Distributions of Sentence Labels across Different Domains

The bar plots 4.12 represent the normalized frequencies of all sentence labels per domain for the three different text types. These distributions reveal domain-specific emphasis on certain kinds of sentences, as well as systematic human- and AI-authorship differences. Instead of analyzing each field in isolation, grouping fields by shared discourse profiles reveals different label distributions.

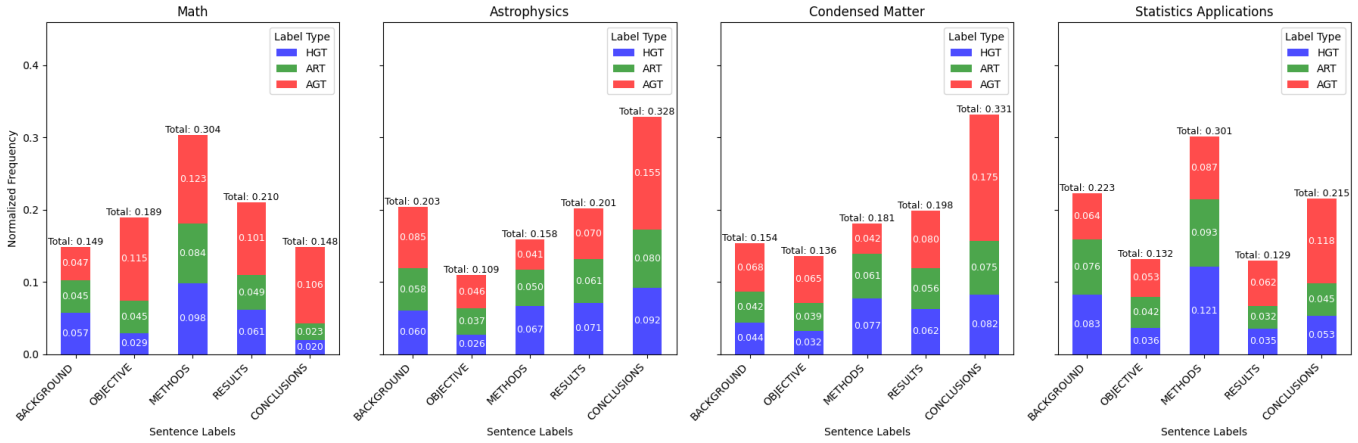
1. Area Studies, Business & Economics, PubMed, and Social Sciences (see Figure 4.12a) consistently have high rates of Background sentences, frequently over 30% and reaching up to 43% in Business & Economics and Social Sciences. This signals a strong preference for contextual framing. Objective sentences remain the least used (5% or less for each text type), falling below 14% across all four fields, with PubMed and Social Sciences both below 8%. For each domain in the Conclusion sentences, HGT and ART are below 7%, yet AGT is almost doubling it. These domains lean heavily on Background sentences.
2. Mathematics and Statistics Applications (see Figure 4.12b) have a very high focus on Methods, exceeding 30% overall. Background and Conclusions remain muted, together contributing less than 25% of total sentence types. Conclusion sentences appear more frequently in AGT outputs than in their HGT and ART counterparts, suggesting that AGT tends to favor interpretive closure.
3. Astrophysics and Condensed Matter (see Figure 4.12b) have the reversed profile, with lean Background sections but hugely inflated Conclusions sections (more than 30% overall). This shows that AGT, under scientific generative writing, overgenerates summarizing and interpretive material in the Conclusion.
4. Computational Language and Nonlinear Sciences (see Figure(4.12c) present more evenly spread distributions, with each sentence label tipping at around 15-20%. Notably, AGT Conclusions in Nonlinear Sciences peak above 14%, clearly exceeding the level of HGT and ART. Suggesting again, the general trend for pure generation of closing statements.

Collectively, these clusters reveal two general AI-induced deviations: (1) excessive reliance on Conclusions, in almost all disciplines; and (2) underrepresentation of Objectives statements, most acutely in context-sensitive disciplines (e.g., such as Social Sciences and Biomedical Fields).

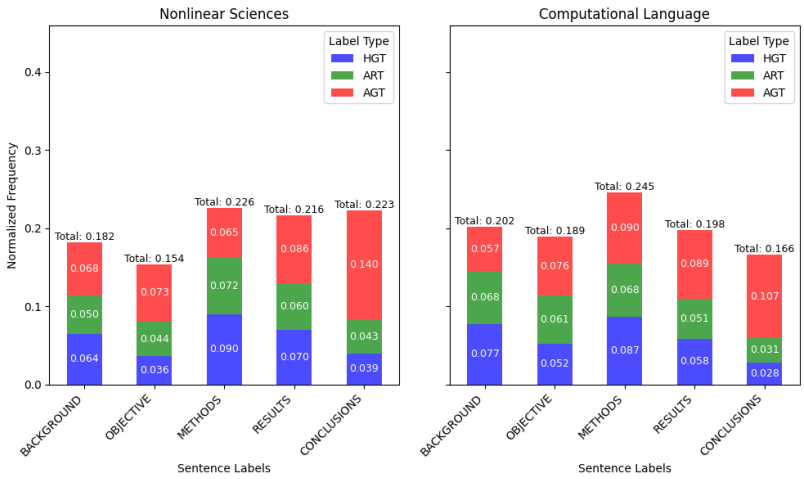
For more domain-specific frequency analyses, such as common words and common adjacent sentence transitions frequencies, see the visualizations in Appendix B.



(a) Area Studies, Business & Economics, PubMed, Social Sciences



(b) Mathematics, Astrophysics, Condensed Matter, Statistics Applications



(c) Nonlinear Sciences, Computational Language

Figure 4.12.: Sentence-level label frequencies (Background, Objective, Methods, Results, Conclusions) across ten domains and three different text types (HGT, ART, AGT). Each sub-figure groups related fields for ease of comparison.

Building on the numerical associations reported in Section 4.4.3, the PMI heatmap in Figure 4.13 confirms and enhances our understanding of how different sentence labels intersect with different domains. Rather than reproducing the PMI definition (Fano and Hawkins, 1961), we point out here the notable differences. For a detailed interpretation of the PMI values (i.e., distinctions between positive and negative associations), please refer to the itemized explanation in Section 4.4.3. Warmer cells denote stronger-than-chance co-occurrence, and cooler cells indicate the opposite.

PMI heatmap confirms and complements the findings derived from sentence label frequency distributions. The focus of Background sections is highly variable across domains, ranging from extremely positive values for PMI to negative associations. This highlights the context-specific nature of background material, which is extremely focused on in Social Sciences (social) and Business & Economics (b&e) but far less so in technically based disciplines such as Condensed Matter (cond-mat) and Astrophysics (astro-ph). Overall, the PMI values follow the trends in the sentence label frequency distributions. Background sentences, for instance, occur more frequently across most domains. Objectives are too rare across context-driven domains, and Methods sections take center stage across mathematical domains such as Mathematics and Statistics Applications. Furthermore, for a more detailed comparison, the PMI analysis for each text type (HGT, ART, and AGT) can be found in Appendix D.

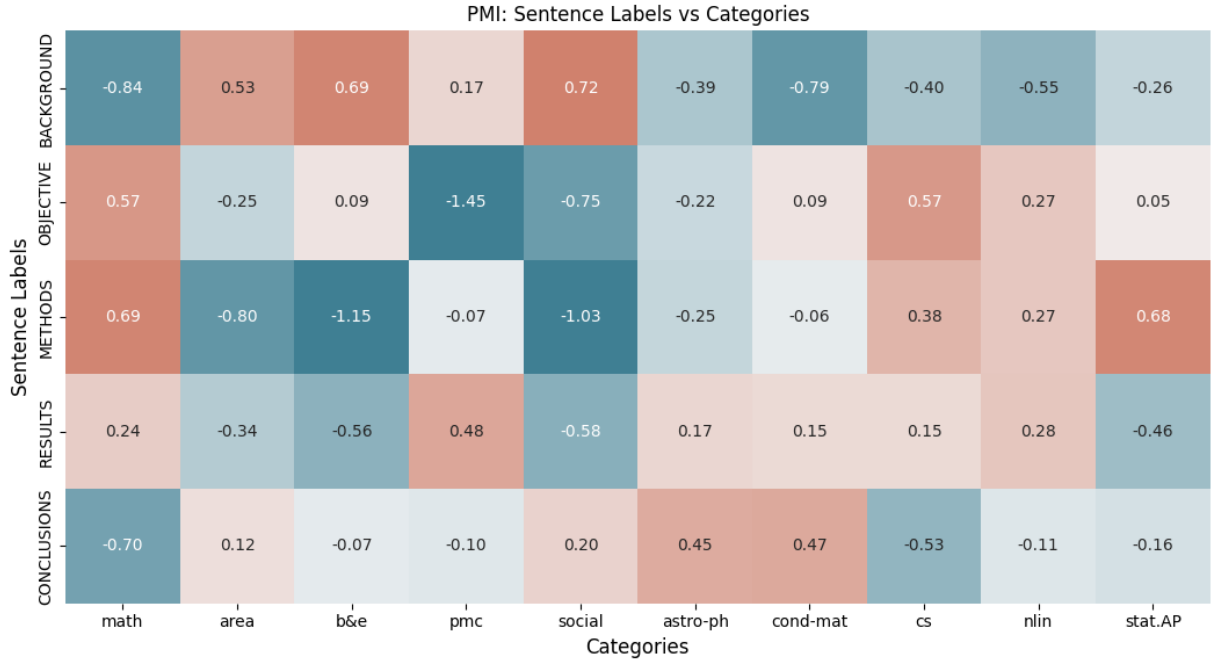


Figure 4.13.: Heatmap of Pointwise Mutual Information (PMI) values between sentence labels and academic categories, where red cells indicate stronger-than-expected associations and blue cells show weaker-than-expected associations across scientific domains

KL Divergence across Domain-Specific Abstracts

Figure 4.14 presents Kullback-Leibler (KL) divergence heatmaps comparing sentence label distributions across scientific domains within individual abstract types. This analysis assesses how sentence label distributions vary between domains for each text type separately. Every heatmap compares each research field (Mathematics, Area Studies, Business & Economics, PubMed, Social Sciences, Astrophysics, Condensed Matter, Computational Language, Nonlinear Sciences, and Statistical Applications) with themselves. KL divergence values reveal consistent patterns among the three types of texts.

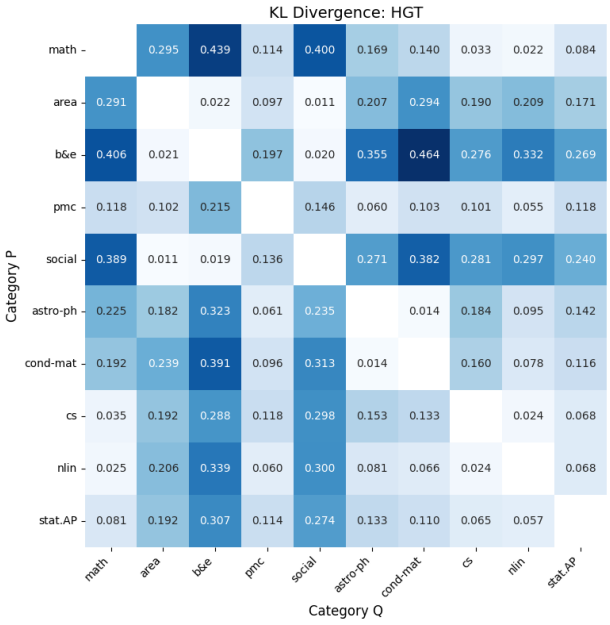
The HGT heatmap in Figure 4.14a exhibits the highest inter-domain distinctions. The KL divergence values are large between Mathematics and other humanities/social science domains (e.g., Mathematics and Business & Economics: 0.400; Mathematics and Social Sciences: 0.389). These high values indicate substantial differences in how sentence labels are distributed, suggesting that the structure of abstracts in Mathematics differs from those in fields such as Social Sciences. Technical and mathematical domains have minimal KL divergence between one another, with most values clustering near zero. This suggests that these domains share highly similar lexical structures.

The ART heatmap in Figure 4.14b generally shows a reduction in KL divergence values compared to the HGT heatmap, with the majority of comparisons showing between 0.1 and 0.3 divergence. This illustrates that AI revision has a bias towards homogenizing sentence label distributions across domains. Certain domain pairs show relatively high divergence, which shows that ART does not eliminate domain-specific stylistic divergence from HGT. For instance, Mathematics continues to show high divergence from multiple fields.

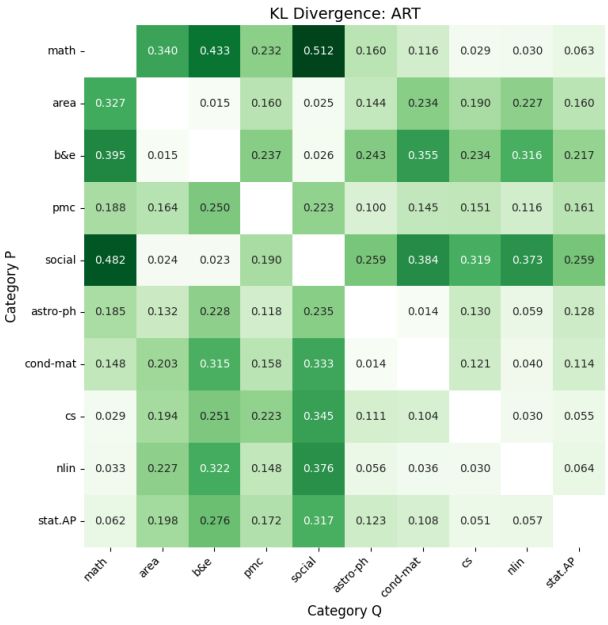
The AGT heatmap in Figure 4.14c shows the lowest KL divergences overall, with the lowest divergences indicating greater similarity in sentence label distribution between domains. Mathematics indicates the highest divergences found between and among different of the other disciplines. However, these values are still much lower than the values with the HGT and ART abstracts. Since AGT displays the least variation across domains, it suggests a strong tendency toward uniform structural patterns, regardless of disciplinary context.

The KL divergence heatmaps present a clear pattern: HGT abstracts have the most domain-specific variation in sentence label distributions, and AGT abstracts are more uniform. ART has an intermediate effect, partially reducing inter-domain variation but still maintaining some degree of stylistic variation.

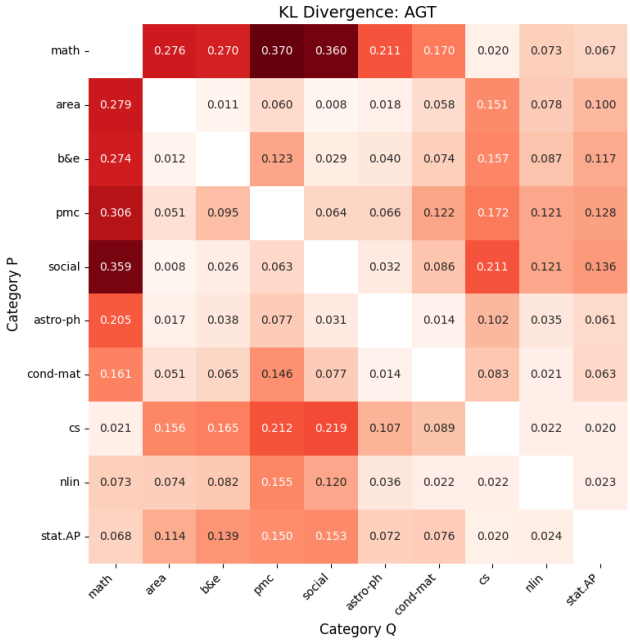
Additionally, a combined KL Divergence analysis was performed in which the different text types were considered as a unit. The identical evaluations were identified and can also be found in the Appendix D.



(a) HGT



(b) ART



(c) AGT

Figure 4.14.: KL Divergence Heatmaps comparing sentence label distributions across scientific domains for HGT (a), ART (b), and AGT (c).

5. Experiment

In this chapter, we give an overview of the technical details of the experiments. We start with a description of the experimental setup. This includes the dataset, the models, and the evaluation criteria for the results. Afterward, we move on to the results and provide a brief discussion of their implications.

5.1. Setup

In the following section, we discuss the technical details of the experimental setup. This includes both the dataset we used and all the LLMs we fine-tuned. The hardware details are stated as well in the Appendix E.

5.1.1. Dataset

From the dataset that we've already introduced in Chapter 3, we structured our experiments around two distinct classification tasks to evaluate the stylistic and structural differences between HGT, ART, and AGT abstracts.

First, we used the whole abstract as the input unit for a supervised classification task. The goal was to train a model capable of distinguishing between the three text types based on their overall structure and linguistic features. The dataset was split into training (70%), validation (15%), and test (15%) sets. We ensured that all three versions of an abstract (HGT, ART, AGT) from a single paper were included in the same split, using the article ID as the key. This avoids data leakage and ensures that the classifier cannot rely on lexical patterns unique to individual papers, thus encouraging a deeper understanding of stylistic differences across text types. The dataset is described in detail in Table 3.1.

Following, we transformed the dataset by extracting individual sentences from each abstract and retaining their corresponding text type labels. This allowed us to train models to distinguish between HGT, ART, and AGT at a finer granularity. The same 70/15/15 split was applied based on article ID to maintain consistency with the abstract-level task and to ensure that all sentences from a given paper remained in the same subset. This setup presents a more challenging classification problem, as it removes document-level context and forces the model to rely only on sentence-level stylistic cues. Detailed statistics for this sentence-level dataset are provided in Table 3.3.

5.1.2. Baseline & LLM

We fine-tuned multiple pre-trained language models. As a Baseline, we selected SciBERT (Beltagy et al., 2019), an encoder-only transformer-based model specifically pre-trained on a large corpus of scientific publications. SciBERT served as a strong starting point due to its familiarity with domain-specific vocabulary and syntactic structures.

To further explore the capabilities of larger and more flexible architectures, we also fine-tuned several open-source LLaMa models[1]. Given their substantial size and memory requirements, we applied quantization to each LLaMa model to reduce computational overhead and enable training on an NVIDIA A100 40GB GPU. This optimization allowed us to conduct efficient fine-tuning while maintaining model performance.

5.2. Approach

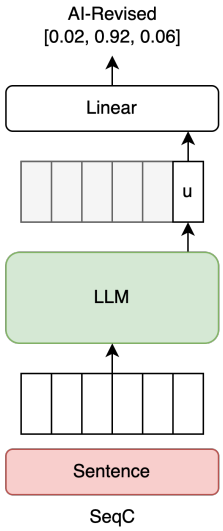


Figure 5.1.: Sequence-Based Classification Approach (SeqC): The sentence is processed by the LLM, the hidden state $\boldsymbol{u}$ is extracted, and a linear layer maps it to the output classes.

SciBERT (Beltagy et al., 2019) is an encoder-only transformer model that is commonly used for text classification by encoding input sequences and applying additional classification layers. LLaMa models and most LLMs are decoder-only models (Touvron et al., 2023; Grattafiori et al., 2024), they are typically used as generative models (to generate output text containing labels). Sequence-Based Classification (SeqC) fine-tunes decoder-only LLMs for classification by repurposing them as encoding models. Ruan et al. found that SeqC is more effective than the generation-based approach and the other two encoding-based approaches evaluated in her study. Based on these findings, we applied the SeqC approach for our classification task. The input for this approach is the text/sentence we want to classify. The LM processes the input to understand context and semantics. After the processing, we take the LM's hidden state $\boldsymbol{u}$, which captures the main meaning of the input. A classification Layer maps the LM's output to class probabilities. The overall process is illustrated in Figure 5.1.

[1] `https://www.llama.com/docs/model-cards-and-prompt-formats/`

5.3. Evaluation

Initially, our approach focused on using the entire abstract as input to train the language models. We used the input length with a Max-Token value of 512, as the abstracts typically exceeded 256 tokens but remained within 512. Early results from this experiment showed unexpectedly high classification performance. The models were able to distinguish between HGT, ART, and AGT abstracts with extremely high accuracy (see Table 5.2).

We suspected that the strong performance may have been influenced by the SciBERT model's domain-specific pretraining on scientific literature. To verify this, we conducted additional experiments using other language models such as BERT (Devlin et al., 2019) and RoBERTa (Liu et al., 2019). Surprisingly, these models also achieved similarly high classification scores, particularly in differentiating between HGT and ART. While these results are impressive, they raised concerns regarding the task's difficulty level because they appeared "too good to be true". As an additional experiment, we briefly trained a binary classifier to distinguish only between HGT and ART abstracts. The goal was to evaluate whether the revision process introduces sufficiently detectable features that even a reduced classification task could capture. Once again, the model achieved near-perfect performance, reinforcing our assumption that the abstract-level classification task might not accurately reflect the subtleties of human-AI revision. A summary of these outcomes can be found in Table 5.2.

Following these results, we decided to introduce a more challenging classification task by transitioning from abstract-level to sentence-level classification. This setup removes higher document context and forces the model to rely only on fine-grained stylistic and syntactic cues. The shift to sentence-level inputs also allowed us to decrease the Max-Token threshold to 256 because sentences rarely exceeded this limit.

During the data creation phase, we encountered a class imbalance issue. The fact that AGT abstracts were longer and more wordy relative to HGT and ART meant the extracted sentences were accordingly larger in number for AGT. This imbalance had the potential to skew training dynamics and lead to biased model predictions. To address this issue, we implemented class weighting during model training. This approach adjusted the loss function to account for the unequal distribution of the sentence types. The introduction of class weights resulted in a consistent performance improvement of 2–3% on all the evaluation metrics, leading to more balanced and realistic classification outcomes (see Table 5.1).

Model SciBERT	**Accuracy**	**F1-Score**	**Precision**	**Recall**
Class Unbalanced	78.82%	78.86%	79.91%	78.82%
Class Balanced	80.79%	81.30%	82.57%	80.79%

Table 5.1.: Baseline SciBERT Balanced vs Unbalanced Class

Since our LLM output is categorical label-based, we rely primarily on classical machine learning evaluation metrics. These metrics offer robust and interpretable ways to assess model performance. The evaluation metrics (Hossin and Sulaiman, 2015) were used to validate our models for both abstract-level and sentence-level classification tasks. All the reported metric values are class-weighted to correct for class imbalance, with class weighting applied during training and evaluation. This leads to conditions under which accuracy and recall values are identical.

Accuracy

Accuracy is the most intuitive and straightforward metric, representing the ratio of correctly predicted instances over the total number of predictions. It is defined as:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

- TP (True Positives): Correctly predicted positive cases
- TN (True Negatives): Correctly predicted negative cases
- FP (False Positives): Incorrectly predicted positive cases
- FN (False Negatives): Incorrectly predicted negative cases

Precision

Precision calculates the ratio of correctly predicted instances to all predicted positive instances. It is defined as:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

Recall

Recall (also known as sensitivity or true positive rate) measures the model's ability to correctly identify all actual positive instances. It is defined as:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

F1-Score

The F1 Score is the harmonic mean of precision and recall, offering a single metric that balances both false positives and false negatives. It is especially useful when we need a trade-off between precision and recall:

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

5.4. Results

Classification Reports

Abstract-Level

Model	Class Type	Accuracy	F1-Score	Precision	Recall
SciBERT	3-Class: HGT/ART/AGT	96.59%	96.59%	96.66%	96.59%
RoBERTa	3-Class: HGT/ART/AGT	98.00%	98.00%	98.03%	98.00%
BERT	3-Class: HGT/ART/AGT	95.63%	95.63%	95.57%	95.63%
SciBERT	2-Class: HGT/ART	96.56%	96.55%	96.78%	96.56%
RoBERTa	2-Class: HGT/ART	99.56%	99.56%	99.56%	99.56%

Table 5.2.: Model performance (Accuracy, F1-Score, Precision, Recall) across 3-Class (HGT, ART, AGT) and 2-Class (HGT, ART) classification tasks using weighted evaluation metrics.

Table 5.2 presents the initial results of classifying entire abstracts as either HGT, ART, or AGT. We began with a Three-Classification (3-Class) task, where the models were trained to distinguish between HGT, ART, and AGT. This was followed by a more focused Two-Classification (2-Class) task, being specifically designed to separate between HGT and ART. As the table illustrates, all models, including SciBERT-3-Class, RoBERTa-3-Class, and BERT-3-Class, achieved exceptionally high scores across all evaluation metrics. Values consistently exceeded 95%, with some models even reaching near-perfect performance (e.g., 99.56% for RoBERTa-2-Class).

Based on these observations, we shifted our approach from classifying entire abstracts to classifying individual sentences. Sentence-level classification presents a more challenging task, requiring the models to analyze and differentiate based on finer-grained linguistic features rather than overall stylistic patterns. Sentence-by-sentence evaluation allows us a closer estimation of how accurately the models are able to distinguish between HGT, ART, and AGT content under real-world scenarios.

Sentence-Level

At the sentence-level, the SciBERT model predicts scores of about 80% on all metrics in the 3-Class scenario. We used these as our Baseline Model when we tested the performance of the large LLMs in this classification task.

Before we compare the final LLM configurations to this baseline, we first compared various LLaMa models with varying parameter sizes and configurations and identified the best-performing configuration for the classification task. Table 5.3 presents the performance comparison across various LLaMa models with different parameter sizes and configurations. These results are trained on Epoch 6, since too long training epoch is worse for the metrics (see the Llama-3.1-8B-Instruct-Epoch-10 result in the Table 5.3) due to overfitting.

Model	Accuracy	F1-Score	Precision	Recall
Llama-2-7B (2023)	82.31%	82.18%	82.99%	82.31%
Llama-2-13B (2023)	83.96%	83.95%	84.46%	83.96%
Llama-3-8B[1]	85.59%	85.80%	86.19%	85.59%
Llama-3.1-8B-Instruct[1]	**86.47%**	**86.63%**	**86.88%**	**86.47%**
Llama-3.2-3B (2024)	84.18%	84.36%	84.80%	84.18%
Llama-3.2-3B-Instruct (2024)	84.94%	85.28%	86.13%	84.94%
Llama-3.1-8B-Instruct-Epoch-10[1]	84.85%	85.18%	85.95%	84.85%

Table 5.3.: Comparison of LLaMA models with varying sizes and configurations using weighted metrics. Results are based on training up to Epoch 6, where Llama-3.1-8B-Instruct performed best. Longer training (e.g., Epoch 10) showed reduced performance due to overfitting.

Next, we fine-tuned Llama-3.1-8B-Instruct (Grattafiori et al., 2024) using QLoRA (Dettmers et al., 2023), systematically varying LoRA (Low-Rank Adaptation, Hu et al. 2022) rank (r) and alpha (α) parameters, which are designed to optimize the model fine-tuning by adjusting the number of parameters and scaling factors. From Table 5.4, the best LoRA configuration was r=64 and α=64 with accuracy of 87.19%, F1-Score of 87.31%, precision of 87.48%, and recall of 87.19%. The table indicates that finding the correct configurations for LoRA provides better scores.

LoRA Config	Accuracy	F1-Score	Precision	Recall
r=128, α=128	86.47%	86.63%	86.88%	86.47%
r=256, α=256	83.24%	83.57%	84.50%	83.24%
r=64, α=64	**87.19%**	**87.31%**	**87.48%**	**87.19%**
r=64, α=128	86.00%	86.23%	86.67%	86.00%

Table 5.4.: LoRA configuration performance for fine-tuning Llama-3.1-8B-Instruct, with the best results achieved at $r = 64$ and $\alpha = 64$.

Learning rate is a significant hyperparameter in model training, influencing how fast the model updates its weights during optimization. Table 5.5 shows the comparison of model performance at different learning rates. Our findings indicate that the optimal performance is obtained with a learning rate of 2e-4. Decreasing the learning rate to 1e-4 slightly decreases the metrics, and increasing it to 3e-4 further degrades performance. Learning rate 5e-4 results in an overall decrease of all the metrics of evaluation, and also the accuracy falls to 72.01%. These findings emphasize the importance of tracking the learning rate with accuracy to make sure that convergence speed and solution quality are in balance. A learning rate that is too high can produce unstable training and prevent the model from converging to an optimal solution.

These experiments collectively demonstrate the effectiveness of LLaMA models and the significance of model configurations for the accurate classification of HGT, ART, and AGT.

[1] `https://github.com/meta-llama/llama3`, `https://www.llama.com/models/llama-3/`

Learning Rate	Accuracy	F1-Score	Precision	Recall
1e-4	87.02%	87.17%	87.43%	87.02%
2e-4	**87.19%**	**87.31%**	**87.48%**	**87.19%**
3e-4	86.03%	86.21%	86.53%	86.03%
5e-4	72.01%	72.57%	74.00%	72.01%

Table 5.5.: Model performance at different learning rates, with optimal results at 2e-4. Higher and lower rates degrade performance.

Comparison Baseline vs Fine-Tuned LLM

As shown in Table 5.6, a closer examination of the performance of classification indicates that the best-performing LLaMA model (Llama-3.1-8B-Instruct) with LoRa performs better than the Baseline SciBERT model across all evaluation metrics. Our SciBERT Baseline Model has an overall accuracy of 80.79% and the best fine-tuned LLaMA Model with the best possible params has an accuracy of 87.79%. We gained a 6.5% improvement, which is great.

Overall Performance				
Model	**Accuracy**	**F1-Score**	**Precision**	**Recall**
Baseline SciBERT	80.79%	81.30%	82.57%	80.79%
Llama-3.1-8B-Instruct	87.19%	87.31%	87.48%	87.19%
Per-Class Metrics				
Text Type	**Model**	**Precision**	**Recall**	**F1-Score**
Human-Generated	Baseline	93%	83%	88%
	LLM	98%	93%	95%
AI-Revised	Baseline	63%	79%	70%
	LLM	76%	80%	78%
AI-Generated	Baseline	88%	80%	84%
	LLM	88%	88%	89%

Table 5.6.: Comparison of Baseline SciBERT and LlaMA-3.1-8B-Instruct in overall performance and per-class evaluation. LLM consistently outperforms across all categories.

If we take a closer look at Per-Class Metrics in Table 5.6, then both models have high performance in HGT. Llama increased the precision and recall by 5% and 10%, resulting a high F1-Score of 95%. When looking at the AGT, both models have equal precision, but the LLM improved the recall by 8%, lifting the F1-Score by 5%. The hardest class to classify is the ART class. LLM got a major improvement in precision, but a slight recall improvement.

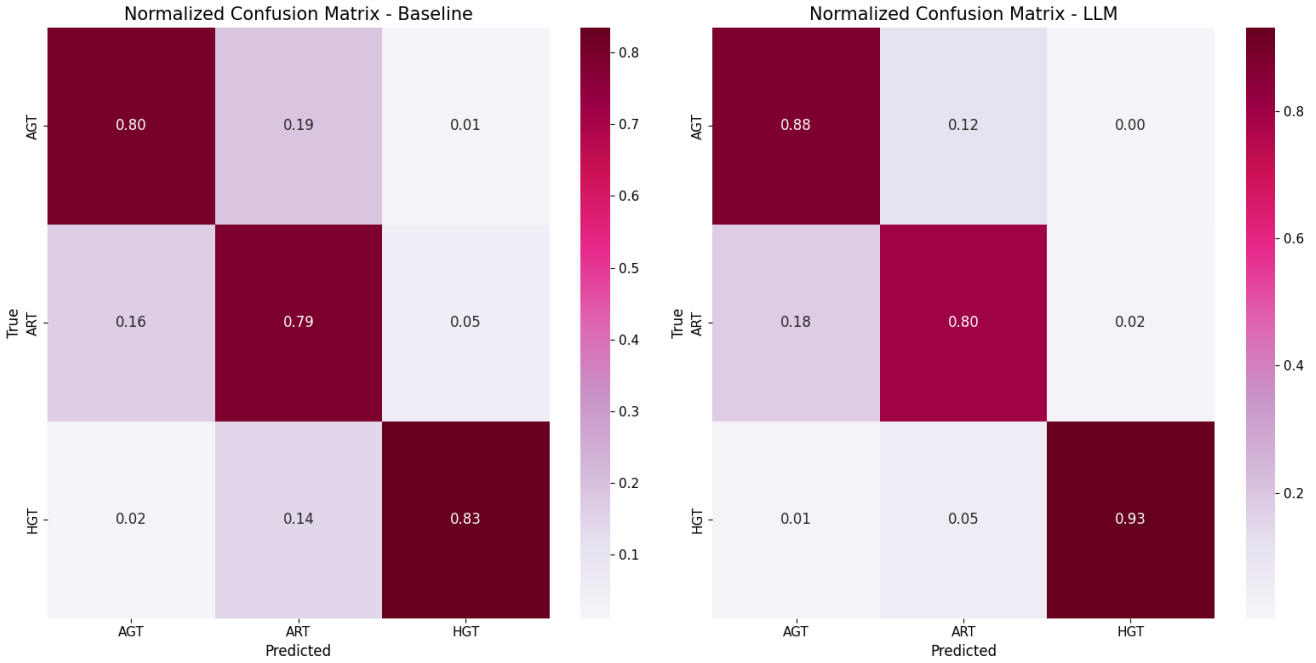


Figure 5.2.: Normalized confusion matrices for the Baseline model (left) and fine-tuned LLM (right), illustrating classification accuracy and misclassification across all classes.

The normalized confusion matrices in Figure 5.2 further demonstrate the superiority of the fine-tuned LLM by visualizing how well each model distinguishes between the three text types. In confusion matrices, darker diagonal elements (higher values) indicate better classification performance for each class. The LLM matrix shows darker diagonal elements and lighter off-diagonal elements compared to the Baseline, indicating reduced misclassifications and stronger class separation. Key observations include:

- For the AGT class, the LLM increases true positive rate from 0.80 to 0.88 and reduces misclassifications as ART from 0.19 to 0.12. Misclassification as HGT are nearly eliminated.
- The ART class maintained a similar true positive rate (0.79 Baseline vs 0.80 LLM), but the LLM reduces misclassifications as HGT from 0.05 to 0.02, while it increases the confusion with AGT slightly (0.16 to 0.18).
- The HGT class showed significant improvement, with its true positive rate rising from 0.83 (Baseline) to 0.93 (LLM). Misclassification as ART drops notably from 0.14 to 0.05, and the confusion with AGT, already low, is further reduce by 1%.

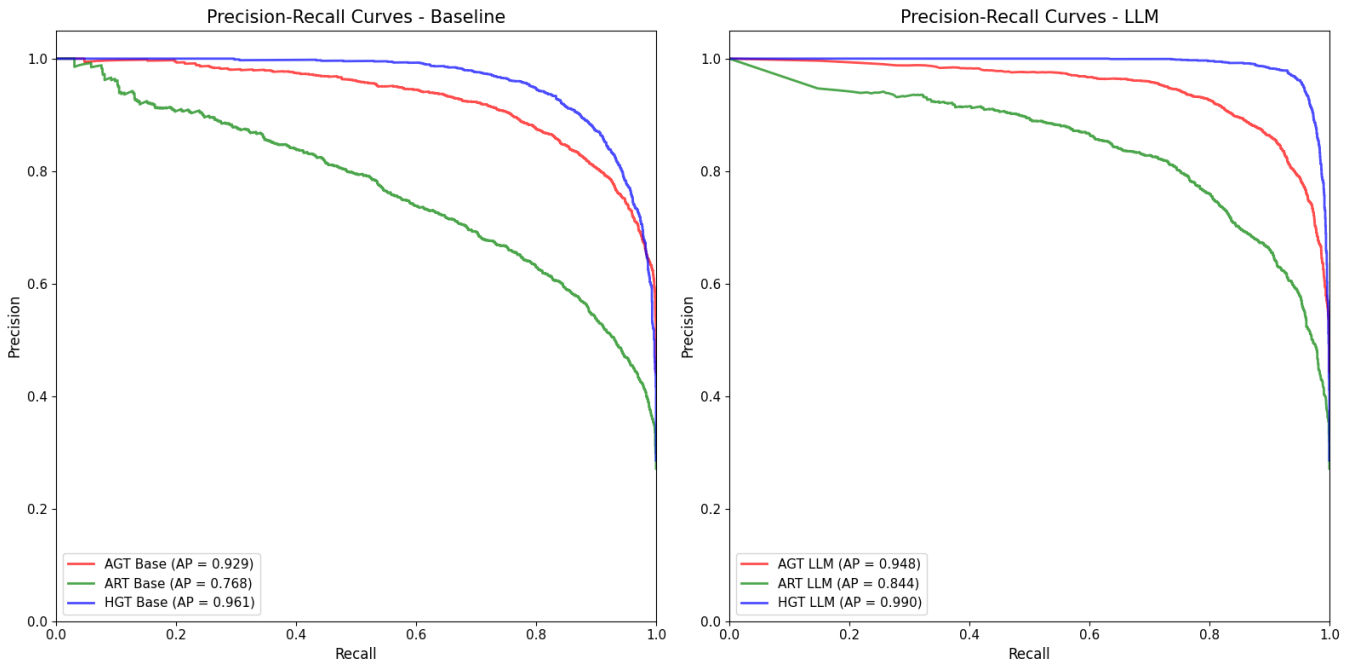


Figure 5.3.: Precision-Recall curves comparing Baseline model (left) and fine-tuned LLM (right) across different classes, demonstrating improved average precision and more consistent high precision at increasing recall levels with the LLM.

The Precision-Recall (PR) in Figure 5.3 illustrates the classification performance of both models across varying recall thresholds. The PR curve of an ideal model would maintain high precision (near 1.0) even as recall increases, forming a curve that stays in the upper-right corner of the graph. The fine-tuned LLM shows this desired pattern more prominently, particularly for the ART class, where the curve maintains higher precision values across all recall thresholds compared to the Baseline model.

Overall, the fine-tuned LLM consistently outperforms the Baseline model across all three text classes, showing higher average precision (AP) scores and more stable high-precision behavior throughout.

- The AGT class AP improved from 0.929 (Baseline) to 0.948 (LLM).
- The ART class AP showed notable growth from 0.768 (Baseline) to 0.844 (LLM).
- The HGT class AP increased from 0.961 (Baseline) to 0.990 (LLM).

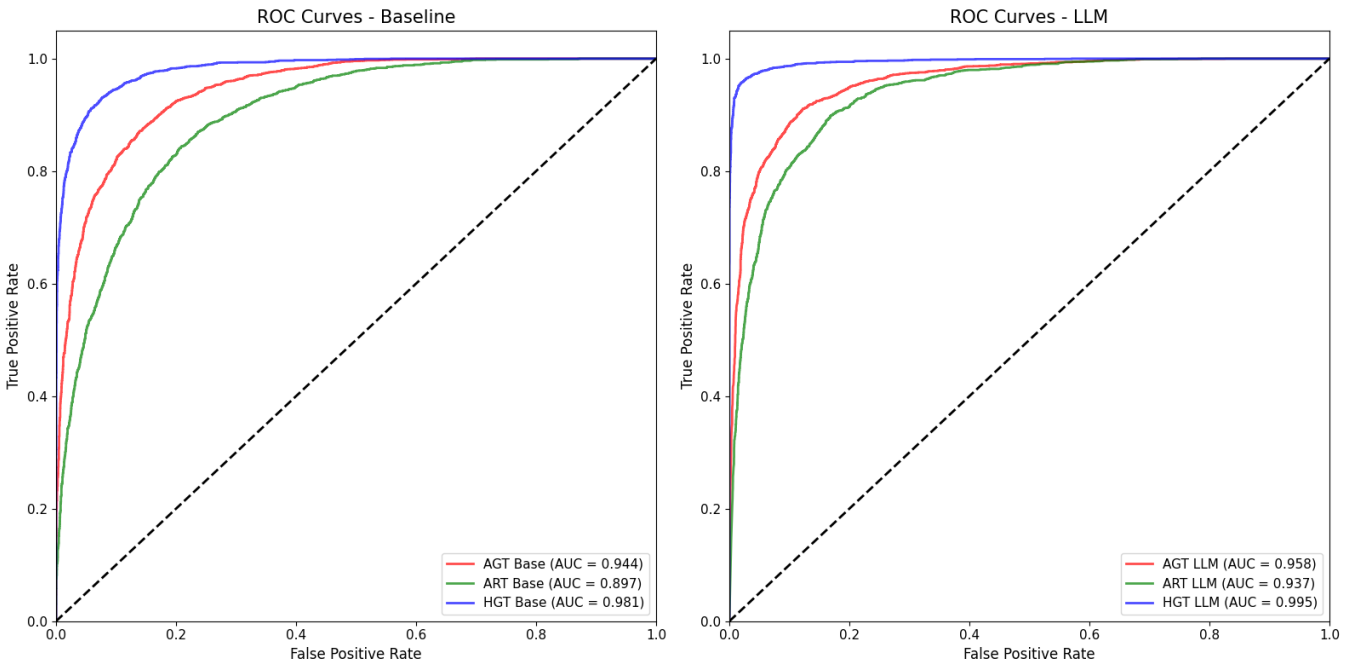


Figure 5.4.: Receiver Operating Characteristic (ROC) curves and Area Under Curve (AUC) scores for Baseline model (left) and fine-tuned LLM (right), highlighting consistently higher AUC values and superior true positive rates at low false positive rates with the LLM.

The Receiver Operating Characteristic (ROC) curves in Figure 5.4 visualize the trade-off between true positive rate (sensitivity) and false positive rate across different thresholds. In ROC curves, better performance is indicated by curves that push toward the top-left corner of the graph. This represents a model's ability to achieve high true positive rates while maintaining low false positive rates. The Area Under Curve (AUC) measures the overall ability of a model to distinguish between classes. AUC values range from 0 to 1, where 1.0 indicates perfect classification and 0.5 means random guessing.

- The AGT class AUC improves from 0.944 (Baseline) to 0.958 (LLM).
- The ART class AUC increases from 0.897 (Baseline) to 0.937 (LLM).
- The HGT class AUC rises from an already high 0.981 (Baseline) to an exceptional 0.995 (LLM).

The LLM's curves are more "bowed" toward the ideal top-left corner, particularly for the HGT class. The steeper initial rise of the LLM curves also indicates that it achieves higher true positive rates at lower false positive rate thresholds.

Model Performence Metrics

To quantitatively assess the performance of the LLM and Baseline models, we examine two key metrics: Hamming Loss (Hamming, 1950) and Jaccard Similarity (Jaccard, 1901). Hamming loss measures the fraction of labels that are incorrectly predicted. It represents the ratio of instances where the model predicts the wrong class. A lower Hamming loss indicates better performance and is defined as:

$$\text{Hamming Loss} = \frac{\text{Number of incorrect predictions}}{\text{Total predictions}}$$

Jaccard Similarity, also known as the Jaccard Index, measures the similarity between the predicted and true sets of labels. For each class, it is calculated as the size of the intersection divided by the size of the union of the predicted and true labels. A higher Jaccard Similarity indicates better performance for that specific class. The Jaccard Similarity scores provide insights into how well each model performs on each class individually.

$$\text{Jaccard} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives} + \text{False Negatives}}$$

The LLM demonstrates a Hamming Loss of 0.085, while the Baseline model has a higher Hamming Loss of 0.128. This indicates that the LLM makes fewer overall misclassifications than the Baseline model. Comparing the Jaccard Similarity scores, the LLM outperforms the Baseline model across all three classes. This suggests that the LLM is better at correctly identifying instances of each class. See Table 5.7 for the comparison.

Metric	**Baseline**	**LLM**
Hamming Loss	0.128	0.085
Jaccard Similarity (AGT)	0.719	0.783
Jaccard Similarity (ART)	0.543	0.639
Jaccard Similarity (HGT)	0.787	0.911

Table 5.7.: Comparison of Hamming Loss and Jaccard Similarity scores for Baseline and LLM models.

Error Analysis

Table 5.8 details the frequency and average confidence of the various error types made by two models. The two most frequent errors for both models involve misclassifying AGT as ART and vice versa. Specifically, the Baseline model misclassifies AGT as ART 670 times, while the LLM does so 420 times. Reversed, the Baseline model misclassifies ART as AGT 356 times, whereas the LLM makes this error 400 times. These errors suggest that distinguishing between AGT and ART is inherently challenging, possibly due to shared stylistic or structural resemblance.

The LLM in all instances is more confident than the Baseline model. For instance, when the Baseline model misclassifies AGT as ART, the average confidence rate stands at 75.2%, while the LLM demonstrates a significantly higher average confidence of 91.3%. Similarly, in ART-to-AGT misclassifications, the Baseline model reaches an average of 75.5%, as compared to the LLM's 92.6%.

Error Type		Count		Avg Confidence	
True	**Predicted**	**BL**	**LLM**	**BL**	**LLM**
AGT	ART	670	420	75.2%	91.3%
ART	AGT	356	400	75.5%	92.6%
HGT	ART	328	126	74.9%	90.3%
ART	HGT	102	39	75.8%	89.4%
HGT	ART	52	33	66.2%	88.8%
AGT	HGT	40	12	71.4%	83.2%

Table 5.8.: Error type comparison between the Baseline model and fine-tuned LLM, showing frequency and average confidence for each misclassification.

Interestingly, the LLM makes fewer errors overall in misclassifying HGT as either ART or AGT. This suggests that the LLM is better at recognizing features indicative of human authorship. However, the LLM's higher confidence in its errors raises concerns about potential overconfidence, which could impact the reliability of its classifications in practical applications. Further evaluation of each model's confidence is necessary for a better understanding.

Now we are going to talk about the error rates for each sentence label (Background, Objective, Methods, Results, Conclusions) for both the LLM and Baseline model (see Figure 5.5). For all sentence labels, the Baseline model has higher error rates than the LLM. Specifically, the Baseline model has the highest error rate for Methods and Background sentences, while the LLM consistently maintains lower error rates across all sentence labels. Both LLM and Baseline models have the lowest error rates for Conclusions sentences compared to other types. The LLM model achieves a notably lower error rate (approximately 0.10) compared to the Baseline model (approximately 0.15).

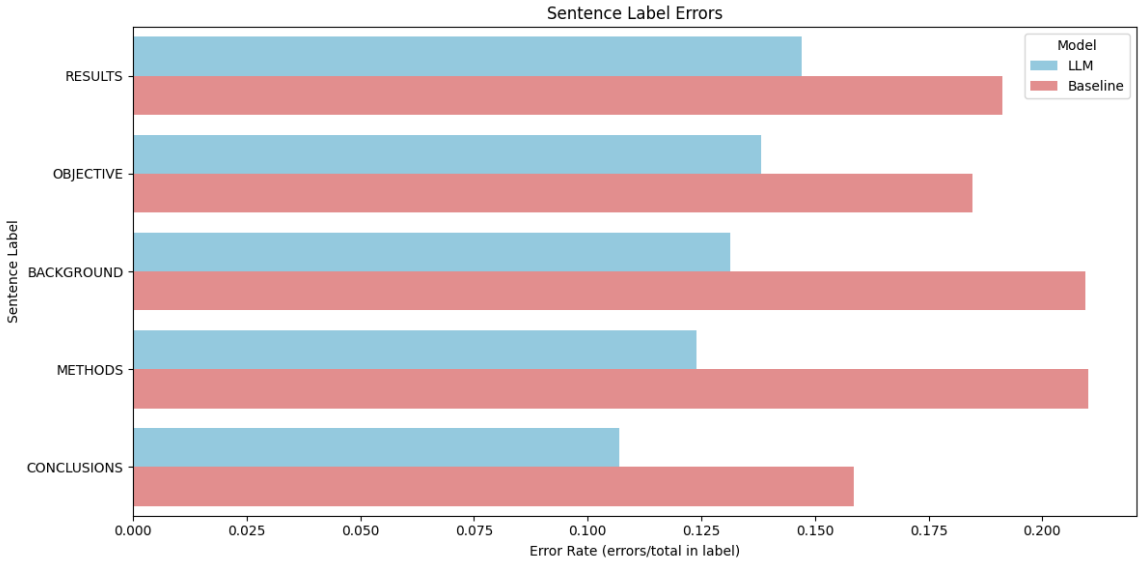


Figure 5.5.: This figure shows the error rates by sentence label (Background, Objective, Methods, Results, Conclusions) for the Baseline and the fine-tuned LLM.

Confidence Distribution

The confidence distributions of the Baseline and LLM models, as visualized in Figure 5.6, reveal interesting differences in how each model expresses certainty in its predictions and how this relates to prediction correctness. This section presents a comparative analysis of confidence distributions of our Baseline model and the best fine-tuned Llama Model (LLM) implementation.

For the Baseline model (left panel), the model confidence distribution is quite broad, from approximately 0.4 to 1.0. Correct predictions (orange bars) are distributed across this range but become especially prominent as confidence increases, resulting in a sharp spike at the highest confidence value of 1.0, where the majority of correct predictions are concentrated. Incorrect predictions (blue bars) are scattered throughout the confidence range, even at high confidence levels, indicating that the Baseline model is occasionally overconfident in its incorrect predictions.

In contrast, the LLM model (right panel) displays a much more concentrated confidence distribution. Nearly all predictions, both correct and incorrect, are made at the highest confidence level, with a dramatic spike at 1.0. The number of correct predictions at this level is roughly double that of the Baseline model, demonstrating the LLM's strong tendency to assign maximal confidence to its predictions. Incorrect predictions, while far fewer, are also clustered at high confidence, reflecting that when the LLM is wrong, it is likely to be very confident in the error. There is lower activity at lower confidence levels, suggesting that the LLM model is not just more decisive but also less hesitant in predictions compared to the Baseline.

The Baseline model distributes its confidence more broadly, reflecting greater uncertainty or caution, while the LLM model is maximally confident in nearly all its predictions. While such decisiveness can be advantageous, the presence of incorrect predictions at maximal confidence in both models suggests that further calibration may be necessary to ensure that confidence scores more accurately reflect true predictive reliability.

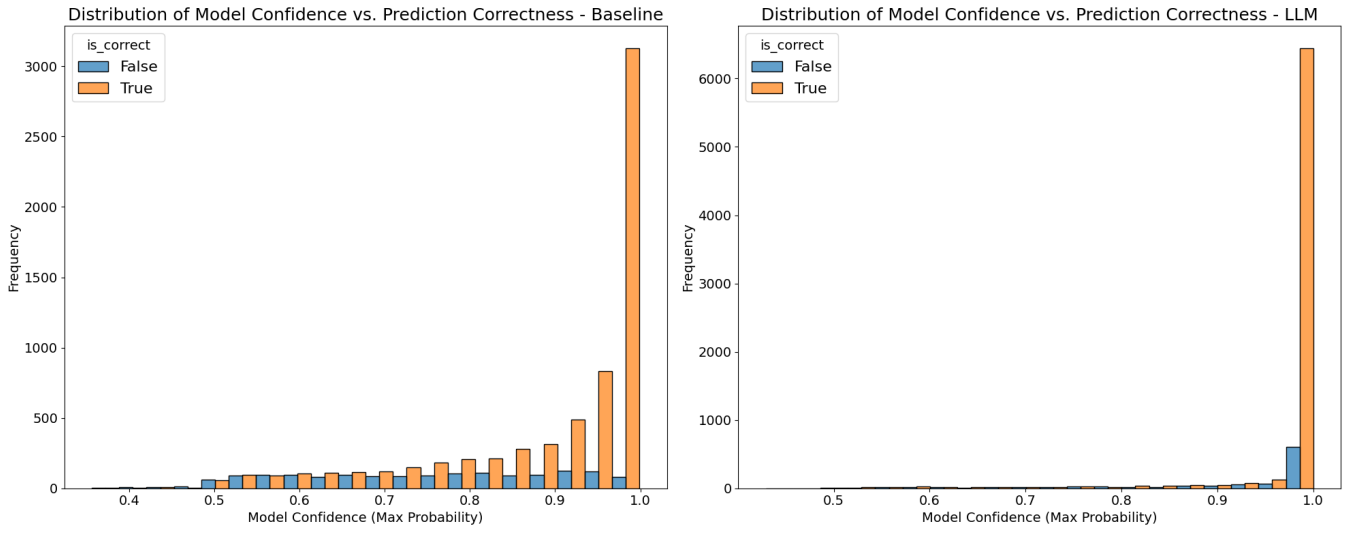


Figure 5.6.: Confidence distributions of the Baseline model (left) and the fine-tuned LLM (right), segmented by prediction correctness. Orange bars indicate correct predictions, blue bars indicate incorrect ones.

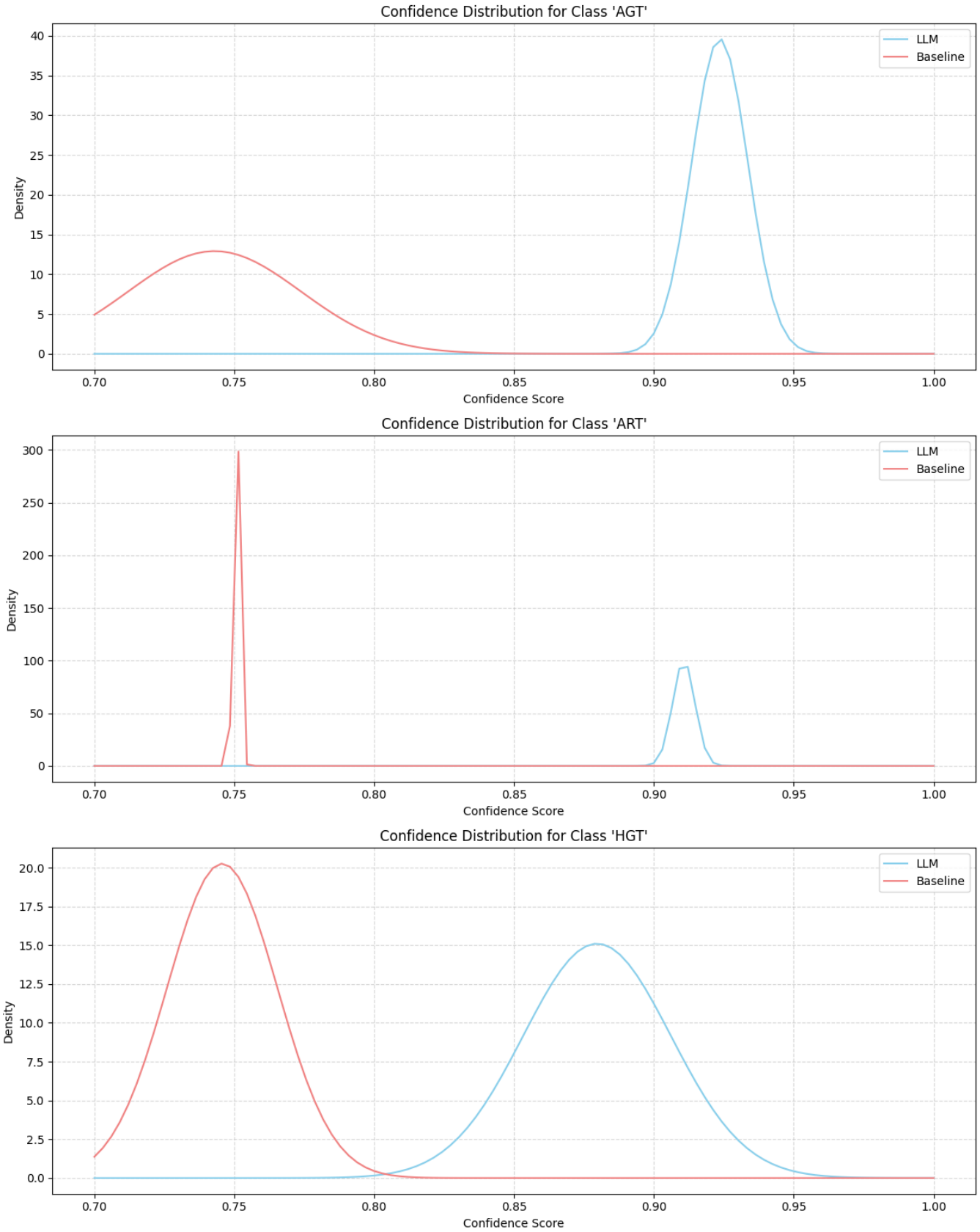


Figure 5.7.: This figure illustrates the kernel density of confidence score distributions for each class, comparing the fine-tuned LLM and Baseline models.

The Figure 5.7 displays kernel density plots of the confidence score distributions for each class (AGT, ART, HGT), comparing the fine-tuned LLM and Baseline Model. Each subplot corresponds to one class, with the x-axis representing the confidence score and the y-axis showing the density of predictions at each confidence level.

- For the AGT class, the LLM model's confidence scores are tightly clustered around 0.92-0.93, indicating very high and consistent confidence. The Baseline model, by contrast, has a broader and lower confidence distribution centered around 0.75.
- For the ART class, the Baseline model's predictions are highly concentrated around 0.75, while the LLM model again shows a higher and narrower peak near 0.91.
- For the HGT class, the Baseline model's confidence is again lower and more spread out, whereas the LLM model maintains a higher average confidence, peaking around 0.89.

These plots indicate that the LLM model is not only more confident overall but also more consistent in confidence across all classes. However, this consistency comes at the risk of being overconfident, as can be seen in the previous figure, where some errors are made with high certainty.

Text Length vs. Error Analysis

Figure 5.8 shows the relationship between text length (word count) and prediction correctness for both models. Both models follow a similar pattern, with the most common predictions occurring when text length is between approximately 15 and 40 words. The number of correct predictions (True, orange) generally decreases as text length increases beyond this range. Incorrect predictions (False, blue) are more prevalent in the range of 10 to 40 words. Graphically, the distributions of correct and incorrect predictions across different text lengths are extremely close between the Baseline and fine-tuned LLM, indicating that text length influences both models in similar ways. The slight differences suggest that the LLM might have a marginally improved ability to handle longer texts, but the overall trend remains consistent across both models.

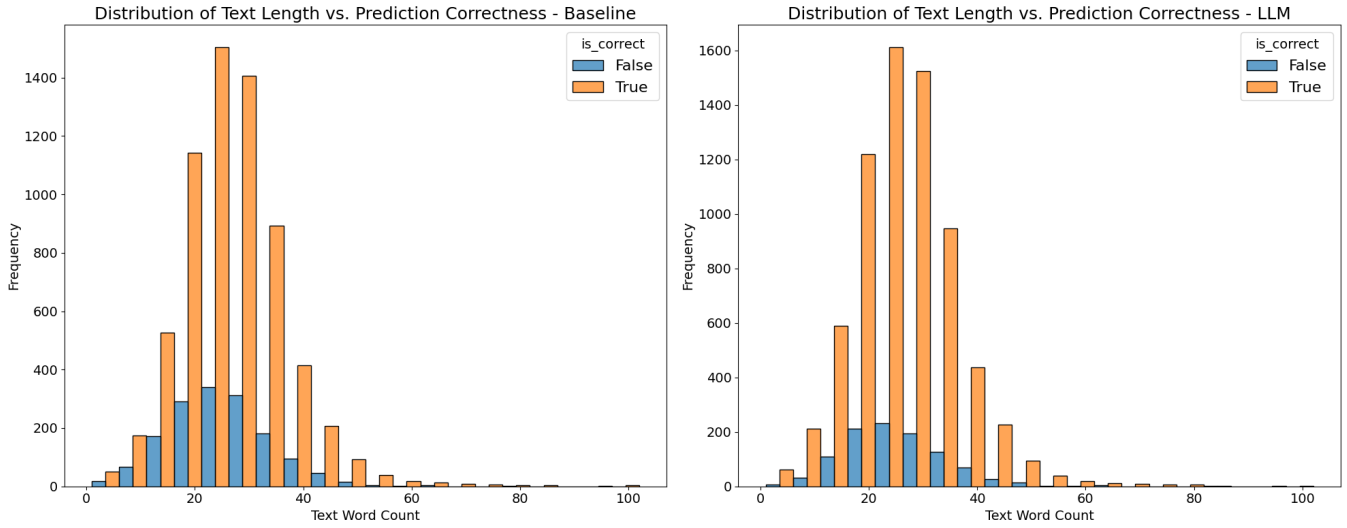


Figure 5.8.: This figure shows the relationship between text length (word count) and prediction correctness for Baseline and fine-tuned LLM.

False Positive / Negative

A comprehensive evaluation of model errors is crucial for understanding the strengths and weaknesses of both the Baseline and LLM models. The attached figures provide a detailed comparison of false positives and false negatives across all classes, highlighting not only the counts but also the associated model confidence for each type of error.

Figure 5.9 visualizes the percentage reduction of false positives and false negatives when switching from the Baseline model to the fine-tuned LLM. The left bar plot shows that the LLM achieves substantial decreases in false positives for ART and HGT (over 40% and 60% respectively), whereas there is a slight increase in false positives for AGT. The right bar plot demonstrates a marked reduction in false negatives in all classes, most notably for HGT with nearly 60% and AGT with nearly 40%. These results confirm that the fine-tuned LLM achieves significant improvements in both types of error for most of the classes, particularly in HGT separation.

The Figure 5.10 offers a more granular breakdown of error counts and mean confidence for both false positives and false negatives, separated by class and model. The top row is a comparison of absolute error counts. The LLM model consistently produces fewer false positives and false negatives across all classes, with the greatest improvement in the 'HGT' class. The bottom row displays the mean confidence associated with these errors. The LLM's false positives are made with much higher confidence than the Baseline, while for the false negatives, both models show relatively low confidence, with the LLM being slightly lower. This pattern only further supports the earlier observation that the LLM predicts with higher confidence, regardless of correctness.

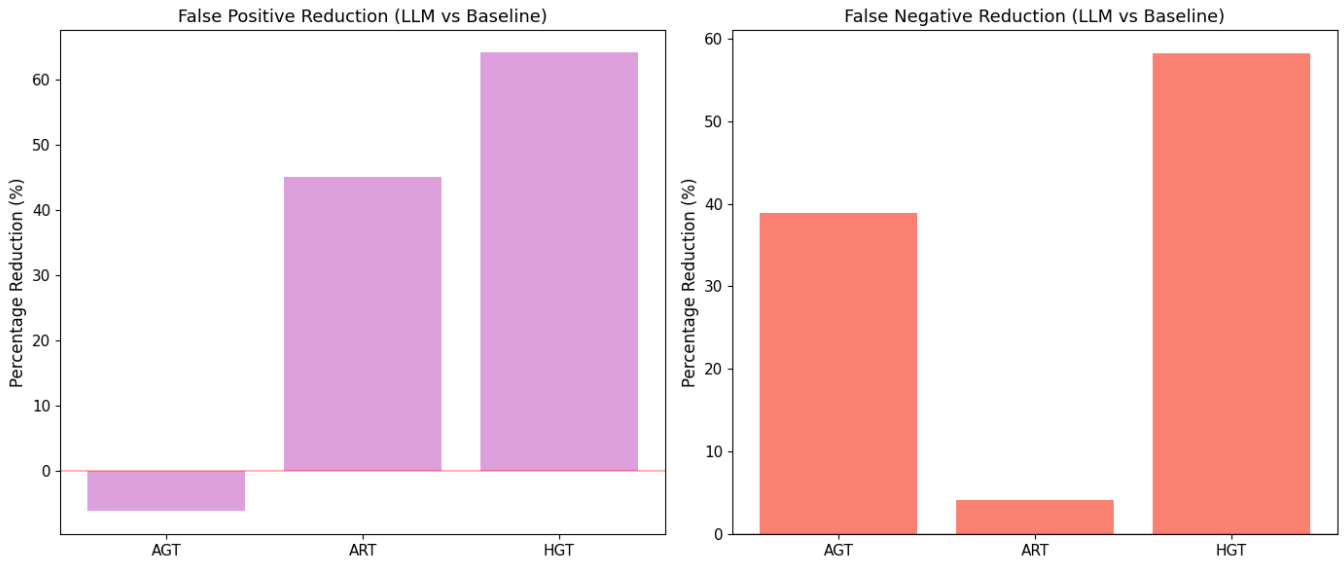


Figure 5.9.: This figure shows bar plots percentage reduction in false positives and false negatives when moving from the Baseline to the fine-tuned LLM model.

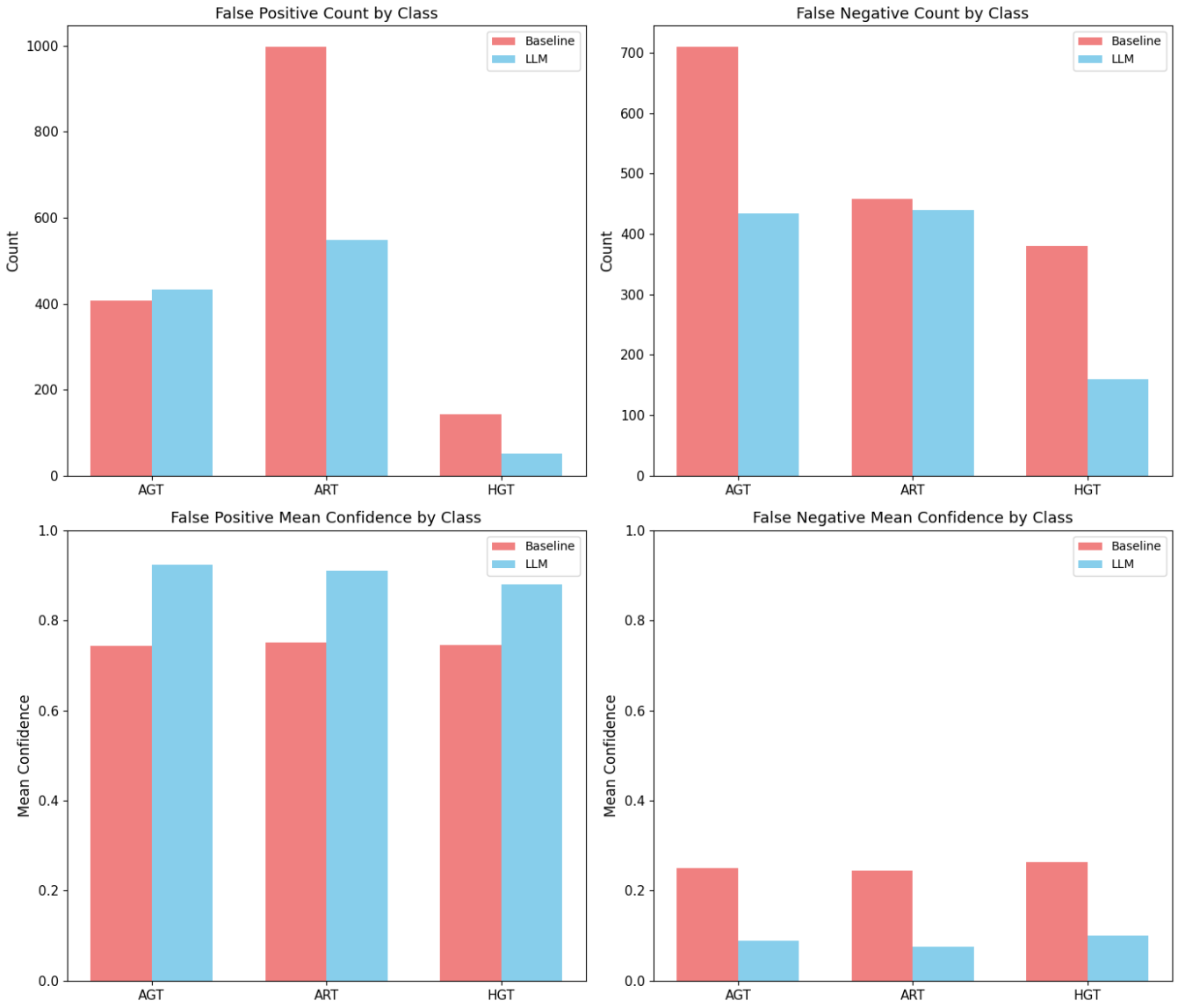


Figure 5.10.: This figure shows bar plots of error counts (top row) and mean confidence scores (bottom row) for false positives and false negatives by class and model.

5.5. Discussion

Our experiments indicate that classifying entire abstracts as HGT, ART, and AGT is relatively easy for modern classifiers, but labeling individual sentences as this is a much more challenging task. In the abstract setting, SciBERT (Beltagy et al., 2019), RoBERTa (Liu et al., 2019), and even general-purpose BERT (Devlin et al., 2019) models all achieved over 95% accuracy.

This unexpected ease can be explained through a number of findings. First, KL divergence analysis at the abstract-level (see Table 4.10) indicates large differences in the word distributions across the three text types. Our data analysis further shows that AI revisions tend to reduce human variability, apply more standardized academic phrasing, and improve coherence. Since abstracts are longer and contain multiple sentences, these rising lexical and stylistic differences become even more pronounced, making it easier for classifiers to distinguish between the text types.

To achieve even finer distinctions with less information, we shifted the strategy over to use sentence-level classification. But during preparation, we also had a significant class-level sentence imbalance: AGT abstracts produced more sentences than their corresponding HGT and ART counterparts. Adding class weights to the loss function corrected the bias, delivering consistent 2–3% gains in precision, recall, accuracy, and F1-score. This adjustment is necessary to ensure equal exposure to all three classes is essential while training on fine-grained units.

The state-of-the-art Llama-3.1-8B-Instruct model posted the top of 87.19% accuracy, and thereby surpassed that of the 80.79 percent of SciBERT Baseline by over six percentage points. Sentences out of context have fewer discriminative features, forcing classifiers to rely more on finer cues in syntax, word choice, and micro-coherence.

Different elements resulted in performance increases seen with instruction-fine-tuned LLaMA. Quantizing every model to a size that would fit within an NVIDIA A100 40 GB GPU allowed us to try higher parameter counts without exorbitant resource costs, and utilizing PEFT (Xu et al., 2023) through LoRA (Hu et al., 2022; Dettmers et al., 2023) allowed for efficient adaptation. It is interesting to note that neither the scaling up of the model size (e.g., Llama-2-13B vs. Llama-3.1-8B) nor doubling LoRA rank promised better results (see Table 5.4), highlighting the need for careful tuning of both architecture and fine-tuning methodology.

Error analyses reveal that the most frequent confusions are between AGT and ART sentences. However, HGT is relatively unambiguous. This is as expected according to our linguistics that AI revisions preserve a lot of the original human structure and heavily depend on generative patterns.

The experimental results show that while whole-abstract classification can achieve near-optimal performance. Robust sentence-level AI revision detection requires models of sufficient size and highly optimized.

6. Conclusion

The increasing integration of AI into academic writing presents a challenge for maintaining transparency, academic integrity, and authorship authenticity. In this study, we address this challenge by distinguishing between HGT, ART, and AGT in academic writing. Existing detection systems and datasets remain largely limited to binary classification and fail to account for the hybrid nature of modern scholarly texts.

To address these issues, we have developed a parallel dataset, SciHRA-Detect, consisting of 9.000 research abstracts on ten academic domains, categorized as HGT, AGT, and ART. The dataset was carefully built on an academic manuscript published before 2020 to minimize prior AI influence, and was balanced by domain and text length. The creation process included rigorous cleaning, overlap filtering, and prompt engineering to maintain high quality and consistency across text types. This allowed us to compare directly between the different text types that have the same topic/content.

The analysis revealed that HGT uses a more varied and flexible vocabulary, while ART and AGT rely more on standardized academic language. Readability and sentiment analyses confirmed this, with HGT being more readable and emotionally balanced, and AGT and ART leaning towards more complex and positively oriented language. Structurally, AGT followed a consistent pattern and emphasized Conclusion sentences are a lot more than HGT and ART. In contrast, ART retained the flexible structure of HGT, suggesting that AI revisions are mostly carried out on the surface level, enhancing clarity without altering the rhetorical flow. In general, HGT is more diverse and natural, AGT is more formal and stiff, and ART is a refined version of HGT that has improved in coherence and clarity.

Experiments with Baseline classifier and fine-tuned LLMs demonstrated that three-way classification of HGT, AGT, and ART is feasible, with LLM-based models achieving up to 87.31% F1-score.

Through comparative analysis of HGT, AGT, and ART, this thesis has illustrated how ART is an outlier and troublesome category, often misclassified as AGT, which can lead to unfair assessment in academic writing. The research confirms that differentiation between the three text types is not just possible but essential to maintaining academic integrity in an era of ”ubiquitous AI assistance”.

7. Limitation & Future Work

In this section, we first identify and discuss the key limitations in our dataset construction, model specifications, and analytical processes. We then provide specific recommendations for future work to address these gaps and increase the robustness and generalizability of our results.

7.1. Limitation

While covering a wide range of paper categories, our dataset has some notable limitations and biases in the creation and revision of text. To begin with, we only use GPT-4o for both AI-generated and AI-revised text. While GPT-4o is both powerful and popular, it is also neither the newest nor possibly the best available model on the market for academic writing tasks. Practitioners will often use a range of models, including newer OpenAI models.

In addition, our study focuses on English-language texts, limiting the generalizability of our results to other languages and writing styles. Another limitation of the methodology arises from our prompt engineering process. We manually engineered prompts that closely matched human-written abstracts and constructed our dataset based on these. As a result, the dataset is heavily dependent on our two prompts, and there is potential for bias. Ideally, a more robust strategy would be to employ multiple prompts and compare the resulting analyses systematically. However, time and resource constraints prevented this broader investigation.

The shift from abstract-level to sentence-level analysis introduced a class imbalance in our dataset, which we attempted to counter by applying class weighting to our classification models. Nevertheless, a more balanced dataset would likely yield better results. In terms of data selection, we employed articles published before 2020 to reduce the chances of AI participation in writing. We cannot completely rule out the possibility that authors utilized previous AI-supported tools (e.g., Grammarly), which can introduce unknown bias. This method also implies that our results might not represent current academic writing habits, where AI use is more common.

Our analysis of top words, bigram, and trigram analysis is subject to small changes with limited scalability, which would affect the robustness of such findings.

We employed only a SciBERT-based classifier in functional labeling on the sentence-level, which would have introduced one-sidedness in the analysis. Similarly, our experimental section is limited to models of LLaMA.

7.2. Future Work

Building on the limitations of this thesis, several ways for future investigation are arising, which could significantly increase the richness and external validity of the findings. One strong area is to improve the diversity of language models used in producing and revising abstracts. While this study employed only GPT-4o, follow-up studies should incorporate additional models. That would allow for a more comprehensive comparison and a greater understanding of how different AI systems influence features of scholarly writing.

Packing more diversity into prompts used to generate and revise abstracts would demonstrate how prompt construction is sensitive to our results. Similarly, incorporating multiple LLMs (notably newer or more domain-specialized architectures) would demonstrate if model choice matters to revision behaviors and accuracy classification. Extending the corpus to include non-English abstracts and publications from after 2020 would show how AI-driven tools have reshaped scholarly writing conventions in recent years. Constructing an evenly balanced sentence-level corpus would eliminate class weighting and enable more fair analysis of stylistic cues. Moreover, the *SciHRA-Detect* dataset was designed specifically across ten distinct academic domains to enable domain-specific investigations. While this study has already done initial analyses in this regard, there remains great potential for more detailed research. Such work could lead to more refined and context-aware detection models that consider domain-specific writing conventions.

In our sentence-level analysis, we used a SciBERT-based classifier to assign 5 rhetorical roles to abstract sentences. SciBERT performs well for scientific text, but employing a single model might be biased. Future studies would need to utilize other classification models to compare label consistency and whether the structural findings are model-dependent.

Finally, future research should not only focus on the technical development but also the ethical and pragmatic aspects of AI in academic research. Setting standards for the open and responsible use of AI tools for academic writing will become increasingly important as these technologies evolve and propagate.

By addressing these directions, future research can help build a more comprehensive, balanced, and ethically informed understanding of AI's role in scholarly communication. Without such effort, authorship may become increasingly ambiguous, and trust in published research could start breaking down. This study sets the foundation for developing fair and robust classification systems that support the responsible and transparent integration of AI in academic writing.

Acknowledgments

I, Mithusan Naguleswaran, declare that I wrote all the content of this thesis myself. After finishing my thesis, I corrected misspellings and grammatical errors with the help of the AI-powered language tool Grammarly, DeepL and ChatGPT.

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A. Prompts for Generating AI Abstracts

Here is the prompt for generating the AI-Revised Texts (ART):

System:
You are an AI academic writing assistant specializing in revising academic abstracts.
Your task is to revise and polish the provided abstract (100–250 words) to improve its clarity, coherence, and alignment with the highest academic standards for the domain.
Focus on improving sentence structure, word choice, and logical flow while maintaining scientific accuracy, the original meaning, and the core messages.
Guidelines:

- Ensure the revised abstract maintains the same structure as the original.
- Write in a clear, scholarly tone that aligns with the voice and conventions of the domain.
- The revised abstract must remain faithful to the original text, capturing its essence while enhancing its clarity and impact.
- Avoid introducing new information or diverging from the original content."

User:
Here is the abstract to revise:

[ABSTRACT]

Provide only the finalized abstract as a single, continuous paragraph.
Do not include headings, bullet points, or additional commentary.

Here is the prompt for generating the AI-Generated Texts (AGT):

System:
You are an AI acting as the author of a research paper.
Your task is to generate a concise, polished abstract (100–250 words) based on the provided full text. The abstract should sound as though it was written by the actual author of the paper and meet the highest academic standards for the domain.
Guidelines:

- Identify and summarize the typical kinds of information found in most abstracts, which may include:

  1. The context or background of the study, explaining its relevance or importance.
  2. The research problem, objective, or central question addressed by the study.
  3. The methods or approach used to conduct the research.
  4. The key findings, results, or arguments derived from the study.
  5. The implications, significance, or contributions of the research to the field.

- Write in a clear, scholarly tone that aligns with the voice and conventions of the domain.
- Maintain a concise structure, balancing brevity with completeness.
- Avoid introducing new information or diverging from the original content.
- Focus on clarity, coherence, and academic rigor, while preserving the original intent and key messages.

User:
Here is the full text:

[FULLTEXT WITHOUT ABSTRACT]

Provide only the finalized abstract as a single, continuous paragraph.
Do not include headings, bullet points, or additional commentary.

For the AGT prompt, additional guidelines on how to write good abstracts were included, as shown in the nested box above. They were meant to direct the language model to produce abstracts highly in line with academic conventions and the other text types. For further information on abstract format and requirements, see Section 3.3 ”What is a Good Abstract?”.

B. Frequency Analysis

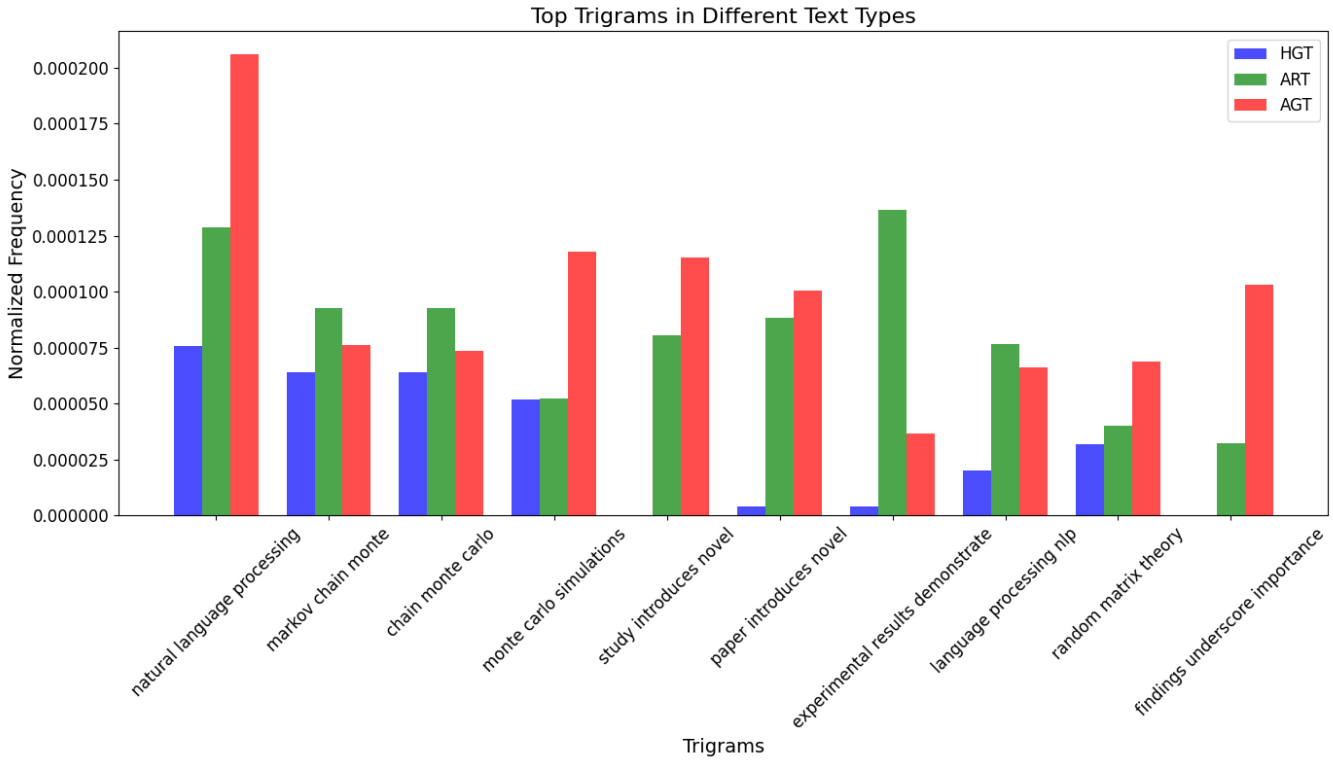


Figure B.1.: Normalized frequency of the top trigram words in different text types.

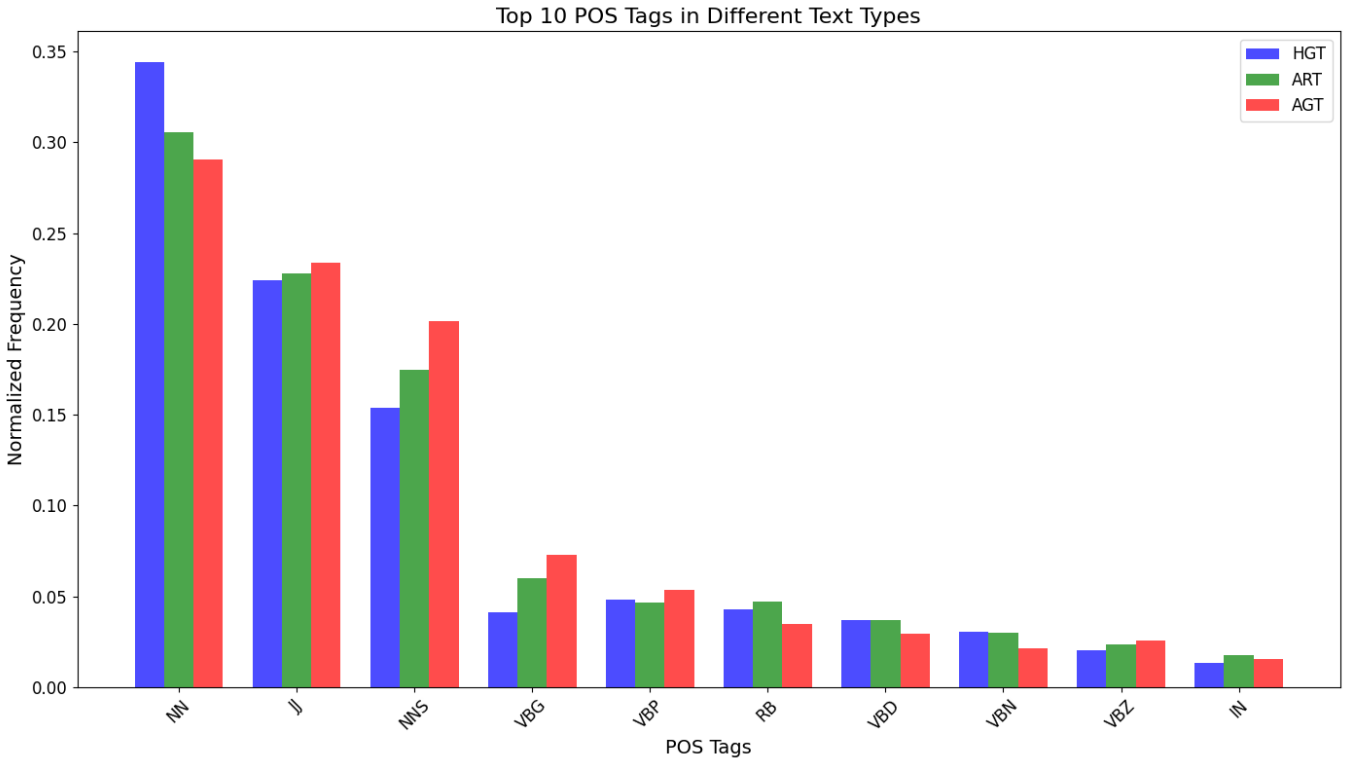
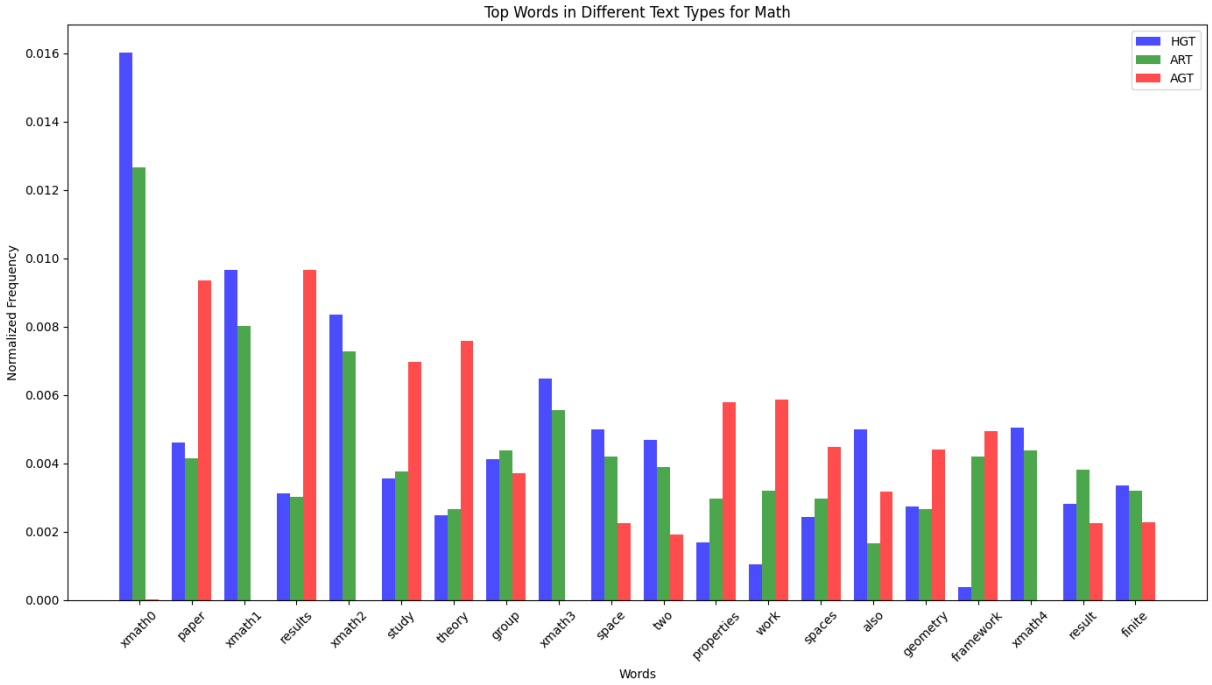
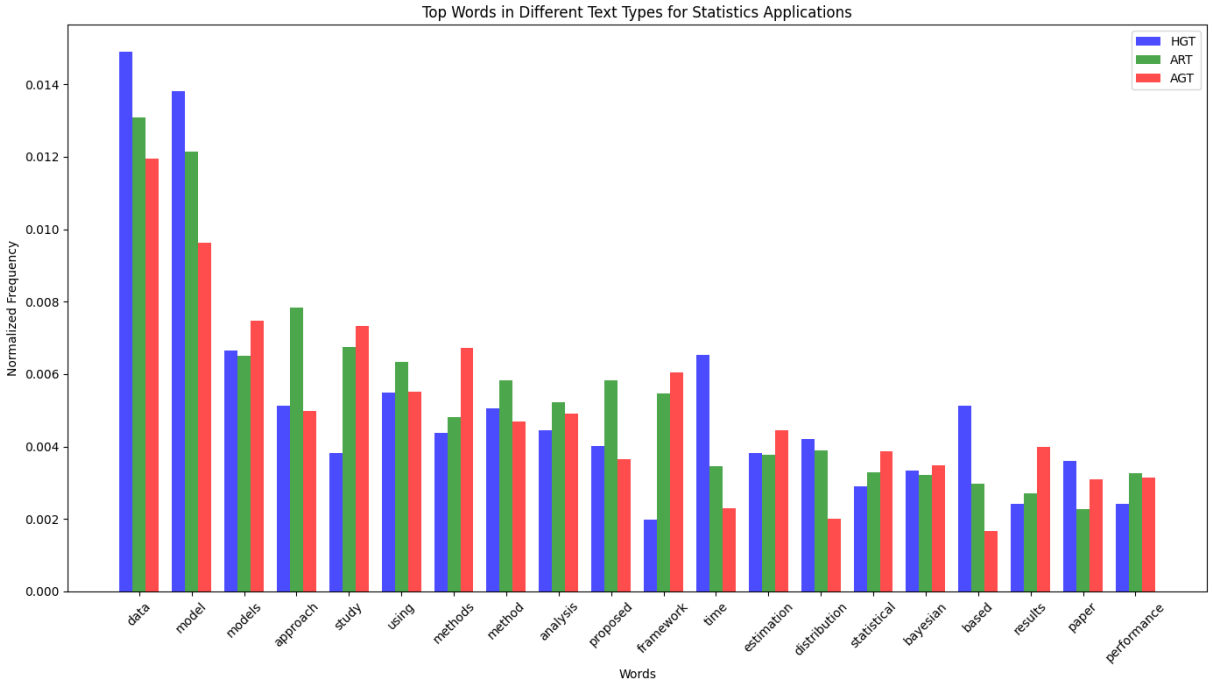


Figure B.2.: Normalized frequency of the top 10 POS Tags in different text types.

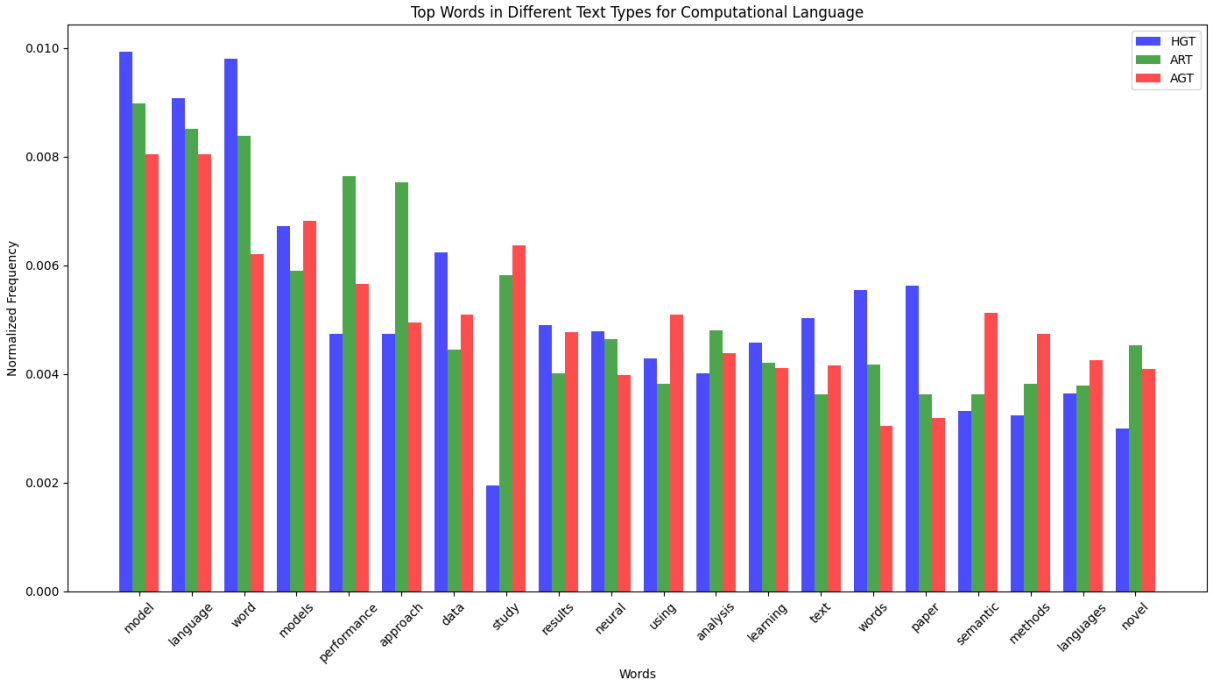


Top words in different text types in domain ”Math”

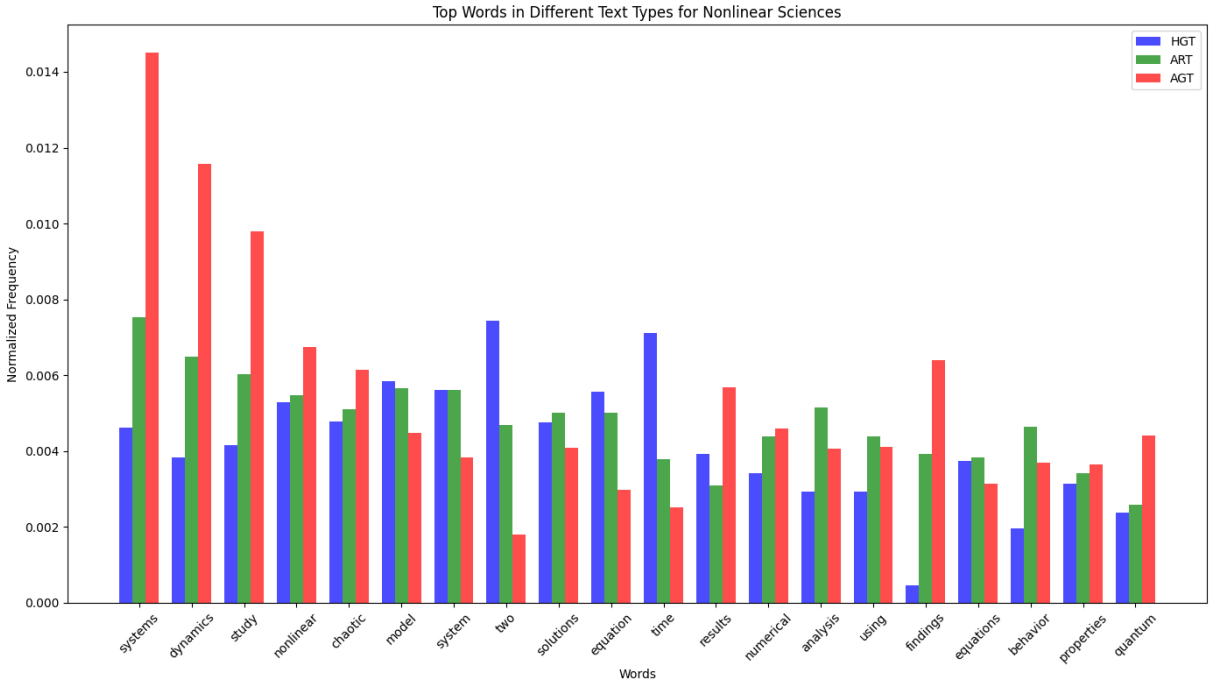


Top words in different text types in domain ”Statistics Application”

Figure B.3.: Normalized frequencies of the most common words across the different text types within ”Math” & ”Statistics Application”

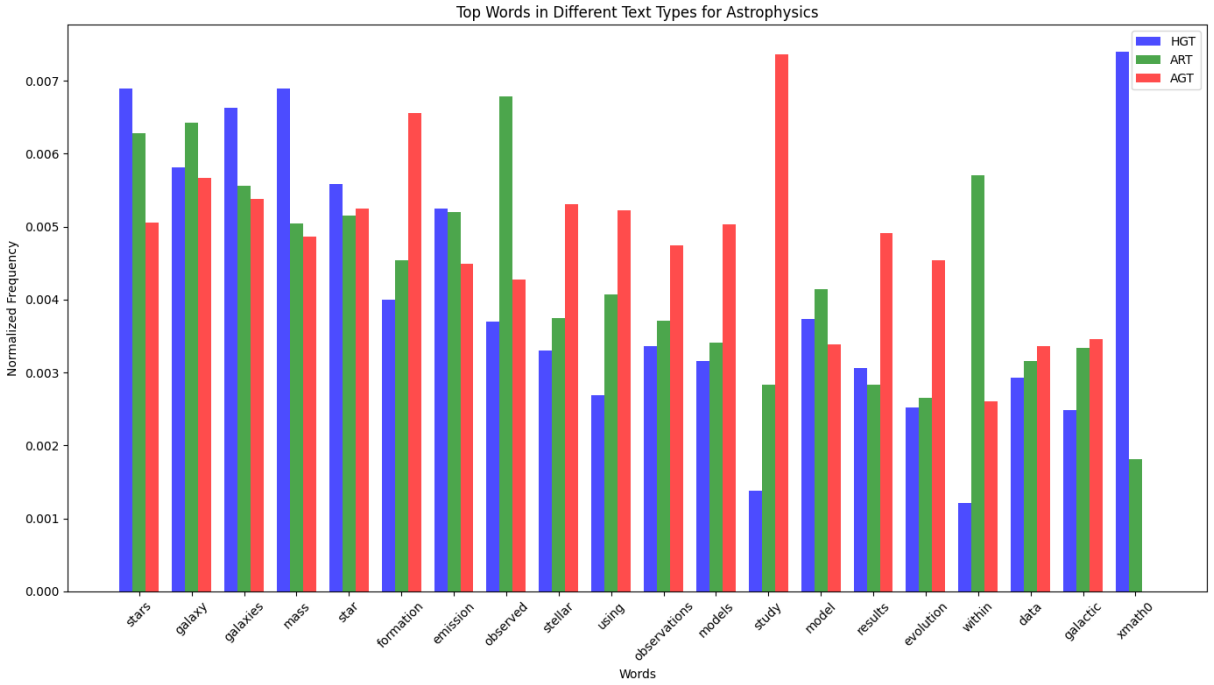


Top words in different text types in domain ”Computational Language”

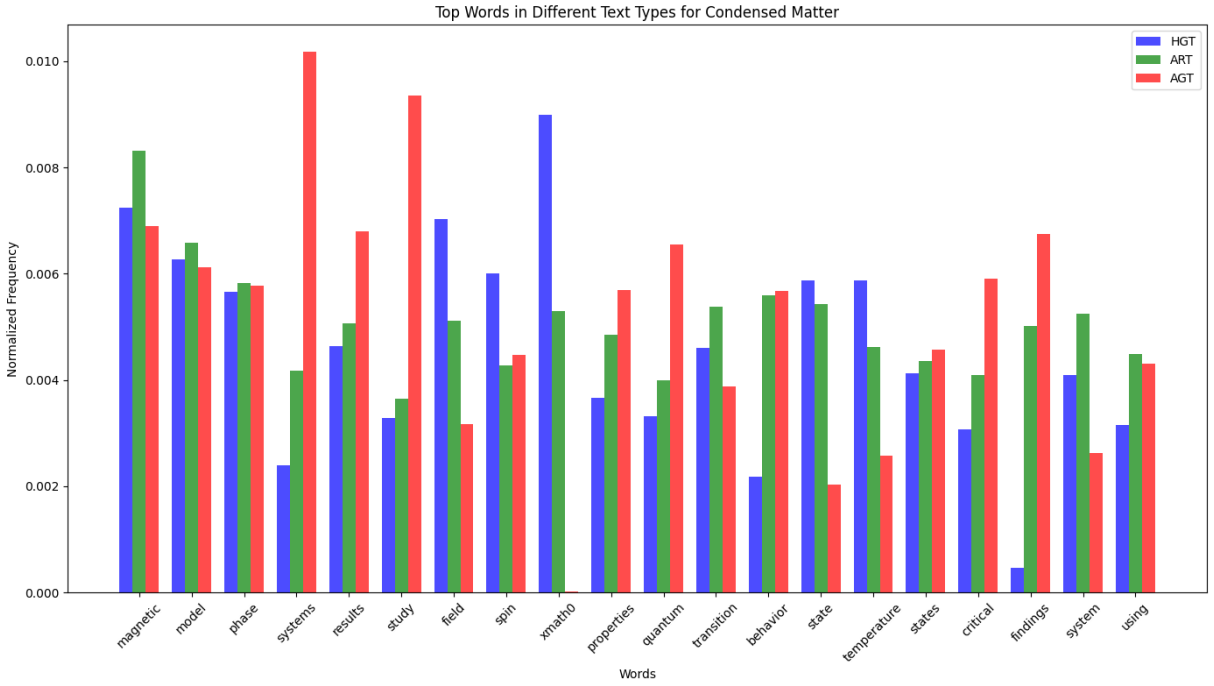


Top words in different text types in domain ”Nonlinear Sciences”

Figure B.4.: Normalized frequencies of the most common words across the different text types within ”Computational Language” & ”Nonlinear Sciences”

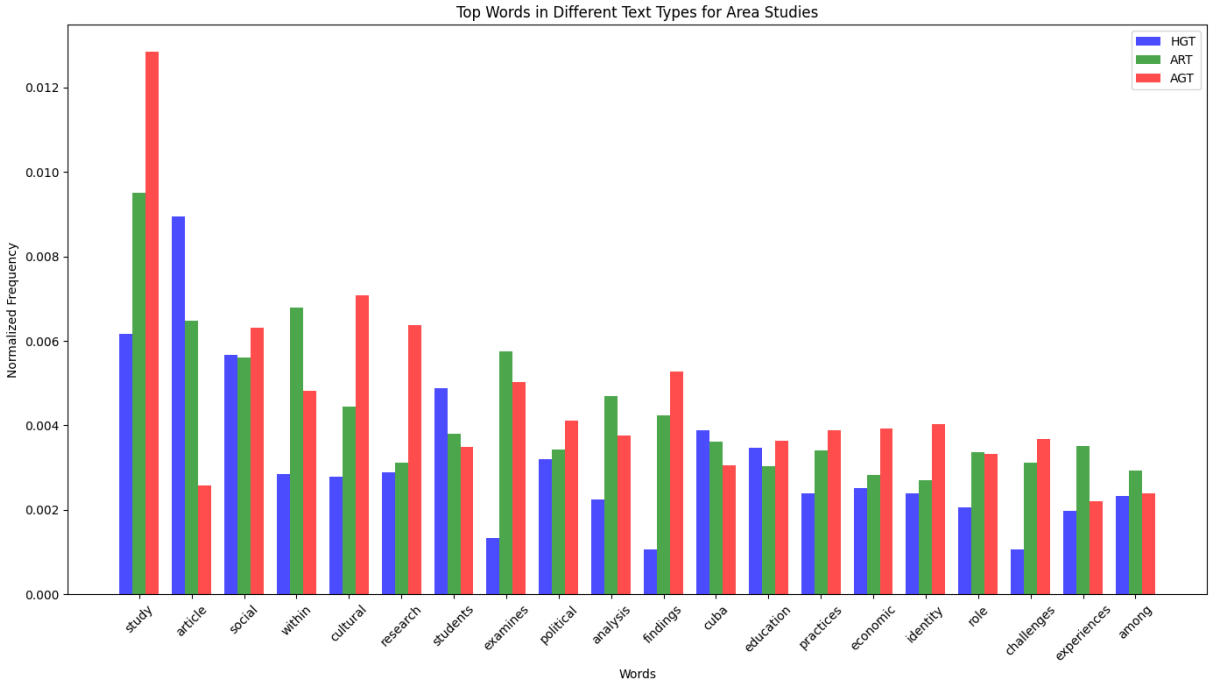


Top words in different text types in domain ”Astrophysics”

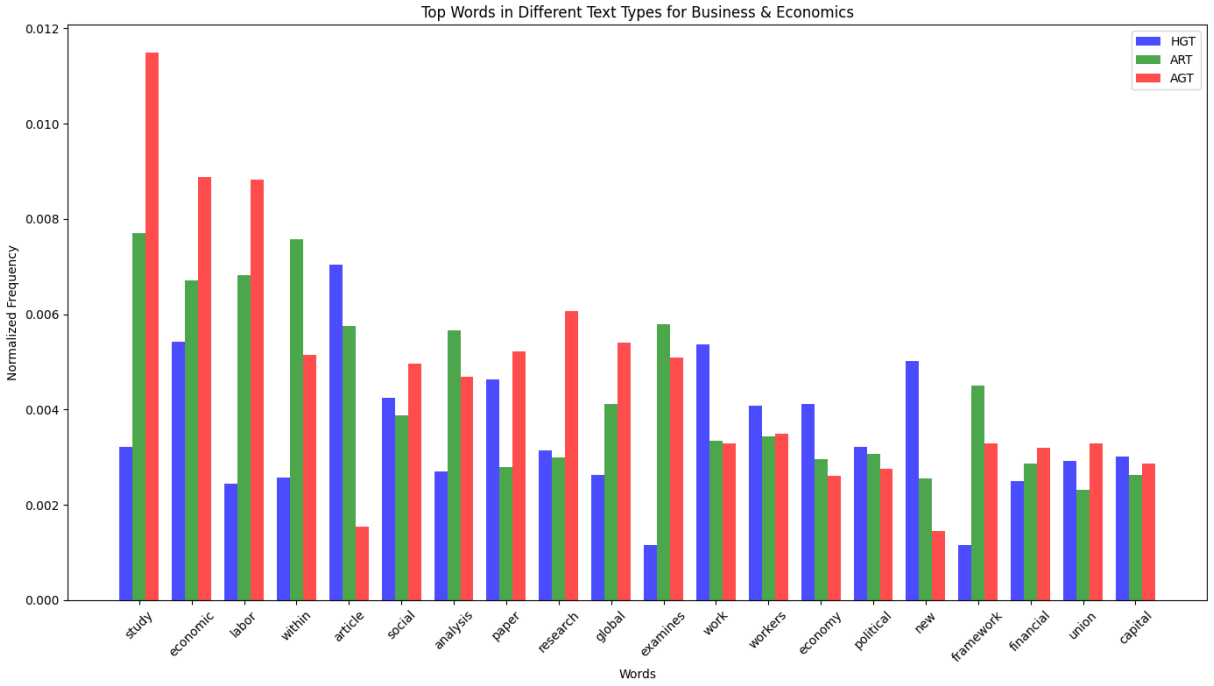


Top words in different text types in domain ”Condensed Matter”

Figure B.5.: Normalized frequencies of the most common words across the different text types within ”Astrophysics” & ”Condensed Matter”

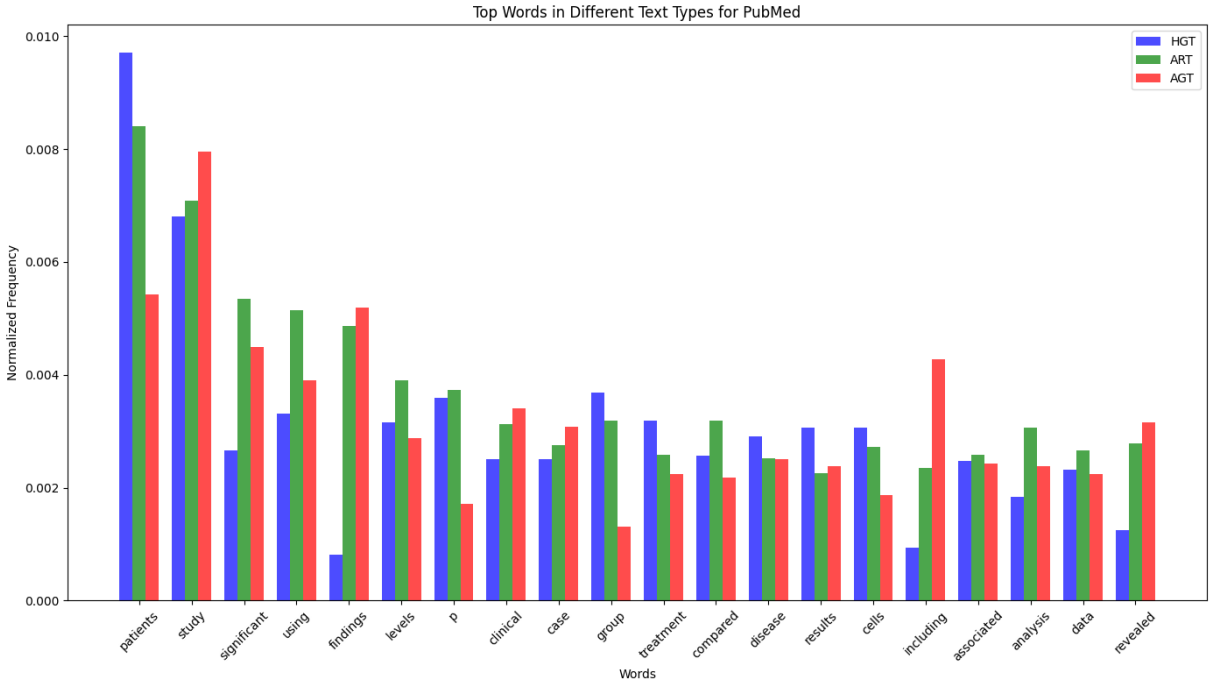


Top words in different text types in domain ”Area Studies”

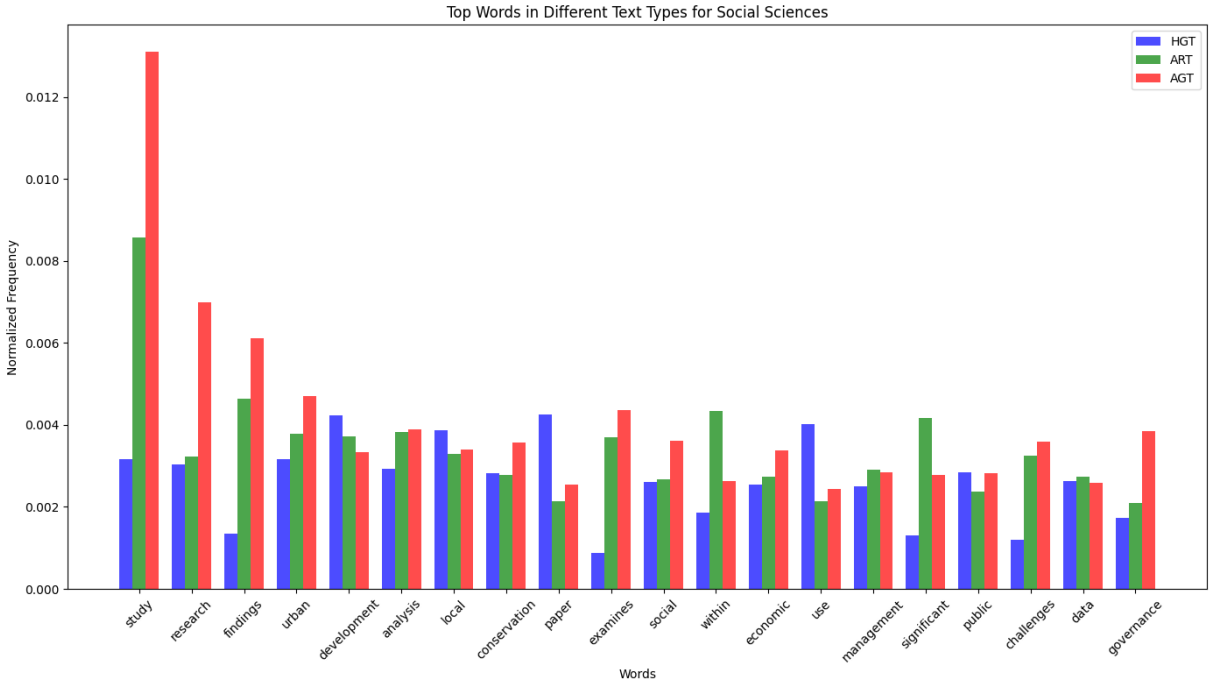


Top words in different text types in domain ”Business & Economics”

Figure B.6.: Normalized frequencies of the most common words across the different text types within ”Area Studies” & ”Business & Economics”

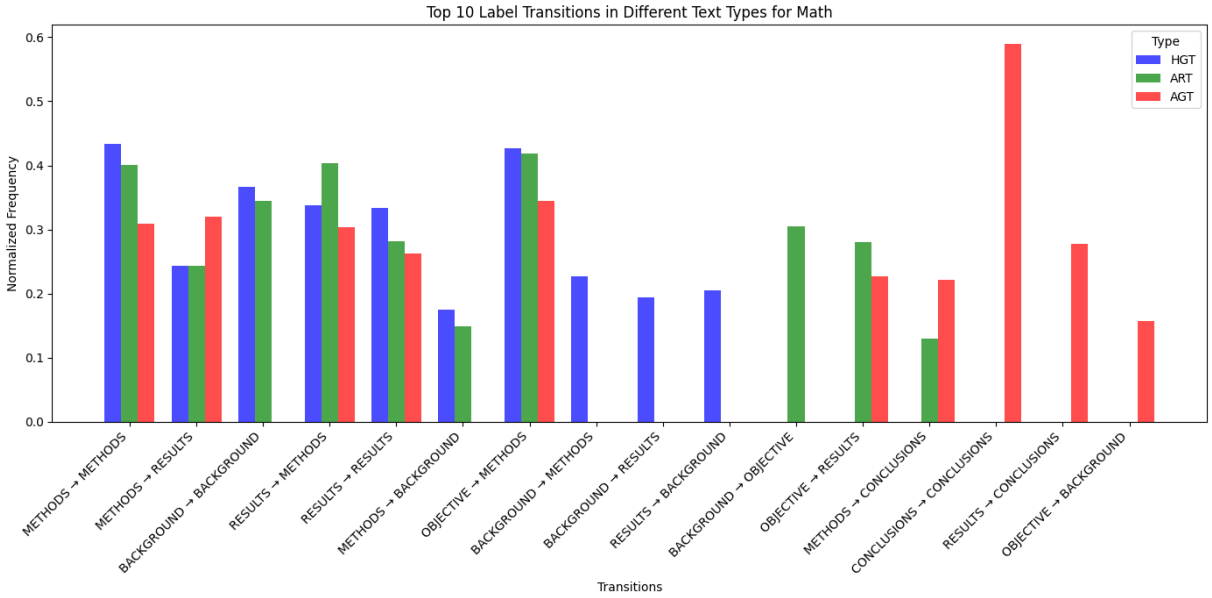


Top words in different text types in domain ”PubMed”

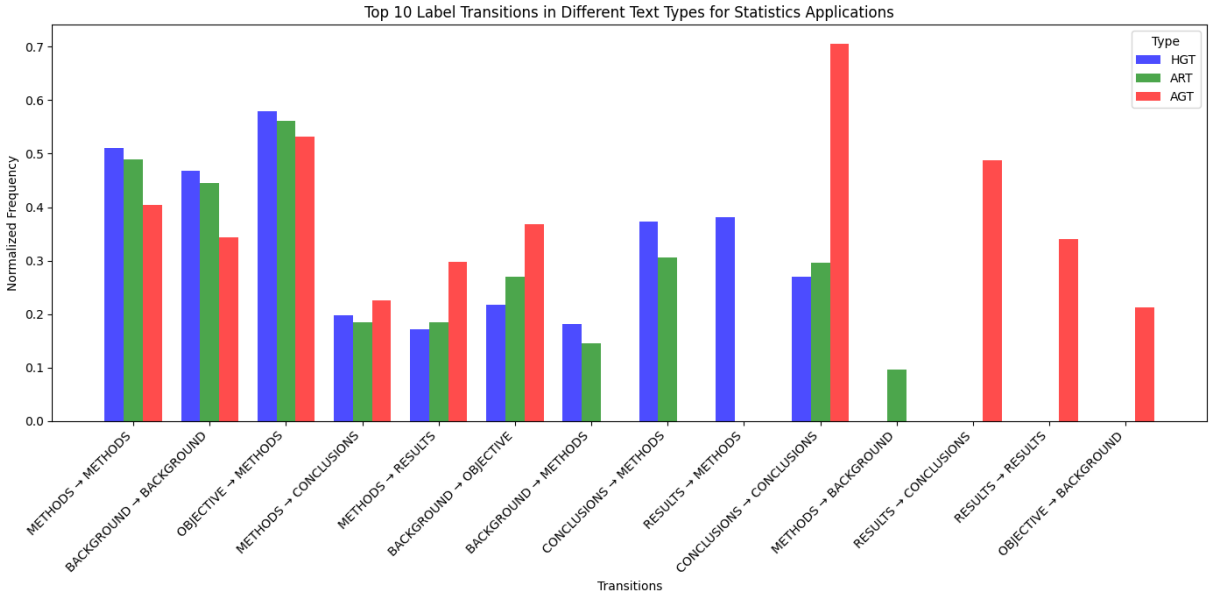


Top words in different text types in domain ”Social Sciences”

Figure B.7.: Normalized frequencies of the most common words across the different text types within ”PubMed” & ”Social Sciences”

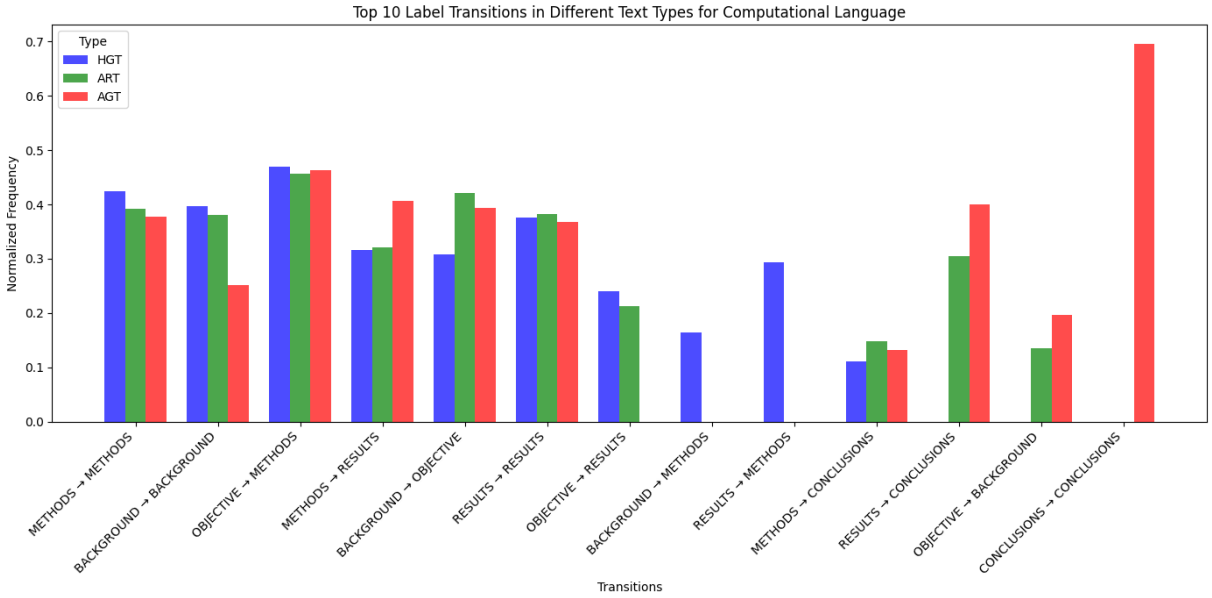


Top transition in different text types in domain ”Math”

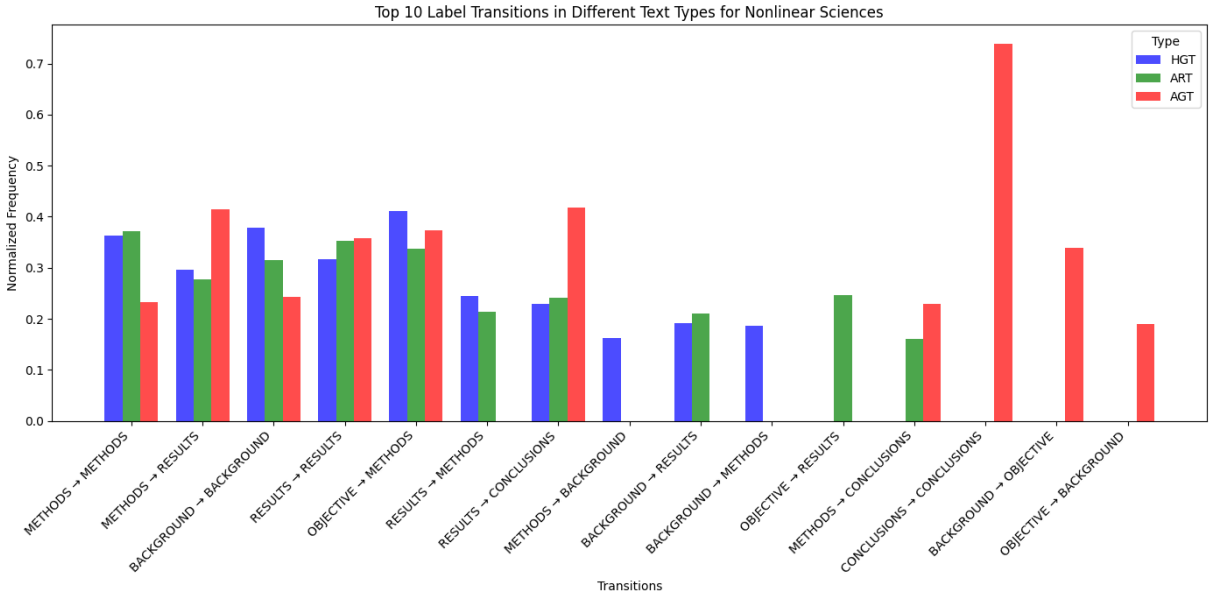


Top transition in different text types in domain ”Statistics Application”

Figure B.8.: Normalized frequency of the top transition adjacent sentences across different text types within the ”Math” & ”Statistics Application” domains.

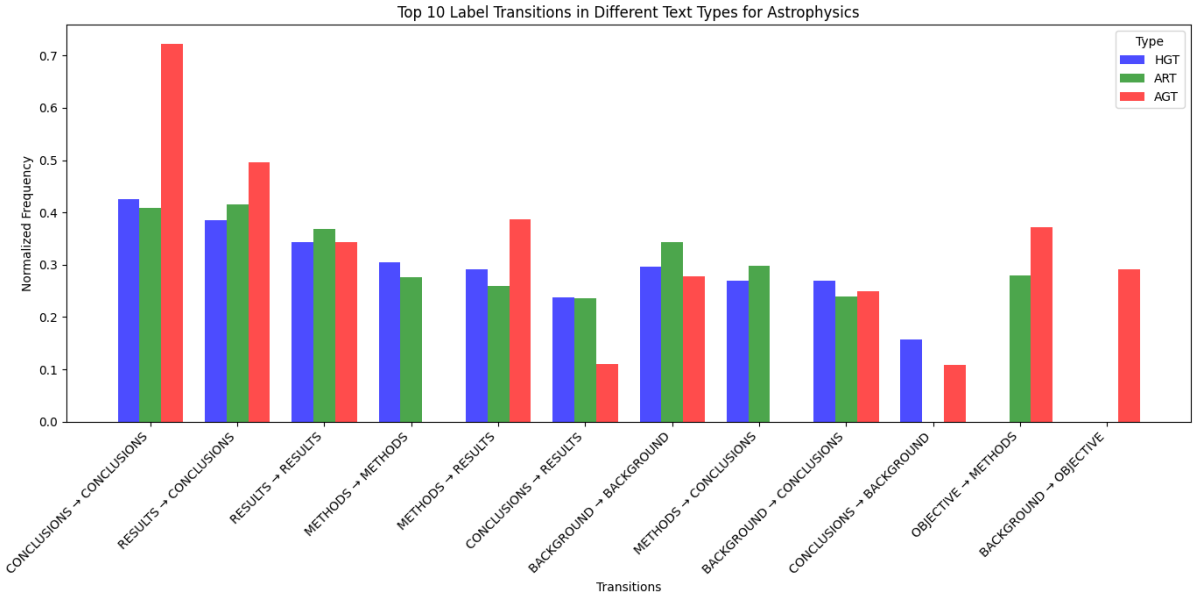


Top transition in different text types in domain ”Computational Language”

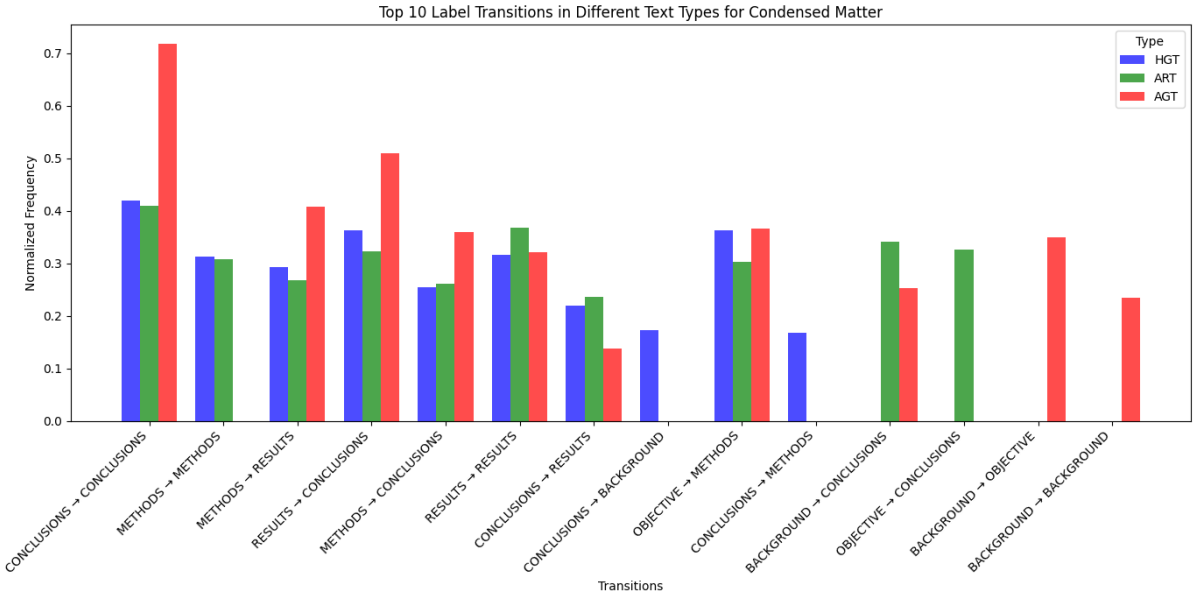


Top transition in different text types in domain ”Nonlinear Sciences”

Figure B.9.: Normalized frequency of the top transition adjacent sentences across different text types within the ”Computational Language” & ”Nonlinear Sciences” domains.

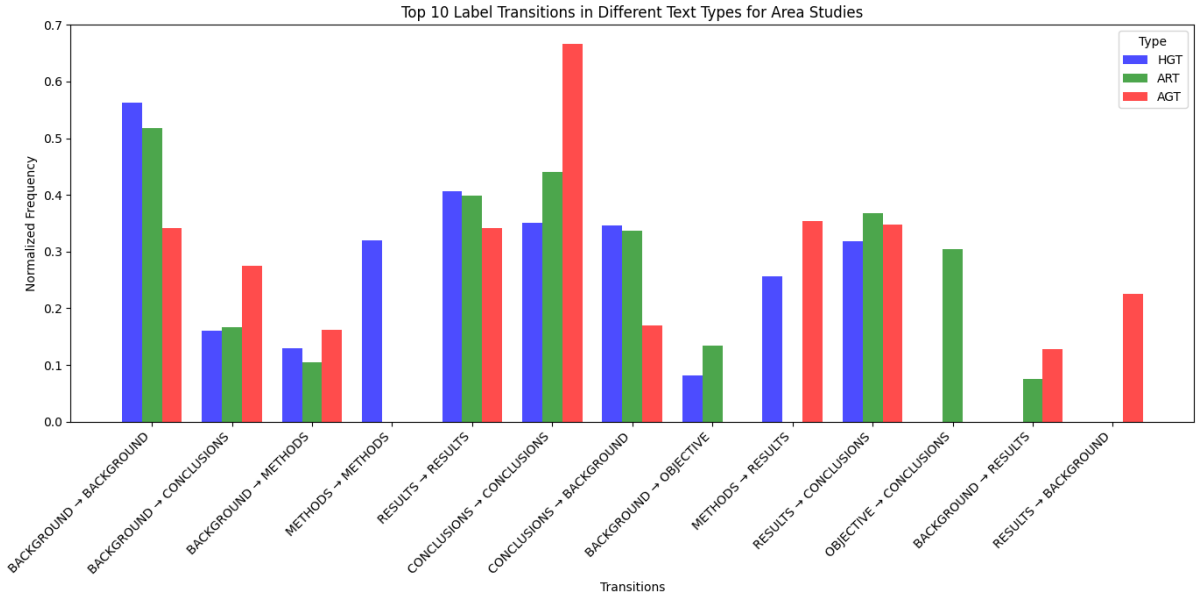


Top transition in different text types in domain ”Astrophysics”

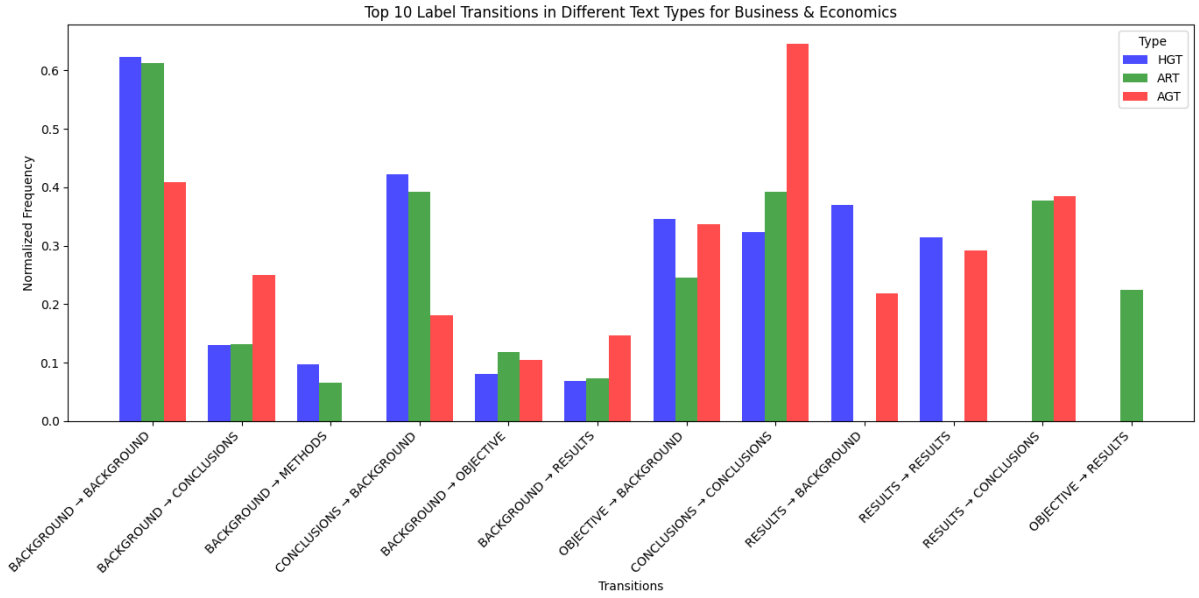


Top transition in different text types in domain ”Condensed Matter”

Figure B.10.: Normalized frequency of the top transition adjacent sentences across different text types within the ”Astrophysics” & ”Condensed Matter” domains.

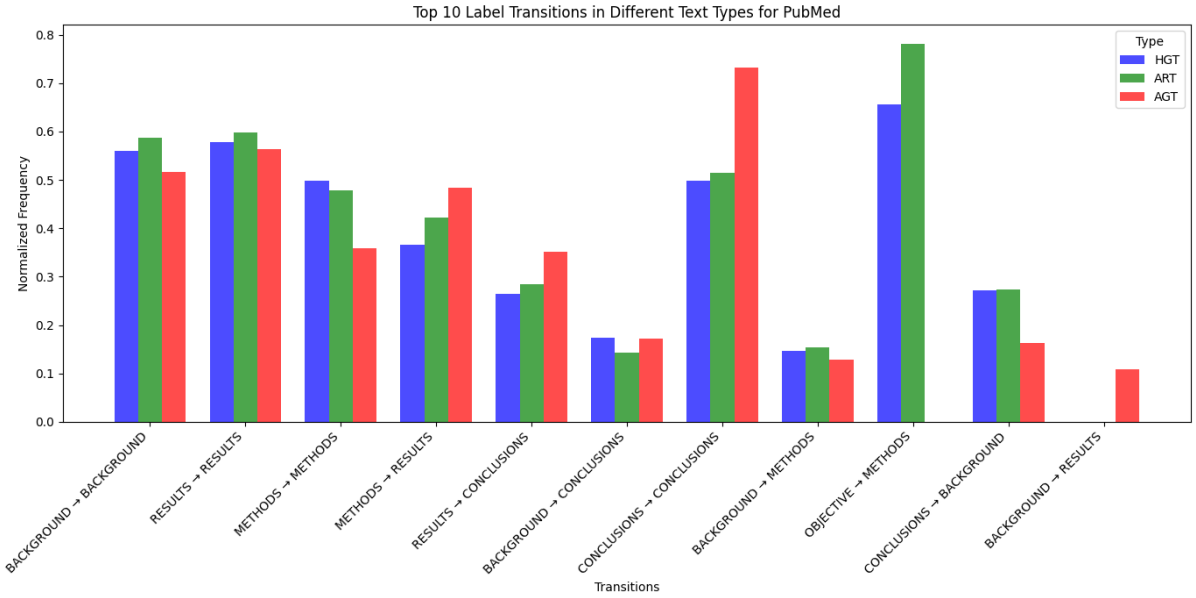


Top transition in different text types in domain ”Area Studies”

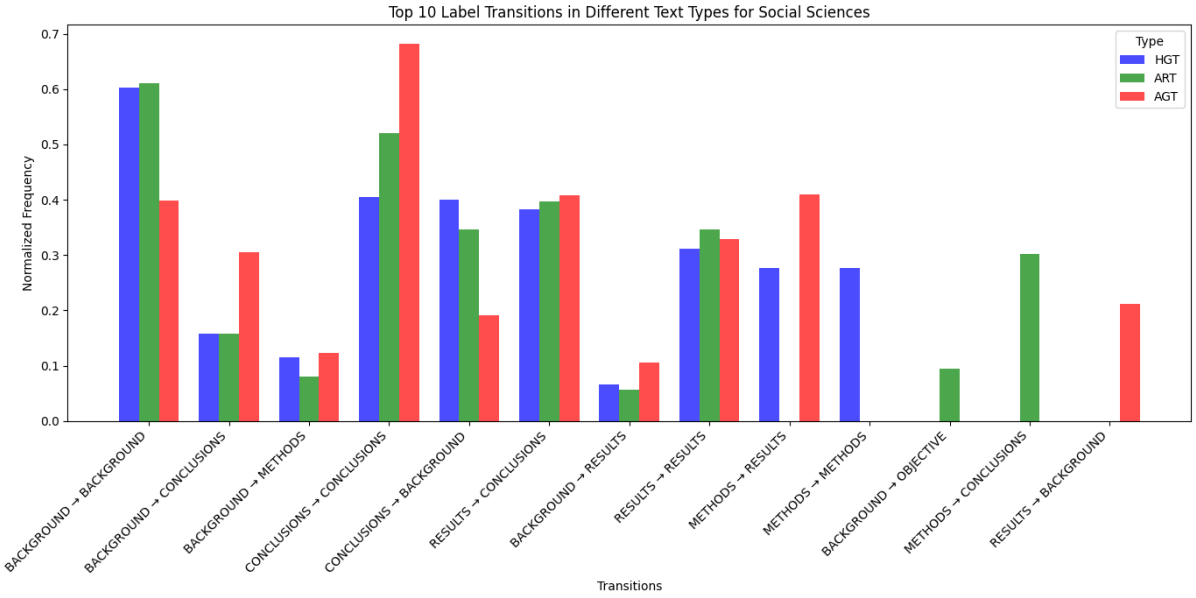


Top transition in different text types in domain ”Business & Economics”

Figure B.11.: Normalized frequency of the top transition adjacent sentences across different text types within the ”Area Studies” & ”Business & Economics” domains.



Top transition in different text types in domain ”PubMed”



Top transition in different text types in domain ”Social Sciences”

Figure B.12.: Normalized frequency of the top transition adjacent sentences across different text types within the ”PubMed” & ”Social Sciences” domains.

C. Comparative Metric Analysis Sentence Label

The tables provided present a comparative overview of linguistic and structural metrics across different models (HGT, ART, AGT) and sentence label (Background, Methods, Objective, Results, Conclusions).

Metric	HGT	ART	AGT
Flesch Kincaid	10.730	12.208	14.582
Flesch Reading Ease	22.084	9.828	4.500
Avg. Sentence Length	19.718	20.083	23.969
Passive Voice Ratio	0.221	0.188	0.113
Ratio Was/Were	0.086	0.037	0.021
Unique Word Ratio	0.582	0.595	0.673
Hapax Legomena	27.605	28.809	32.598
Yule's K	0.011	0.008	0.008
Shannon Entropy	3.733	3.697	4.124
Coherence Score	0.291	0.287	0.314
Burstiness	0.177	0.135	0.131
MATTR	0.453	0.453	0.547
Dugast's Uber Index	55.199	67.832	91.686

Table C.1.: This table presents the metric values for the Background sentence label across the different text types.

Metric	HGT	ART	AGT
Flesch Kincaid	9.754	10.061	12.849
Flesch Reading Ease	22.926	10.293	4.622
Avg. Sentence Length	18.239	17.281	21.944
Passive Voice Ratio	0.315	0.302	0.251
Ratio Was/Were	0.232	0.204	0.187
Unique Word Ratio	0.560	0.513	0.604
Hapax Legomena	21.092	19.679	23.543
Yule's K	0.011	0.007	0.008
Shannon Entropy	3.352	3.006	3.476
Coherence Score	0.203	0.171	0.204
Burstiness	0.168	0.101	0.096
MATTR	0.333	0.269	0.354
Dugast's Uber Index	52.091	56.880	82.503

(a) **Methods** section

Metric	HGT	ART	AGT
Flesch Kincaid	7.924	10.579	14.966
Flesch Reading Ease	14.189	8.492	5.589
Avg. Sentence Length	15.576	18.402	26.487
Passive Voice Ratio	0.107	0.068	0.030
Ratio Was/Were	0.035	0.008	0.004
Unique Word Ratio	0.441	0.532	0.685
Hapax Legomena	12.753	16.102	25.269
Yule's K	0.005	0.006	0.007
Shannon Entropy	2.357	2.848	3.929
Coherence Score	0.075	0.125	0.289
Burstiness	0.034	0.047	0.082
MATTR	0.099	0.151	0.425
Dugast's Uber Index	44.790	63.677	93.722

(b) **Objective** section

Table C.2.: This table presents the metric values for the Methods and Objective sentence labels across the different text types.

Metric	HGT	ART	AGT
Flesch Kincaid	8.723	9.078	14.291
Flesch Reading Ease	20.164	10.287	6.344
Avg. Sentence Length	17.971	16.926	24.907
Passive Voice Ratio	0.189	0.148	0.154
Ratio Was/Were	0.216	0.157	0.194
Unique Word Ratio	0.490	0.463	0.671
Hapax Legomena	17.876	17.670	29.779
Yule's K	0.008	0.007	0.008
Shannon Entropy	2.919	2.724	4.073
Coherence Score	0.176	0.164	0.299
Burstiness	0.103	0.080	0.129
MATTR	0.248	0.228	0.524
Dugast's Uber Index	46.159	49.737	89.957

(a) **Results** section

Metric	HGT	ART	AGT
Flesch Kincaid	8.954	11.176	18.344
Flesch Reading Ease	18.690	7.776	3.940
Avg. Sentence Length	16.265	17.660	27.785
Passive Voice Ratio	0.200	0.117	0.072
Ratio Was/Were	0.045	0.022	0.041
Unique Word Ratio	0.502	0.558	0.803
Hapax Legomena	17.558	19.677	46.348
Yule's K	0.008	0.006	0.010
Shannon Entropy	2.876	3.125	5.371
Coherence Score	0.174	0.205	0.434
Burstiness	0.108	0.088	0.193
MATTR	0.253	0.270	0.821
Dugast's Uber Index	49.423	68.091	97.016

(b) **Conclusions** section

Table C.3.: This table presents the metric values for the Results and Conclusions sentence labels across the different text types.

D. Comparative Metric Analysis across Domains

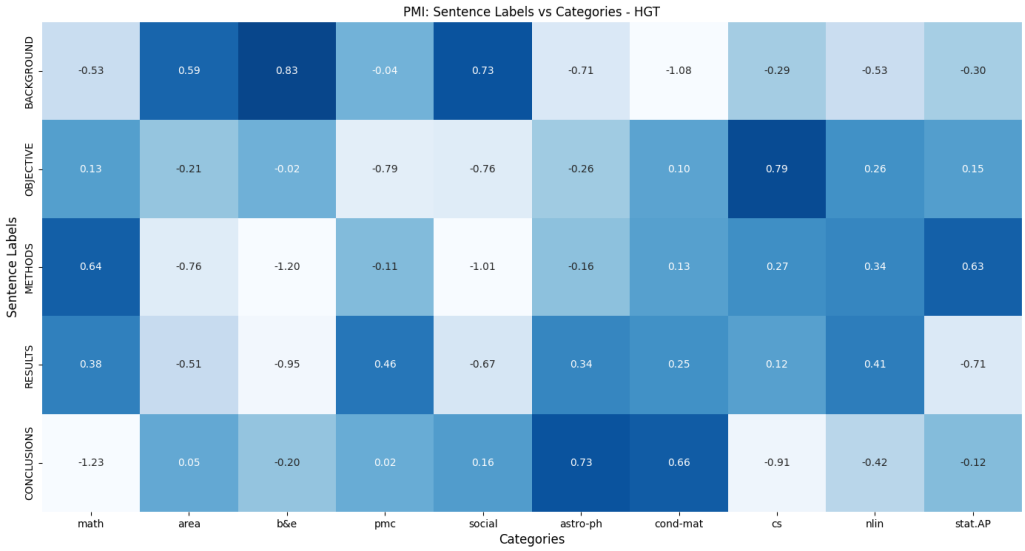


Figure D.1.: Pointwise Mutual Information values between sentence labels and scientific categories for HGT.

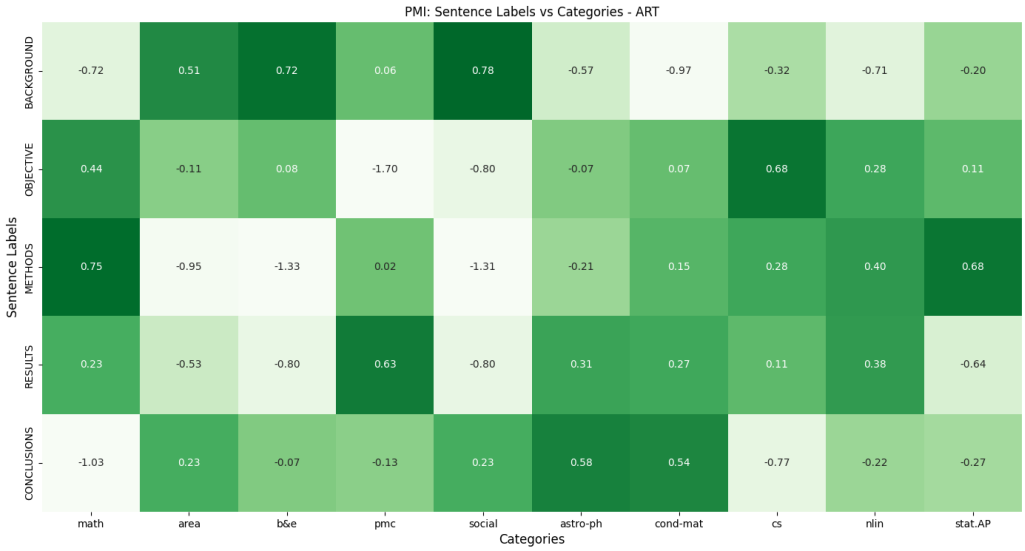


Figure D.2.: Pointwise Mutual Information values between sentence labels and scientific categories for ART.

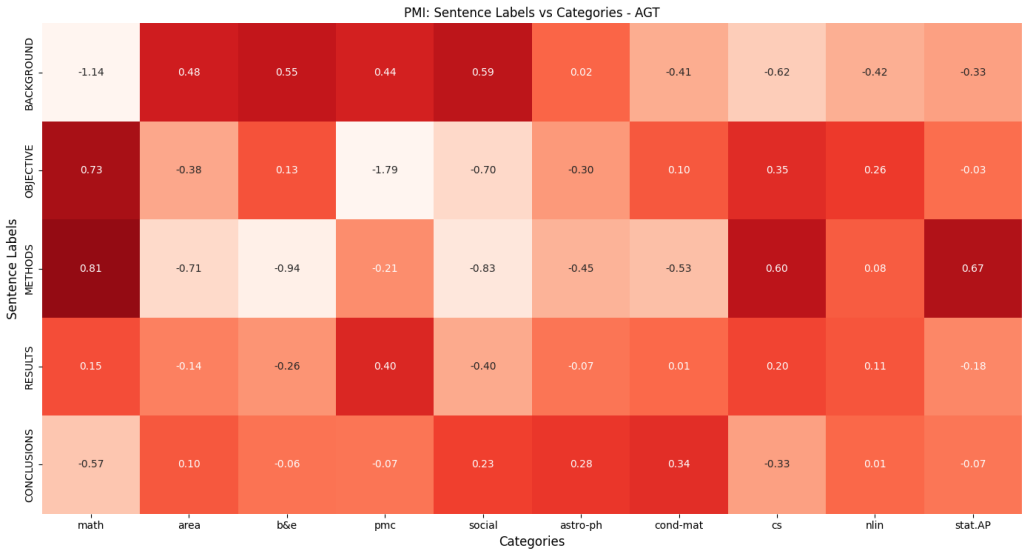


Figure D.3.: Pointwise Mutual Information values between sentence labels and scientific categories for AGT.

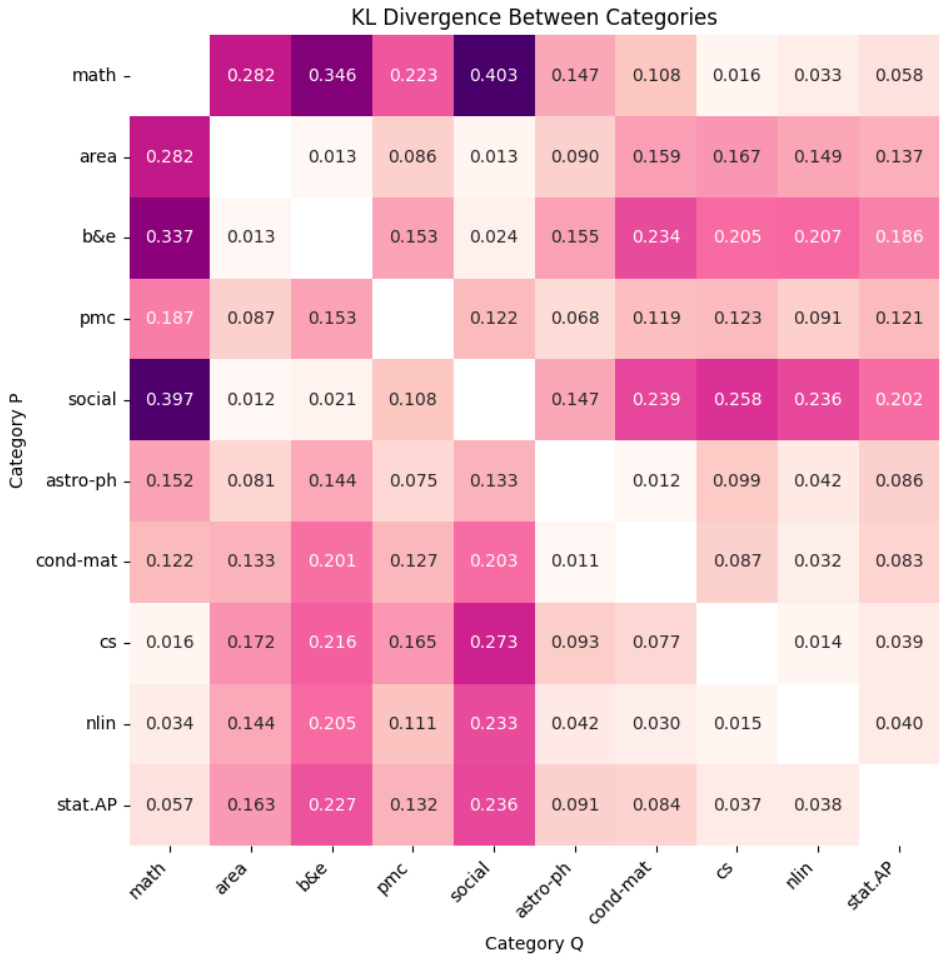


Figure D.4.: KL Divergence heatmap of word distributions across scientific categories, combining all text types.

E. Model Settings

This section provides the fine-tuning settings used for training the SciBERT-based sentence label classifier. Table E.1 lists the training parameters.

Parameter	Value
Pretrained Model	`allenai/scibert_scivocab_uncased`
Tokenizer Used	AutoTokenizer from pretrained model
Max Token Length	512 tokens
Batch Size (train/eval)	32
Number of Epochs	10
Learning Rate	2×10^{-5}
Optimizer	AdamW (default)
Gradient Accumulation Steps	1 (default)

Table E.1.: General fine-tuning settings for SciBERT sentence classification.

In Table E.2, the training parameters for the SciBERT-based sentence classifier (Baseline Model) are outlined, including the model, tokenizer, and hyperparameters.

Parameter	Value
Pretrained Model	`allenai/scibert_scivocab_uncased`
Tokenizer Used	AutoTokenizer from pretrained model
Max Token Length	256
Batch Size (train/eval)	32
Number of Epochs	10
Learning Rate	2×10^{-5}
Optimizer	AdamW
Weight Decay	0.01
Gradient Accumulation Steps	1 (default)

Table E.2.: Training Parameter for fine-tuning the Baseline Model.

Table E.3 summarizes the hyperparameters and configurations used for fine-tuning the large language model (LLM).

Parameter	**Value**
Pretrained Model	`LLama-3.1-8B-Instruct`
Tokenizer Used	AutoTokenizer from pretrained model
Max Token Length	256
Train Batch Size	32
Eval Batch Size	8
Number of Epochs	6
Learning Rate	2×10^{-4}
Learning Rate Scheduler Type	cosine
Tokenizer Padding side	right
Bits and bytes quantization	4-bit
LoRA rank	64
LoRA alpha	64
LoRA dropout	0.1
Weight Decay	0.001
Max Grad Norm	0.3
Warm Up Ratio	0.03
Target Modules	all-linear
Task Type	SEQ_CLS

Table E.3.: Training Parameter for fine-tuning the LLM.

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